

Estimation of voters' preferences

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Abstract

This paper develops a method to recover voters' preferences from aggregate election returns under spatial voting. The method exploits the convex partitions of the policy space induced by candidate positions, and requires only that a voter's utility decrease in the distance between the candidate's position and the voter's ideal point. I show that this method allows for separate identification of voters' ideal-point distributions and dimension weights. I apply it to study the evolution of preferences (ideal points and weights) among voters of the populist radical-right party Alternative for Germany (AfD), using municipal-level Bundestag election results from 2013–2025. The policy space is two-dimensional, defined by left–right and anti-elite positions. The results indicate that over time the AfD electorate shifted from relatively stronger anti-elite positions towards ideological alignment with the party's radical-right platform. Counterfactual simulations suggest that a moderate hardening of Left–Right positions, and immigration policy in particular, could allow establishment parties to capture up to 40% of the AfD electorate in the latest 2025 elections.

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1 Introduction

Many contemporary electoral developments, such as the rise of populist parties in Europe and partisan realignment in the United States, reflect shifts in political equilibrium. Explaining these patterns requires understanding the motivations underlying both the supply side and the demand side of politics. This paper focuses on the latter and seeks to recover the voters’ preferences that rationalize observed electoral outcomes.

Voting is distinctive because the key object of interest – individual vote choice – cannot be observed directly, as neither actual votes nor reliable stated preferences are typically available. This paper develops a nonparametric revealed-preference method to recover voter preferences from *aggregate* election returns within a spatial voting framework. The approach relies solely on the geometry of choice implied by the model, imposes no additional distributional assumptions, and is broadly applicable across electoral settings in which voters face variation in choice environments.

I apply this approach to examine the evolution of populist voters’ preferences over time, focusing on supporters of the Alternative for Germany (AfD), a populist radical-right party in Germany. A central debate in the literature is whether populist support is driven more by *thin* ideology or *thick* ideology, that is, by anti-elite appeals and rhetoric or by broader ideological stances¹. The results of this paper suggest that the reasons for populist support could evolve, and if earlier AfD support was more mixed, later AfD support is strongly rooted in party’s radical right stance. I further show that these patterns differ between East and West Germany, suggesting that populist voting varies across time and space and is shaped by institutional and historical contexts.

In the first part of the paper, I develop a nonparametric approach to recovering voters’ preferences from aggregate election returns that relies only on the spatial voting model and imposes no additional assumptions on preference heterogeneity. The key observation is that, given aggregate election outcomes, there are generally infinitely many preference distributions that are consistent with the data. This fundamental underidentification can be addressed in at least two ways. One approach is to add structure, either by strengthening assumptions on the choice model or by imposing a structure on unobserved heterogeneity.² Another approach is to answer the questions of interest without attempting to recover a single best-fitting preference distribution, at the cost of obtaining bounds on the parameters rather than point estimates. This paper adopts the latter approach.

Specifically, I focus on bounding *target parameters*, namely interpretable functionals of the

¹This debate is centered around two different conceptual understandings of the phenomenon of populism – see [Mudde and Kaltwasser \(2017\)](#); [Vries and Hobolt \(2020\)](#) for a discussion of the approaches used to conceptualize the phenomenon, or [Guriev and Papaioannou \(2022\)](#) for an overview of political economy literature.

²This is the approach taken, among others, by [Merlo and De Paula \(2017\)](#); [Iaryczower, Montero and Kim \(2024\)](#).

underlying preference distribution (e.g., moments). For example, a target parameter could be the share of individuals whose ideal points lie in a given region of the policy space, such as the mass of voters with left-leaning views along the Left–Right dimension. Methodologically, the analysis builds on [Tebaldi, Torgovitsky and Yang \(2023\)](#), who develop a nonparametric revealed-preference framework to estimate demand for health insurance plans in the California marketplace.

In the second part of the paper, I apply the framework to Germany, where the electoral landscape has undergone a pronounced transformation over the past decade. The central empirical object in this paper is the electoral base of the very successful populist party – Alternative for Germany (AfD), a party that combines radical-right positions with an explicitly anti-establishment (anti-elite) appeal.

The analysis leverages rich within-federal states variation in political supply. In each election year (2013, 2017, 2021), voters in each of Germany’s 299 electoral districts face a district-specific set of party-affiliated candidates. I estimate candidate positions on two dimensions—Left–Right and Anti-Elite—using the Bundestag candidate surveys and a complementary candidate questionnaire dataset. Positions are constructed from multiple survey items per dimension (e.g., redistribution and immigration for Left–Right; trust in institutions and referendums for Anti-Elite), standardized, and aggregated into indices.

On the voters’ side, the unit of analysis is an age–gender–education group within a state and year (e.g., female, 18–24, higher education, Saxony, 2021). I combine municipality-level party-list vote shares for the six main parties (AfD, CDU, FDP, Greens, Left, SPD) with municipality demographics in 24 age–gender–education bins. Together, these data allow the model to relate observed vote shares to latent preference heterogeneity across demographic groups.

Two institutional features are important for implementation. First, Bundestag elections are held under a mixed electoral system: voters cast one ballot for a district candidate and one for a party list. I focus on party-list vote shares and treat them as sincere among the six major parties, while using district candidates’ positions as proxies for locally perceived party positions.

Second, the identifying variation comes from differences in candidate positions across electoral districts within a state and year. The identifying assumption is that within-state variation is captured by the observed age–gender–education composition of municipalities and is unrelated to unobserved voter heterogeneity across districts. I view this assumption as plausible because, in practice, single-member district candidates are approved by the state party leadership, which is typically more concerned with advancing preferred candidates to the Bundestag than with responding to specific local voter demands. I test the robustness of my results to this assumption in two ways: first, by focusing on small municipalities, which are least likely to be decisive for candidates’ electoral success, and second, by restricting the analysis to non-contender AfD candidates, who have weaker incentives

to cater to the local electorate.

To characterize who AfD voters are, I bound target parameters defining a distribution of preferences of AfD voters. In particular, I compare the share of AfD supporters who would prefer a counterfactual party that is slightly more pro-elite (holding the radical-right stance fixed) versus a counterfactual party that is less right-wing (holding anti-elite positioning fixed) in a hypothetical scenario of AfD split. This delivers an interpretable measure of whether AfD support is anchored more in anti-elite sentiment or in ideological radical-right preferences, and how this composition evolves across elections. The results indicate a clear time trend: AfD support becomes increasingly rooted in the radical-right dimension, while the anti-elite component is strongest around 2017 and declines markedly by 2021. This aggregate pattern hides substantial regional heterogeneity. In Western Germany, AfD support is never unambiguously more anti-elite than ideological, whereas in Eastern Germany AfD voters are estimated to be substantially more anti-elite in earlier years, followed by a pronounced shift toward radical-right alignment—especially between 2017 and 2021.

Although the model does not allow for point identification of the weights voters attach to different ideological dimensions, it does deliver set identification, and the precision of the set is ultimately an empirical question. In practice, the model yields informative sets for relative weights. Estimating the identified sets separately for East and West Germany, I find that the left–right dimension was consistently more important for AfD voters than the anti-elite dimension across time and space, with the exception of Eastern Germany in 2013, when the anti-elite dimension is estimated to be slightly more important.

These exercises help trace the evolution of voters’ preferences, but they do not determine which dimension matters more for the AfD’s success, since that depends jointly on preference weights, bliss-point distributions, and the positions of competing parties. To address this, I use a Shapley value decomposition, a method widely used in computer science to quantify the contribution of variables of interest in complex models, and ask: *How much of AfD support is explained by its right-wing position versus its anti-elite stance?* The results indicate that the AfD’s right-wing stance was of great importance in 2021, accounting for up to 80% of its support, whereas support in earlier years reflects a more mixed contribution of the two dimensions.

Next, I decompose the change in AfD support between 2017 and 2021 into compositional change, supply-side position change, and within-group preference evolution. The results suggest that a large share of the change is driven by within-demographic-group preference evolution, while also highlighting a role for party positions and, in the case of right-wing voters in Western Germany, compositional demographic change.

Finally, I benchmark these findings against competing approaches. Replicating the same coun-

terfactual using German Election Study survey data yields broadly similar time patterns in ideal points distribution but also highlights survey issues: AfD vote choice is underreported, and—among self-reported AfD voters—anti-elite motivations appear disproportionately large, consistent with social-desirability pressures that may depress stated right-wing preferences. This suggests that survey-based approaches may be informative, once their usual limitations are properly taken into account. Comparing to a standard parametric spatial-choice model further shows that parametric functional-form and distributional assumptions can materially affect inference: the parametric estimates lie outside the nonparametric identified set, underscoring the value of the revealed-preference, assumption-lean approach for recovering the voters’ preferences.

Related Literature. This paper is related to several strands of literature. First, to the growing literature that estimates structural models of voting in elections (Coate and Conlin, 2004; Ujhelyi, Chatterjee and Szabó, 2021; Iaryczower, Montero and Kim, 2024; Kawai and Sunada, 2025; Berry, Cox and Haile, 2025; Longuet-Marx, 2025). This paper differs from these in that its underlying framework is a multi-dimensional spatial voting model, in which individuals vote for the candidate closest to them in the weighted policy space.³

The paper is closely related to Degan and Merlo (2009), who first formalize partitioning the policy space under the spatial model of voting using Voronoi cells, and to Merlo and De Paula (2017), who explore nonparametric identification of voter preferences under such partition. Conceptually this paper is closest to the latter. Merlo and De Paula (2017) obtain identification under a large-support assumption on candidates’ positions and estimate the model with a sieve minimum-distance estimator. This approach fixes the curvature of the bliss-point distribution and recovers that distribution and the weights jointly, so *both* can be biased if the imposed distribution is too inflexible relative to the true one.⁴ This paper instead uses Voronoi cells and their superimposition, requiring neither a full-support assumption on candidates’ positions for identification nor estimation of the exact voter preference distribution. It separates the problem into recovering (i) the weights and (ii) the bliss-point distribution; this comes at the cost of potentially not obtaining the single best-fitting weight and bliss-point distribution, but identified sets for each.

Second, this paper is related to the literature that exploits different methodologies to retrieve voters’ preferences including the use of survey answers to the questions on political attitudes, self-positioning and mapping them to ideologies (Fiorina and Abrams, 2008; Draca and Schwarz, 2024;

³Of these papers, only Iaryczower, Montero and Kim (2024) estimates a spatial voting model, and only in one dimension. As noted in Merlo and De Paula (2017), estimation of a single-dimensional model is considerably easier, since the problem of separately identifying *weights* from bliss points does not arise.

⁴The flexibility of the estimator is limited by the amount of data, so the problem is especially acute in small samples. I provide a Monte-Carlo simulation of weight estimation using the approach of this paper and the approach from Merlo and De Paula (2017) in the Appendix Section B.

Algan et al., 2017; Gethin, Martínez-Toledano and Piketty, 2022; Hangartner et al., 2019; Desmet, Ortuno-Ortín and Wacziarg, 2024). The common shortcomings of these methods are that the views voters held do not necessarily translate into voting choices, as well as that some views might be stigmatized or demanded and not expressed truthfully. In other words, survey-based stated preferences can differ significantly from revealed preferences inferred from electoral outcomes.

Thirdly, this paper contributes to the large and growing literature on the causes of the rise and success of the populist and anti-establishment parties and leaders in the Western World. There is an extensive literature examining the various drivers of populist voting, including economic factors such as trade and the 2008–2009 Global Financial Crisis (Autor et al., 2020; Colantone and Stanig, 2018*b,a*; Algan et al., 2017), particularly in the German context (Dippel et al., 2022), as well as immigration (Dustmann, Vasiljeva and Piil Damm, 2019; Alesina and Tabellini, 2024) and cultural factors (Margalit, 2019; Algan et al., 2018; Gennaioli and Tabellini, 2025), among others. While these drivers of populist support could be neatly mapped into different ideological dimensions (see, e.g. Rodrik (2021) for a clear formalization of this logic and meta-analysis for mechanisms related to globalization shocks), they do not allow us to understand the relative importance of each, or their interplay. This paper addresses this gap by estimating the preferences of the populist voters, and exploring its evolution. In this respect, it is closely related to Danieli et al. (2023), which studies the rise of radical-right populist parties in Europe. Unlike this paper, however, that study relies on survey-based self-reported vote choices and issue preferences; moreover, while its conceptual framework can be cast as a bliss-point model, it cannot separate a strengthening association between individual characteristics and ideal points on a given dimension from a change in the weight on that dimension.

Lastly, this paper contributes to the debate on the importance of ideological and populist (anti-establishment) dimensions in populist voting. A large part of the literature emphasizes the role of host ideologies in shaping populist support Acemoglu, Egorov and Sonin (2013); Mudde (2013); Norris and Inglehart (2019); Guenther and Nunnari (2025), whereas another strand adopts an *ideational* approach and highlights the role of populist attitudes distinct from ideology Mudde (2004); Akkerman, Mudde and Zaslove (2014); Van Kessel (2015). Several studies have sought to compare the importance of these two factors directly using survey and experimental data (Silva, Fuks and Tamaki, 2022; Neuner and Wratil, 2022; Neuner et al., 2025). This paper shows that populist voting can take different forms, with different factors prevailing across regions and over time.

The rest of the paper is structured as follows. Section 2 provides an overview of the non-parametric model of choice, and Section 3 describes the scope of the approach. Section 4 describes institutional context and data. Section 5 covers estimation and identification details. Section 6 provides the main results of the paper, and Section ?? compares the results of this paper with results one could obtain using surveys or parametrized models of party choice. Section 7 concludes.

2 Nonparametric Model of Party Choice

This section develops a nonparametric model of choice in a voting environment and outlines its identification. It draws on [Merlo and De Paula \(2017\)](#), which formalizes the logic of Voronoi partitions in the estimation of voter preferences, and on [Tebaldi, Torgovitsky and Yang \(2023\)](#), which develops a nonparametric approach for discrete choice models. Relative to the former, the framework differs in two key respects: identification does not rest on the assumption of a large support for candidate positions, and it does not aim to simultaneously recover a single distribution of voter preferences together with weights, but does it separately. Relative to the latter, it shifts attention from a product-valuation model in a price space to a pure characteristics model in a policy space, in which the main object of interest is individuals' preferences⁵, and extends that model by estimating the relative importance of the different dimensions.

The section proceeds in three steps. Subsection 2.1 introduces the primitives, the identifying assumptions, and the target parameters (functionals of bliss point distributions). Subsection 2.2 shows that for known weights the estimation problem for target parameters is equivalent to a finite-dimensional linear problem. Subsection 2.3 establishes the set identification of weights, and combines the two steps into a single algorithm.

2.1 Setup

Individual i chooses a candidate⁶ j from a set $\mathcal{J} = \{1, \dots, J\}$. Each choice environment is characterized by a vector of potentially endogenous candidate positions $Y_i = (Y_{i1}, \dots, Y_{iJ})$. An individual may face one or more such environments at the same time. In an electoral setting, the ballot provides a natural representation of this set. Individual i has a bliss point position X_i known to themselves, but latent from the perspective of the researcher.

I adopt a spatial voting model ([Downs, 1957](#); [Hinich and Munger, 1996](#)) and assume that valuations are a function of distance between the individual's own bliss point and the positions of the candidates:

$$V_{ij} = V(X_i, Y_{ij}) = U(d(X_i, Y_{ij})),$$

where $d(X_i, Y_{ij})$ denotes the Euclidean distance between the two points in space, and $U(\cdot)$ is a strictly decreasing function. Note that this functional form does not presuppose any specific cardinal loss function, and is invariant to strictly monotone transformations of the (weighted) Euclidean distance.⁷

⁵See [Berry and Pakes \(2007\)](#) for an extensive discussion of the differences between these models.

⁶*Candidate* is used here to denote any choice option appearing on the ballot, including parties, individual candidates, or referendum choice options, among others.

⁷See [Kendall, Nannicini and Trebbi \(2015\)](#); [Cruz et al. \(2024\)](#) for the estimation of the functional form of the loss function. The analysis is not invariant, however, to arbitrary alternative notions of distance. This is because the identification argument relies on the geometry of the regions of the ideological space consisting of points closest

For simplicity of exposition, in this and the next subsection I consider the unweighted Euclidean distance ($d(X_i, Y_{ij}) = \sqrt{(X_i - Y_{ij})'(X_i - Y_{ij})}$). Subsection 2.3 extends the analysis to the weighted Euclidean distance and discusses separate identification of the weights.

The policy space $\mathcal{R}$ is assumed to be nonempty and compact. For the purposes of this paper I focus on the two-dimensional policy space, $\mathcal{R} \subset \mathbb{R}^2$, and, hence, $X_i, Y_{ij} \in \mathcal{R}$. An individual i 's choice is:

$$\text{Choice}_i = \text{Choice}(X_i, Y_i) = \arg \max_{j \in \mathcal{J}} V_{ij} \quad (1)$$

The primitive object of interest is the conditional distribution of bliss points X_i , given candidate positions Y_i , market (region) M_i , and demographic covariates D_i . I assume throughout that this conditional distribution is absolutely continuous, so that, for each realization $Y_i = y$, $M_i = m$, and $D_i = d$, it admits a density $f(x \mid y, m, d)$; since the set of bliss points indifferent between any two candidates has Lebesgue measure zero, ties between choices then occur with probability zero. Let $\mathcal{F}$ denote the set of all such conditional density functions.

Conditioning on Y_i allows for the possibility that the choice environment faced by an individual is *endogenous*: candidate positions may reflect voter characteristics beyond the observed demographics d and market m , in which case $f(x \mid y, m, d)$ varies with y . I assume there exists an observed variable Z_i (an instrument) such that, for all (z, m, d) and all y in the support of Y_i given $(Z_i, M_i, D_i) = (z, m, d)$, the conditional density of bliss points satisfies

$$f(x \mid y, z, m, d) = f(x \mid z, m, d) \quad \text{for all } x. \quad (2)$$

The leading case is when candidate positions are unrelated to voters' unobserved characteristics conditional on (m, d) alone; the assumption then is $f(x \mid y, m, d) = f(x \mid m, d)$ for all y . To save on notation, I proceed under this special case throughout; when a non-degenerate instrument is required, every argument below applies with z added to the conditioning set (m, d) , and objects conditional on (m, d) alone are recovered by averaging over the observed distribution of the instrument, $f(x \mid m, d) = \mathbb{E}[f(x \mid Z_i, m, d) \mid M_i = m, D_i = d]$. The assumption has identifying content only if the instrument is *relevant*, i.e., if candidate positions vary with the instrument. Section 3 discusses institutional settings, and identification strategies from the literature, under which the Assumption 2 is plausible.

Common quantities of interest can be written as functionals of f . For instance, one could be interested in estimating a share of individuals with given demographics with bliss points that are

to each candidate, i.e. Voronoi cells. Alternative metrics may generate different choice regions. For example, in a multidimensional space, using the sum of absolute coordinate-wise differences between a bliss point and a candidate position corresponds to imposing Manhattan, rather than Euclidean, geometry.

left-leaning on a given dimension, $\underline{\mathcal{X}} = \{x \in \mathcal{R} : x_1 \leq \underline{x}_1\}$ for a known threshold $\underline{x}_1$:

$$\theta(m, d; f) = \int_{\underline{\mathcal{X}}} f(x \mid m, d) dx$$

Denote values of interest similar to the one just above as *target parameters*. Other quantities of interest could characterize support for political alternatives or the consequences of institutional rules – for instance, the share of candidate j 's supporters with bliss points in a given region, or counterfactual vote shares under alternative positions or rules. The goal is to infer the values of $\theta(m, d; f)$ that are consistent with both the observed data and the assumptions imposed.

The first assumption is that the support of f is contained in a known subset of $\mathcal{R}$. In the case of the voting application, this naturally translates into assuming a bounded policy space, preventing unreasonably extreme ideal points:

$$\int_{\bar{\mathcal{X}}} f(x \mid m, d) dx = 1 \quad \text{for all } (m, d), \text{ where } \bar{\mathcal{X}} \text{ is a known subset of } \mathcal{R} \quad (3)$$

In voting contexts, only vote shares aggregated over specific administrative units (e.g. municipalities, precincts) are observed; I refer to these units as *submarkets* of a market and index them by s , with $\mathcal{S}(m)$ denoting the submarkets of market m . For each submarket a researcher observes its demographic composition; denote $q_{s,d}$ to be the share of group with $D_i = d$ in submarket s , and $\sum_d q_{s,d} = 1$. The observed data for the submarket s in market m with candidate positions y are then:

$$s_j(y, s) = \sum_d q_{s,d} \mathbb{P}[\text{Choice}_i = j \mid Y_i = y, M_i = m, D_i = d]$$

Each density of bliss points $f \in \mathcal{F}$ also implies a set of vote shares:

$$s_j(y, s; f) = \sum_d q_{s,d} \int_{\mathcal{X}_j(y)} f(x \mid m, d) dx$$

where $\mathcal{X}_j(y) = \{x \in \bar{\mathcal{X}} : V(x, y_j) > V(x, y_k) \ \forall k \neq j\}$ is the attraction region of candidate j under positions y , with y_j the position of candidate j in configuration y . A density f is consistent with the observed choice shares, if

$$\underbrace{s_j(y, s)}_{\text{observed}} = \underbrace{s_j(y, s; f)}_{\text{predicted}} \quad \text{for all } j \text{ and all pairs } (y, s) \text{ observed in the data} \quad (4)$$

The sharp identified set of bliss-point densities is defined as the subset $\mathcal{F}^*$ of $f \in \mathcal{F}$ that satisfy (2), (3), (4). The main object of interest in the paper throughout is the sharp identified set for the target parameter, constructed as:

$$\Theta^* = \{\theta(f) : f \in \mathcal{F}^*\}$$

2.2 Equivalent finite-dimensional problem

How could one compute Θ^* ? Note that $\mathcal{F}$ is an infinite-dimensional object, which does not allow solving the problem via grid search. [Tebaldi, Torgovitsky and Yang \(2023\)](#) develop an approach for checking whether there exists some $f \in \mathcal{F}^*$ that would generate a given conjectured value $t = \theta(f)$ for the target parameter. Their paper proves that the existence of such an f is equivalent to the existence of a solution to a finite-dimensional system of linear equations.

The finite-dimensional system is constructed by first partitioning the policy space into the smallest number of sets needed to describe choice behavior under all relevant candidate positions. Let $\mathcal{Y}$ denote a finite set of candidate position vectors that contains the observed candidate positions, as well as any additional positions relevant for evaluating the target parameter. The Minimal Relevant Partition (MRP) is the coarsest partition of the policy space $\mathcal{R}$ with the following property: any two bliss points in the same subset produce the same choice behavior for all $y \in \mathcal{Y}$, while any two bliss points in different subsets produce different choice behavior for at least one $y \in \mathcal{Y}$. Formally, the MRP is the coarsest partition $\mathbb{X}$ of $\mathcal{R}$ such that

$$x, x' \in \mathcal{X} \text{ for some } \mathcal{X} \in \mathbb{X} \text{ implies } \text{Choice}(x, y) = \text{Choice}(x', y) \text{ for all } y \in \mathcal{Y}.$$

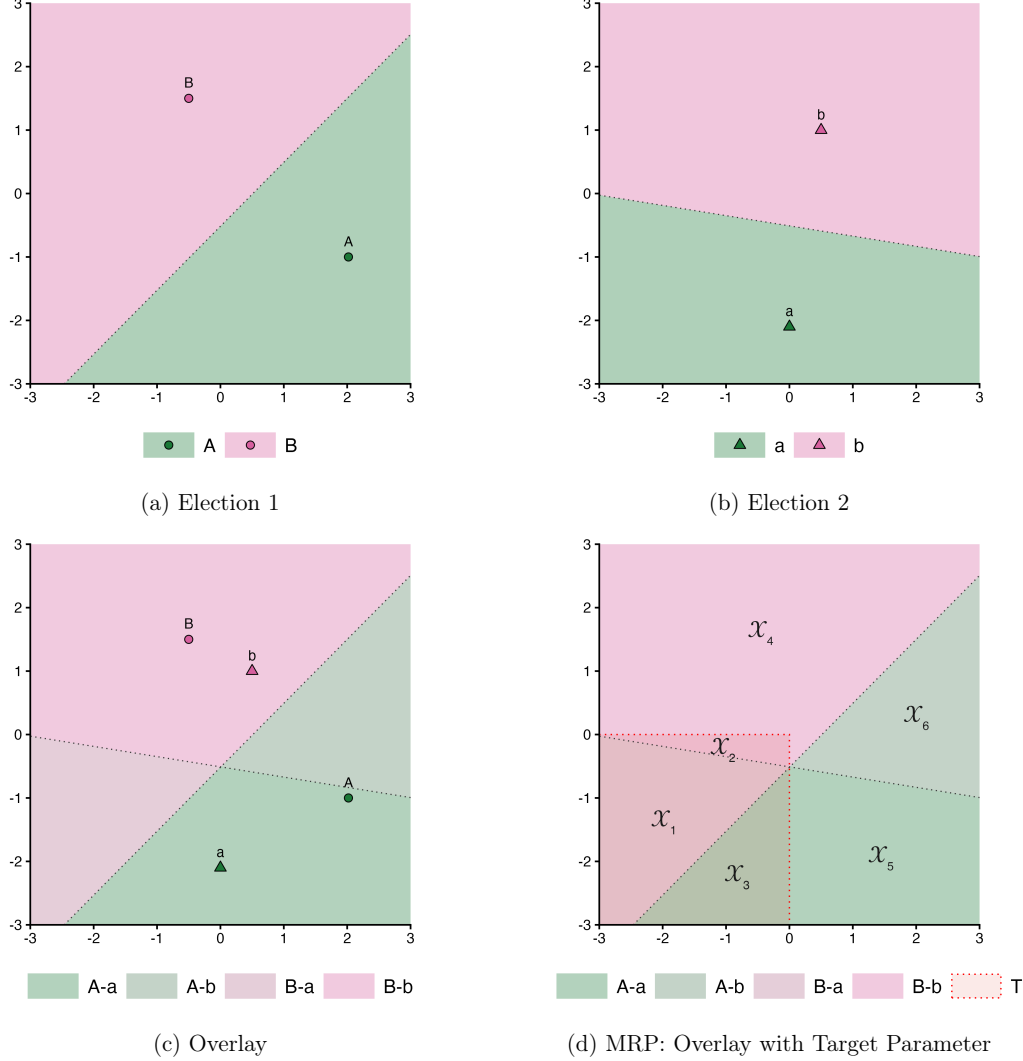
In other words, whenever two different bliss points x, x' belong to the same set from the MRP, this implies that x, x' yield the same choice behavior across all choice environments in $\mathcal{Y}$. Note two distinctive features of this setup. First, it does not require researchers to normalize any of the options. Intuitively, this is because individuals' valuations of the options are well defined relative to one another (see discussion in [Merlo and De Paula \(2017\)](#) on identification of the spatial voting model). Second, because only relative comparisons between candidates are exploited, the framework does not naturally accommodate the turnout decision, which would require an absolute utility threshold for abstention⁸.

Figures 1(a) and 1(b) illustrate the partitions of the policy space generated by two hypothetical position vectors, y_1 and y_2 , in a setting where individuals face different candidate choice sets across elections. For consistency with the application that follows, the illustration uses a two-dimensional policy space. Figure 1(c) shows the two partitions overlaid.

This mirrors the structure of the application ahead, where voters face different choice environ-

⁸In the application ahead, I account for the differential turnout rates across demographic groups across markets.

ments across geographic areas: individuals in the same market m confront different candidate sets (y_1 and y_2 in Figure 1) depending on the electoral district to which they belong. The identifying variation comes from the differences in choice environments of individuals with plausibly similar preferences (same market m , demographics d).



Notes: The figure illustrates a hypothetical setting with elections involving two candidates in each contest, (A, B) and (a, b) . In the two upper panels, the shaded pink region denotes the attraction area of candidates B and b , the green region – of candidates A and a in Elections 1 and 2, respectively. Panel (c) shows the overlay of the partitions in panels (a) and (b). Panel (d) presents the same overlay together with the region corresponding to a target parameter. $\mathcal{X}_n$ denote unique cells that characterize observed behavior in Elections 1 and 2 or a target parameter.

Figure 1. Example of partitions of the policy space in two elections with two parties ((a), (b)) and their overlay (c) with a target parameter (d).

The MRP is obtained by partitioning the continuum of bliss points into sets that generate identical choice behavior for all positions in $\mathcal{Y}$. When the object of interest is behavior under these candidate positions, attention can therefore shift from densities to *mass functions defined over the*

sets of the MRP.

To illustrate this, consider the target parameter defined above, share of voters with left-leaning bliss points. Figure 1 (d) graphically represents this quantity as the share of voters whose bliss points lie in area T , where $T = \{x \in \bar{\mathcal{X}} : x \leq (0, 0)\}$. In the MRP representation, evaluating such a target parameter requires refining the partition by the boundary of T , so that T can be represented as a union of MRP cells. In the Figure 1 (d) T consists of three subsets, $\mathcal{X}_1, \mathcal{X}_2, \mathcal{X}_3$. Hence, the target parameter can be written as:

$$\theta(m, d; f) = \int_{\mathcal{X}_1} f(x | m, d) dx + \int_{\mathcal{X}_2} f(x | m, d) dx + \int_{\mathcal{X}_3} f(x | m, d) dx,$$

which depends not on f per se, but only on the mass that f places on $\mathcal{X}_1, \mathcal{X}_2, \mathcal{X}_3$. Focusing on these masses transforms the intractable infinite-dimensional problem of searching over all $f \in \mathcal{F}$ into a tractable finite-dimensional problem of searching over a finite set of non-negative numbers that sum to unity.

Denote the set of conditional mass functions on $\mathbb{X}$:

$$\Phi = \{\phi \in \mathbb{R}_+^{d_\phi} : \sum_{\mathcal{X} \in \mathbb{X}} \phi(\mathcal{X} | m, d) = 1 \ \forall (m, d) \in \text{supp}(M_i, D_i)\},$$

where d_ϕ is the cardinality of $\mathbb{X} \times \text{supp}(M_i, D_i)$. Denote $\mathbb{X}_j(y)$ the sets in the MRP associated with choosing j under y . Then $\phi \in \Phi$ is consistent with some f that satisfies 4 only if it satisfies:

$$\underbrace{s_j(y, s)}_{\text{observed}} = s_j(y, s; \phi) = \underbrace{\sum_{\mathcal{X} \in \mathbb{X}_j(y)} \sum_d q_{s,d} \phi(\mathcal{X} | m, d)}_{\text{predicted}} \quad \forall j \text{ and observed } (y, s). \quad (5)$$

Similarly, $\phi \in \Phi$ is consistent with an f that satisfies 3 only if it satisfies:

$$\sum_{\mathcal{X} \in \bar{\mathbb{X}}} \phi(\mathcal{X} | m, d) = 1 \text{ for all } (m, d) \quad (6)$$

where $\bar{\mathbb{X}}$ is the subset of cells of $\mathbb{X}$ that are contained in $\bar{\mathcal{X}}$ (the partition can always be refined so that every cell lies either inside or outside $\bar{\mathcal{X}}$).

For any $f \in \mathcal{F}$, define $\bar{\phi}(f) \in \Phi$ as

$$\bar{\phi}(f)(\mathcal{X} | m, d) = \int_{\mathcal{X}} f(x | m, d) dx \quad \text{for all } \mathcal{X} \in \mathbb{X}, (m, d) \in \text{supp}(M_i, D_i).$$

Next, assume that the target parameter is representable in terms of the cell masses: there exists a known function $\bar{\theta} : \Phi \rightarrow \mathbb{R}^{d_\theta}$ such that

$$\theta(f) = \bar{\theta}(\bar{\phi}(f)) \quad \text{for all } f \in \mathcal{F}. \quad (7)$$

This assumption can always be satisfied by choosing $\mathcal{Y}$ (and hence the partition) to include all positions and regions relevant for evaluating the target parameter; in the leading example above, $\bar{\theta}(\phi) = \phi(\mathcal{X}_1 \mid m, d) + \phi(\mathcal{X}_2 \mid m, d) + \phi(\mathcal{X}_3 \mid m, d)$.

The main propositions in [Tebaldi, Torgovitsky and Yang \(2023\)](#) can then be adapted to the voting context as follows:

Proposition 1. *Suppose that 7 is satisfied. Let $\Phi^* \equiv \{\phi \in \Phi : \phi \text{ satisfies 5, 6}\}$, and for any $t \in \mathbb{R}^{d_\theta}$, let $\Phi^*(t) \equiv \Phi^* \cap \{\phi \in \Phi : \bar{\theta}(\phi) = t\}$. Then $t \in \Theta^*$ iff $\Phi^*(t) \neq \emptyset$.*

The nontrivial direction of Proposition 1 is sufficiency: any $\phi \in \Phi^*(t)$ can be extended to a density $f \in \mathcal{F}^*$ with $\theta(f) = t$ by distributing the mass assigned to each cell arbitrarily (for instance, uniformly) within that cell; see Appendix in [Tebaldi, Torgovitsky and Yang \(2023\)](#) for the formal argument.

Proposition 2. *If $\bar{\theta}$ is continuous on Φ and $d_\theta = 1$, then $\Theta^* = [t_{lb}^*, t_{ub}^*]$, where*

$$t_{lb}^* = \min_{\phi \in \Phi^*} \bar{\theta}(\phi), \quad t_{ub}^* = \max_{\phi \in \Phi^*} \bar{\theta}(\phi).$$

Existence of the extrema and the interval form follow because Φ^* is a compact convex polytope and $\bar{\theta}$ is continuous; when $\bar{\theta}$ is linear in ϕ , as in the leading example, both bounds are computed by linear programming.

2.3 Weighted spatial voting model

Assume individual valuations of choice options can be expressed as:

$$V_{ij} = V(X_i, Y_{ij}; W_i) = U(d_{W_i}(X_i, Y_{ij})),$$

where W_i is a weighting matrix whose diagonal elements govern the relative importance of the dimensions, and whose off-diagonal elements determine the interaction between the two dimensions. For computational and identification purposes, I set the off-diagonal elements to zero:

$$W_i = \begin{pmatrix} w_{i1} & 0 \\ 0 & w_{i2} \end{pmatrix}, \quad w_{i1}, w_{i2} > 0.$$

In the two-dimensional setup the distance is $d_{W_i}(X_i, Y_{ij}) = \sqrt{(X_i - Y_{ij})^\top W_i (X_i - Y_{ij})}$. The case considered before is a special case with $w_{i1} = w_{i2}$. Since choices depend only on the relative weights assigned to the dimensions, not on their absolute levels, I normalize $w_{i1} = 1$ without loss of generality and treat $w_{i2} \in \mathbb{R}_+$ as the relative weight of the second dimension.

Throughout this section I impose that the weighting matrix is common across individuals in a given market and denote: $W_i = W = \text{diag}(1, w_2)$ for all i . In the application, w_2 is allowed to differ across region-years but is common within them.

This subsection demonstrates that one can separately identify the weight w_2 and the probability mass function of bliss points ϕ and derive the sharp bounds for the target parameters using those.⁹

The two observational-equivalence conditions 4, 5 for f and ϕ respectively can now be recast as follows. The predicted vote share of candidate j is:

$$s_j(y, s; f, w) = \sum_d q_{s,d} \int_{\mathcal{X}_{j;w}(y)} f(x \mid m, d) dx,$$

where $\mathcal{X}_{j;w}(y) = \{x \in \bar{\mathcal{X}} : V(x, y_j; w) > V(x, y_k; w) \ \forall k \neq j\}$. And it matches the data if

$$s_j(y, s; f, w) = s_j(y, s) \quad \forall j \text{ and observed } (y, s). \quad (8)$$

Likewise, for the equivalent finite dimensional problem, denote by $\mathbb{X}_w$ the MRP generated by $\mathcal{Y}$ under weight w , and by $\mathbb{X}_{j;w}(y)$ its cells on which j is chosen under y ; then

$$\sum_{\mathcal{X} \in \mathbb{X}_{j;w}(y)} \sum_d q_{s,d} \phi(\mathcal{X} \mid m, d) = s_j(y, s) \quad \forall j \text{ and observed } (y, s) \quad (9)$$

The algorithm for obtaining bounds for the target parameter in the presence of an unknown weight is then the following. First, obtain the identified set of weights over a candidate space $\mathcal{W} \subseteq \mathbb{R}_+$,

$$W^* = \{w \in \mathcal{W} : \exists \phi \in \Phi \text{ such that (9) and (6) are satisfied}\}.$$

⁹The closest argument in the literature is [Merlo and De Paula \(2017\)](#), who prove nonparametric point identification of weights and distributions of bliss points in a spatial voting model. Their analysis relies on a large support assumption for candidate positions. In particular, to establish joint identification of bliss points and weights, they rely on the results of [Ichimura and Thompson \(1998\)](#), who prove nonparametric identification in a binary discrete choice model using the Cramer–Wold device: there exists a two-candidate configuration such that the affine hyperplane separating the two candidates delivers different election outcomes under two distinct preference distributions. Building on these results, they extend them to 1) include multiple candidate positions, and 2) introduce weights. Both 1) and 2) are based on the large support assumption, i.e. that one could sample arbitrary candidate configurations from $\mathbb{R}^2$. This assumption is infeasible in political-economy environments, where the set of candidate positions is typically finite and confined to a bounded subset of $\mathbb{R}^2$. Hence, the setup of this paper accommodates the fact that there are just finitely many observed configurations in a bounded policy space.

Second, for each $w \in W^*$ obtain $\Theta^*(w)$ by applying Propositions 1–2 to the partition $\mathbb{X}_w$. The identified set for the target parameter is then $\Theta^* = \bigcup_{w \in W^*} \Theta^*(w)$. Since each $\Theta^*(w)$ is an interval, Θ^* is contained in

$$\left[\min_{w \in W^*} t_{lb}^*(w) ; \max_{w \in W^*} t_{ub}^*(w) \right],$$

and these are the bounds I report. Appendix D describes this estimation procedure in the application.

Note two features of this problem. First, the equations in the finite-dimensional problem (9) are defined by the available partition cells in a minimal relevant partition $\mathbb{X}_{j;w}(y)$, and these cells vary with the weight. Second, given the cells, the distribution of bliss points is matched by fitting $\phi(\mathcal{X} \mid m, d)$ over the available cells. Intuitively, the former is the source of identification for the weights, and the latter the source of identification for the bliss-point distribution: the weight is identified by *which* cells the data require, the mass function by *how much* mass those cells must carry. I now provide intuition for the argument for weight identification.

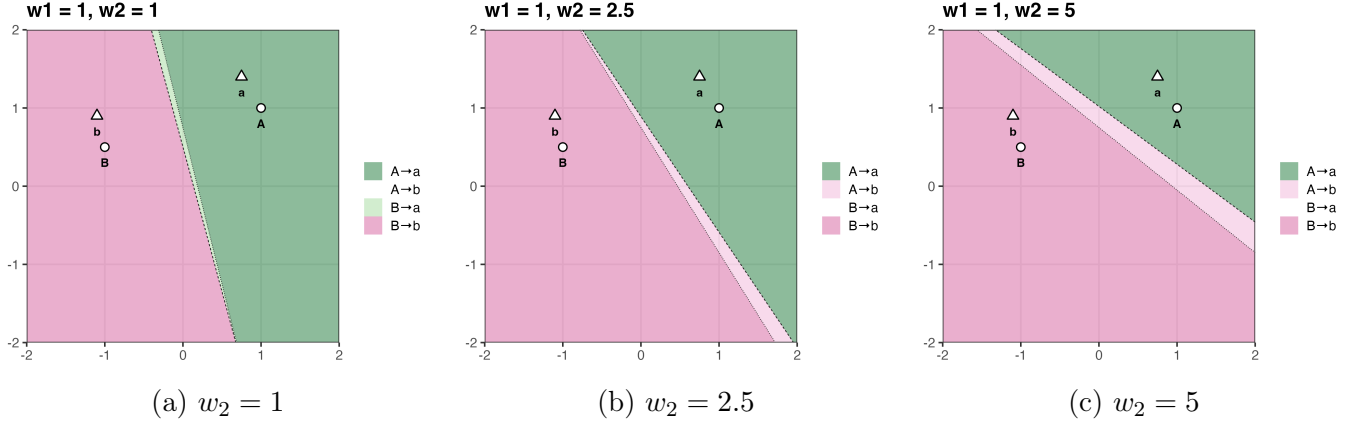
Imagine a researcher observes the results of two elections with two candidates each (candidates A, B and a, b , respectively). The goal is to use these two elections to obtain information about f and W . The observed marginals (vote shares) – s_A, s_B, s_a, s_b – are informative about the joint values $(s_{Aa}, s_{Ab}, s_{Ba}, s_{Bb})$ of interest. Consider the hypothetical scenario from Table 1, in which candidate A and candidate b obtain the majority of votes in respective elections.

	a	b	Total
A	[0, 0.1]	[0.6, 0.7]	0.7
B	[0, 0.1]	[0.2, 0.3]	0.3
Total	0.1	0.9	1

Table 1. Marginals (observed) and Fréchet bounds on the joint cell shares

This scenario implies that the cell A – b hosts at least 60% of all voters. Hence, the minimal relevant partition of those elections must contain a cell that would rationalize such voter behavior. The claim is that such a cell might not appear under every configuration of weights, and that the weights yielding the partition that contains this cell are best suited to explain the behavior of voters. Figure 2 illustrates this point and demonstrates that, in this hypothetical example, the weight configuration $w_2 = 1$ is not well suited to explain the data. Moreover, Figure 2 illustrates the nature of the set-identification of weights in this setup: $w_2 = 2.5$ and $w_2 = 5$ effectively yield the same partition of the policy space and cannot be told apart.

The example shows that weights are identified by the available cells in the minimal relevant partition. Appendix A proves that the same partition can be generated only by weights w_2 from an interval. Thus, w_2 is identified up to an interval on $\mathbb{R}_+$, rather than point-identified. The informative-



Notes: The figure illustrates a hypothetical setting with elections involving two candidates in each contest, (A, B) and (a, b) . Each panel overlays the two elections in the policy space $\mathbb{R}^2$: triangles mark the positions of candidates A and a , circles mark candidates B and b . The shaded regions denote the attraction areas of the joint outcomes: $A \rightarrow a$ (dark green), $A \rightarrow b$ (light green), $B \rightarrow a$ (light pink), and $B \rightarrow b$ (dark pink), where $X \rightarrow y$ denotes voters who prefer X in Election 1 and y in Election 2. The cells present in a given panel constitute the minimal relevant partition for that weight configuration. Across panels, the dimension weight w_2 increases ($w_1 = 1$ throughout).

Figure 2. Illustration of minimal relevant partitions induced by the same candidate configurations $(A, B), (a, b)$ under different values of w_2 , $w_1 = 1$.

ness (length) of this interval is then an empirical question. [Lieber, Tebaldi and Torgovitsky \(2026\)](#) provide a similar argument about the set-identification of relative weights for different characteristics in an additive RUM.

In practice there need not be a unique partition that best explains the data: because the number of feasible cells (over which voter mass is estimated) weakly exceeds the number of observed vote-share marginals, the problem is an under-identified linear system, and several partitions may rationalize the data equally well. Two observations are worth highlighting. First, as [Appendix A](#) shows, each cell is generated only by weights lying in an interval. Identifying power then comes from Fréchet-type necessity constraints¹⁰. Second, additional elections with distinct vote-share configurations weakly increase the number of such constraints. The admissible weights are those lying in the intersection of the intervals associated with every cell required to carry positive mass; since the data were generated at the true weight, every cell the data require is feasible there, so this intersection is a non-empty interval containing the truth, and it weakly shrinks as elections accumulate. The size and shape of the identified set are thus an empirical object.

[Appendix B](#) reports simulations comparing the approach of this paper—which exploits the fact that different weights generate different cells—with that of [Merlo and De Paula \(2017\)](#), who recover

¹⁰For example, if candidate (or coalition) i receives a share of 0.52 in election e and candidate j receives 0.51 in election e' , at least 3% ($0.52 + 0.51 - 1$) of voters must choose the pair (i, j) , so the corresponding cell must be non-empty.

the weights jointly with the distribution of bliss points using a sieve minimum-distance estimator. The simulations suggest that the true weight value belongs to the recovered set under the approach of this paper, whereas the weights recovered jointly with the bliss-point distribution fall outside the identified set. The intuition is that identification of the weights from the cell structure requires no assumption on the distribution of bliss points, so it is not contaminated when that distribution is misspecified; the sieve estimator, by contrast, must approximate the bliss-point density, and any residual misspecification is absorbed into the weight estimate. Appendix B also demonstrate that the identified set for weights is shrinking with a number of elections. Introducing noise in voters’ perceptions of candidate positions leaves this conclusion unchanged: the identified region tightens around the true weight, while the sieve estimates remain outside it.

The identification argument above does not rely on the weight being common across across demographic groups within a market, and a similar argument goes through when W varies across demographic groups too. In practice, however, group-specific weights are considerably more demanding on the data. Informative group-level intervals would require either group-level choice data or far richer variation in candidate configurations than is available in the application.

3 Scope of the Approach

This section discusses the settings in which the identifying assumption of the approach introduced in the previous section is plausible, highlights its broad applicability, and considers possible extensions and generalizations of the model, including the incorporation of strategic voting.

The approach offers a method for estimating voters’ preferences within a spatial voting framework using real-world aggregate voting data. Identification rests on comparing the behavior of voters who share the same preference distribution across different choice environments: either directly, $f(x \mid y, m, d) = f(x \mid m, d)$ for all y , or conditional on an instrument z for candidate positions, $f(x \mid y, z, m, d) = f(x \mid z, m, d)$. While such settings may at first seem uncommon, they are in fact pervasive, and closely related identifying assumptions are already used in the political economy literature and beyond. I organize the discussion around three types of settings.

First, consider simultaneous voting, where the same voters face different choice environments because several elections take place at the same time. The main advantage of this setting is that the identifying condition $f(x \mid y, m, d) = f(x \mid y', m, d)$ holds by construction. Its principal limitation, however, is that the number of simultaneous elections is typically small, resulting in a coarse partition of the policy space. Consequently, the inferences that can be drawn using this approach may be less informative.

Among other cases, simultaneous voting is used in the United States, where elections for multiple offices often take place on the same day; in Australia, where voters cast ballots for both chambers of parliament simultaneously; or in Switzerland, where voters may face multiple electoral ballots as well as referendum questions on the same election day.

Second, *ranked-choice voting (RCV)* provides another setting in which the same voters face varying choice environments. A ranking can be read as a sequence of choices from shrinking choice sets: the ballot contains the same candidates throughout, but the relevant choice environment changes across ranking stages, since a candidate who has already been ranked is no longer available in the voter’s subsequent choice set. As a result, lower-ranked choices can be especially informative for identifying voters’ positions in the policy space. As with simultaneous elections, the identifying condition holds by construction.

RCV is used in national elections in Australia, Ireland, Malta, and Sri Lanka, among others, and has been adopted for state and federal elections in Maine and Alaska as well as for local races in several US cities.

Third, consider the case when *different* voters face *different choice environments across geographic areas*. Such variation is generated by design in most electoral systems, since geographically based representation is a standard feature: examples of first-past-the-post systems across districts include U.S. Senate and House of Representatives elections and House of Commons elections in Canada and the United Kingdom; examples of mixed systems, combining elements of first-past-the-post and proportional representation, include legislative elections in Germany, Hungary, Mexico, Japan, and South Korea, among many others.

In this setting, however, the identifying condition no longer holds by construction. The political economy literature posits that politicians’ and voters’ preferences are closely related, whether because of politicians’ opportunistic incentives, intrinsic preference congruence, or voters’ responsiveness to politicians’ cues (Persson and Tabellini, 2002; Osborne and Slivinski, 1996; Callander and Carbajal, 2022). A researcher should therefore justify the comparison of different choice environments. In practice, this can be addressed in several complementary ways: (1) by identifying contexts or subcontexts in which the condition holds directly; (2) by instrumenting candidate positions; or (3) by relaxing the equality condition and assessing the sensitivity of the results to such relaxation. I now discuss each approach with reference to papers that adopt it.

(1) Contexts where the condition holds directly. Longuet-Marx (2025) exploits congressional district borders within states, following the approach by Spenkuch and Toniatti (2018), and compares only contiguous precinct pairs, which lie in different electoral districts but are very similar otherwise.

In the terminology of this paper, he posits that $f(x | y, m, d) = f(x | y', m, d)$ holds within each precinct pair.¹¹ This implies that the overlay of choice situations should be done only within the pairs of precincts, which limits the available identifying variation. However, by considering many such pairs, a researcher may still be able to recover more about the underlying primitives of interest than from single elections alone. I illustrate this with a simple example in Appendix C.

Merlo and De Paula (2017) and Danieli et al. (2023) treat national-level party positions across European countries as exogenous to voters’ preferences and estimate voter preferences by exploiting cross-country variation in party positions. While the former is closer to the main setup considered here and relies on aggregate voting data, Danieli et al. (2023) combines survey data with party positions. Nevertheless, the same logic of preference estimation applies in this setting as well.

(2) Instrumenting candidate positions. Iaryczower, Montero and Kim (2024) exploit variation in candidates’ positions across Brazilian municipalities to estimate voters’ preferences. Their empirical strategy relies on using the policy positions of mayoral and state-level candidates in the most recent local electoral cycle as an instrument z for current local party positions – the non-degenerate case of Assumption 2.

(3) Relaxing the equality condition. Tebaldi, Torgovitsky and Yang (2023) suggest a way to relax the equality assumption on preference distributions by allowing for pointwise differences in the density. In the voting context, one adaptation of their idea would be to impose $f(x | y, m, d) \in [(1 - \kappa)f(x | y', m, d), (1 + \kappa)f(x | y', m, d)]$ for all x and all y, y' , where κ is a researcher-chosen parameter governing the degree of difference across preference distributions in different regions. Setting $\kappa = 0$ recovers the baseline condition, and the identified set is weakly increasing in κ , so reporting bounds across several values of κ amounts to a sensitivity analysis with respect to the identifying assumption.

The model can also be extended to incorporate other important features of voting behavior, such as strategic voting. Under strategic voting, some individuals may abandon their most preferred candidate if a vote for that candidate is perceived as *wasted*. The key implication is that such behavior generates different effective candidate choice sets across voters. Those who vote for underdog candidates can plausibly be treated as sincere and therefore as facing the full choice set. By contrast, for voters who support viable contenders, it is unclear whether they are choosing their most preferred candidate overall or only their most preferred candidate among viable alternatives. For these voters, the relevant choice set should therefore be restricted to viable candidates. Although this coarsens the partition and may therefore widen the resulting bounds, it makes them more robust to strategic

¹¹In terms of Assumption 2, this amounts to letting the instrument z index the precinct pair: conditional on the pair, the two precincts share the same preference distribution while facing different candidate sets.

voting.

I now turn to describing an application of the model to one of the most widespread electoral regimes: the mixed electoral system.

4 Application: Rise of Populism

One of the central puzzles in the political economy literature is the rise of populism, and in particular radical-right populism, in Western democracies. The empirical literature has largely focused on identifying the causal mechanisms underlying the rise of populist politics, examining their effects on voters' preferences, and analyzing the contextual and individual-level factors that shape these preference responses (see [Guriev and Papaioannou \(2022\)](#) for an overview).

While understanding specific mechanisms is essential, it is equally important to develop a broader understanding of the preferences of populist voters. One recent attempt in this direction is [Danieli et al. \(2023\)](#), who study the rise of populist radical-right parties in Europe using survey evidence over time. To my knowledge, this paper is the first to provide revealed-preference evidence on the preferences of populist voters at large. Focusing on revealed-preference data rather than surveys is important because it avoids problems of survey misreporting, in particular the underreporting of populist vote choice, and makes it possible to recover the relative importance of different dimensions of political competition, which is especially difficult to measure using survey data.¹²

4.1 The Context

Over recent decades, the German electoral landscape has undergone important changes, most notably with the rise of the Alternative for Germany (AfD), a populist radical-right party. For the first time in postwar Germany, a party to the right of the Christian Democrats (CDU/CSU) surpassed the 5% threshold and entered the Bundestag. In the most recent federal election, in 2025, the AfD finished second, winning more votes than several parties of the traditional political establishment, including the Social Democratic Party (SPD), one of Germany's two major parties.

Alternative for Germany was established in 2012 to oppose the policies pursued by the German government to fight the Euro crisis. From the beginning, the party was distinctive in its radical-right and anti-establishment stance. In the run-up to the 2013 Bundestag Elections, its main concerns were Euro Crisis and the Greek bailout, as well as the adherence to the rule of law and stronger participatory democracy ([AfD, 2013](#)). In 2015, the party had experienced an internal split which led

¹²This is because voters tend to adjust their reported positions to align with those of their party, particularly when the issue is not especially important to them. Yet, when relative weights are estimated using survey data, such proximity in reported positions may be mistakenly interpreted as evidence that the dimension is more important to voters.

many of its former members to resign, and leadership was overtaken by conservative, anti-immigrant wing of the party. Since then the party leadership remained in hands of the conservative wing of the party.

Hence, throughout the study period (2013–2025), the AfD was consistently ranked as the most right-wing and anti-elite among Germany’s main parties (Jolly et al., 2022), and it stood out as the most conservative and anti-elite party with a realistic chance of winning seats in the Bundestag. This motivates the choice of the two dimensions of policy space in my application – Left-Right and Pro-Establishment – Anti-Elite (I refer to the latter as just *Anti-Elite* throughout).¹³ I discuss what I mean by those dimensions below in the data section.

The focus of this application is the estimation of AfD voters’ preferences over time. I now turn to the relevant institutional details and data.

My analysis draws on data from German federal parliamentary (Bundestag) elections. Germany is divided into 299 electoral districts across 16 states (e.g. Bavaria, Saxony). Under Germany’s mixed electoral system, each voter casts two votes: a first vote for a direct candidate in a single-member district, and a second vote for a party list. First votes are likely to be heavily shaped by the strategic incentives of first-past-the-post competition, whereas second votes among viable parties can more plausibly be treated as sincere (more on this below). In what follows, I rely exclusively on second-vote party shares.

My analysis is based on the assumption that *candidates’ positions in single member districts shape perceived party positions*. In other words, I use estimated candidate positions (see below) as proxies of parties’ positions in given districts. This allows me to exploit the variation in candidate positions, while preserving the assumption of the sincere voting for the party list. To document the validity of this assumption, I demonstrate that within federal states (which I treat later as separate *markets*) and within parties individual perceptions of party positions correlate with estimated candidate positions on the Left - Right dimension (Table F1). Moreover, the rich survey data from the time around 2021 Bundestag Elections document that voters extensively interact and know their local candidates¹⁴.

Two additional institutional features are important for the analysis. First, state-level (regional) party elites play a substantial role in candidate selection across electoral districts. Although German

¹³This aligns with the literature in political science that highlights that populist parties build their agenda on two main dimensions: ideological and anti-system (See Mudde (2004); Guriev and Papaioannou (2022) among others), and the adjacent paper Nikiforova (2026) that identifies Left-Right and Anti-Establishment as the two central dimensions of voting in Germany in recent years.

¹⁴The phenomenon that candidates’ characteristics affect parties’ perception in mixed electoral systems is referred to as "contamination effect" in the political science literature (Ferrara, Herron and Nishikawa, 2005).

election law formally assigns nominations in single-member districts to local (district) party chapters, state party organizations act as de facto gatekeepers within the region. They can influence nominations both (i) by blocking district candidates they consider unsuitable and (ii) by determining the state party list for Bundestag elections. The latter matters because candidates may enter the Bundestag either through district races or through state lists; in practice, around two-thirds of candidates on state lists also run in single-member districts. Candidate selection is moreover often uncontested, as elites tend to preselect candidates in advance — particularly when incumbents are involved (Detterbeck, 2016; Schüttemeyer et al., 2024). These considerations both motivate the choice of states as separate markets and inform the identification argument.

Section H.1 provides more details and descriptive evidence on the nomination practices employed by the major parties. In particular, it highlights persistence in candidate nominations and patterns of nomination across single-member districts and state lists.

Second, electoral districts are drawn primarily on the basis of population size. Unlike in the United States, redistricting in Germany is nonpartisan and district boundaries are largely stable. A permanent Constituency Commission reviews population changes each electoral cycle and recommends adjustments to the Bundestag. Constituencies should remain within 15% of the average population, must be redrawn if deviations exceed 25%, and must lie entirely within a single state. In practice, only a small share of constituencies changes in any given electoral period.¹⁵

4.2 Data

In the application that follows a state ($N=16$) constitutes a market; markets are subdivided into electoral districts ($N=299$); and each electoral district contains multiple municipalities ($N=11,000$). Voting and demographic data are observed at the municipality level. Each electoral district features a distinct set of candidates. Figure F1 illustrates this division. I now describe the data used for estimation.

The demographic group taken as a unit of analysis consists of age-gender-education bracket in a given market (state), and distribution of preferences is estimated separately for each group within each state in a given year. An example of such a group is *female age 18-24 with higher education in Saxony in 2021*. Hence, to proceed with the estimation I need the data on age-gender-education composition and voting at the same level. I now turn to describing such data.

Municipal-level data. I collect data on turnout and second vote shares for the six main parties at the level of municipalities^{16,17}: *AfD*, *CDU/CSU*, *FDP*, *Greens*, *Left*, *SPD*. I abstract from coalition

¹⁵See, for example, [Die Bundeswahlleiterin](#) (n.d.) for changes made prior to the 2025 Bundestag elections.

¹⁶The cumulative party vote shares for the studied viable parties are 94%, 95%, 91% in years 2013, 2017 and 2021 respectively. This implies that the analysis excludes only a small group of voters who supported non-viable parties

¹⁷The municipality is the smallest unit of analysis in Germany that can be geo-coded.

considerations and treat those parties to be *viable* in the years studied, which implies that if an individual truly prefers one of the considered parties, then there are no strategic considerations not to vote for this party.¹⁸ In turn, this implies that my estimates elicit true voting preferences among the voting population.

I also collect demographic data at a municipal level for each electoral year. I construct municipality-level information on the population composition of eligible voters in 24 age–gender–education brackets, comprising 6 age groups (18–25, 25–35, 35–45, 45–60, 60–70, 70+), two genders, and a binary distinction between individuals with and without higher education.¹⁹ These demographics are measured precisely for 2021, drawing on the 2022 Census. To construct demographics for 2013 and 2017, I use changes in the size of each age–gender group at the district level ($N \approx 400$) and project these changes onto municipalities within each district, using the 2021 municipal composition as the baseline.

I restrict the benchmark sample to municipalities with fewer than 10,000 inhabitants.²⁰ Cross-market variation in the demographic distribution is what identifies preferences separately by demographic group (Berry and Haile, 2021); smaller units retain more distinctive compositions, widening that variation. I test the sensitivity of my results to this threshold value below.

Candidate positions. A key feature of the considered setting is that voters face distinct party-affiliated candidates across single-member districts. To estimate candidates’ ideological positions, I combine two data sources:

- 1) GLES Candidate Study for the Bundestag elections 2013–2025, administered by the German Society for Electoral Research and Leibniz Institute for the Social Sciences (GLES, 2023),
- 2) Wen Wählen (*Whom to Vote For?*), a nonpartisan platform that collects standardized questionnaires from election candidates.

The data sources include candidates from parties with a high likelihood of clearing the 5% threshold. Consequently, candidates from all parties considered in my analysis were contacted for participation in each election year.

First, using the GLES Candidate Study, I construct a left–right score and an anti-elite score for each candidate from multiple survey items corresponding to each dimension (see Table H1 for the

¹⁸It also assumes that voters expected these parties to clear the 5% threshold; the exceptions across years and parties are FDP/AfD (2013), Left (2021) at 4.7–4.9%, and FDP (2025) with 4.3%.

¹⁹To do so, I use data on the age–gender composition of the population at the municipality level and complement them with data on the age–gender–education composition, which are available only for larger municipalities (with more than 10,000 inhabitants), municipal associations (*Gemeindeverbände*), or higher-level administrative units (*Kreise*).

²⁰Note that this restriction effectively captures almost exclusively rural municipalities, making a separate urban–rural distinction unnecessary in this context. It also naturally excludes the city-states (Berlin, Bremen, and Hamburg) from the analysis.

exact wording).²¹ I treat the left–right dimension as a bundle of economic and cultural policies, and construct it from questions spanning economic issues (e.g., whether the government should reduce income differences, or stay out of the economy), cultural issues (e.g., whether same-sex marriage should be banned, or whether women should be free to decide about abortion), and immigration (e.g., whether immigrants should adapt to German culture, or whether immigration benefits the German economy). In line with the literature, I treat the anti-elite dimension as the axis capturing antagonism toward elites and the claim that the people (rather than professional politicians or technocrats) should hold authority (Mudde, 2004; Guriev and Papaioannou, 2022). To quantify a candidate’s position on this dimension, I use items such as whether democracy offers citizens enough opportunity to participate, whether there should be nationwide referendums, and whether citizens have lost trust in politicians. See Table H1 for the list of questions used from GLES Candidate Study.

The same questions are used in each election year. I first recode each item onto a numeric scale, drop responses outside the valid range, and reverse-code items when needed. I then standardize each item using z-scores and compute, for each candidate, the average across all available items within a given dimension to obtain a single index.²²

The participation in the survey is voluntary, and the response rate among the single member district candidates remains moderate – about 35% across the years.

To improve data coverage, I supplement the GLES Candidate Study with a second source, Wen Wählen (Whom to Vote For?), a nonpartisan platform that collects standardised policy questionnaires from election candidates across all major parties. Because the two surveys use distinct question sets, I bridge them using an elastic net regression. The training sample consists of candidates who responded to both surveys across the 2013, 2017, and 2021 federal elections; their estimated ideological positions from the GLES Candidate Study serve as the outcome variable, while their responses to the Wen Wählen questionnaire serve as predictors. Hyperparameters are selected by 10-fold cross-validation. I then apply the trained model to impute positions for candidates who appear only in the Wen Wählen dataset. Figure H5 plots the resulting distributions of candidate positions by party, dimension, and year; and Appendix H.2 provides more details on the estimation of candidates positions.

Complementing both surveys allows me to recover comparable positions for about 65% of candidates across the years. Parties differ by responding rates with CDU having the lowest response rates across the years, and Greens – the highest (37% and 77% respectively). About 50% of AfD candidates are available across the years. Importantly, candidates who respond to the survey and those who do not achieve comparable first-vote shares and have similar winning probabilities across

²¹I use the terms *anti-establishment* and *anti-elite* interchangeably.

²²I prefer this method for its transparency. The correlation with indices based on Item Response Theory exceeds 0.85 across all years and dimensions.

parties. For those candidates who are missing in the data, the median positions across the remaining single member districts are imputed. I test the robustness to this imputation by keeping only electoral districts in which I have estimates for at least 5 out of 6 candidates.

There are two more data sources that I use in the benchmark specifications below.

Representative Electoral Statistics. Federal statistical agency collects a data on voting choices per 12 age-gender groups per states. In particular, I use data on voting choices and abstention per each demographic group per states. This data provides additional moments that help pin down demographic heterogeneity in party choice and account for differences in turnout across groups.

Representative Electoral Statistics uses an official, legally regulated sample of polling stations in which marked ballots allow votes and turnout to be tabulated by gender and age groups. Hence, this data is based on actual ballots, not surveys. For example, in 2021 it covered 2,559 of 94,668 polling stations. Results are published only at level of states to protect secrecy, and are benchmarked to the official state totals, so group-specific estimates are consistent with final results. Figure F3 provides an example of such statistics.

Second-best party choices. I also construct second best party choices using survey data across federal states and across elections. Standard discrete choice models are prone to badly predict substitution behaviour (second choice) of consumers, and second best choices as explicit target moments are known to be very helpful in better fitting substitution patterns (Conlon and Gortmaker, 2023). While stated voting behaviour is biased, I believe that second-best party choices less so. See adjacent paper Nikiforova (2026) for the discussion on this.

Figure F4 provides an evolution of second best party choices of those who indicate AfD as their first choice over time at a national level.

5 Estimation & Identification

5.1 Estimation

Section D describes how the empirical restrictions implied by the spatial voting model are translated into a finite-dimensional optimization problem, and how (i) the weights are computed, and (ii) the bounds for target parameters.

5.2 Identification

The approach in this paper exploits variation in candidate positions across electoral districts within a state in a given year to estimate preferences of AfD voters. Identification assumption then

posits that *in a given year variation in candidates' positions across electoral districts within a state (market) m is unrelated to districts' unobserved voter heterogeneity*, $f(x | y, m, d) = f(x | y', m, d)$. I first discuss the reasons why this assumption may be viewed as plausible, and then discuss how I test validity of this assumption in later sections.

First, survey evidence suggest that age-gender-education brackets explain large share of variation in individuals' self-identification among the voters of one party. To demonstrate this I focus on a more restrictive version of the model using the self-perceived Left-Right position from survey, and estimate an OLS *mean-prediction* regression predicting Left-Right position separately for each demographic group across states within parties. The R^2 of such regressions across federal states takes values $0.3 - 0.85$ with a mean of 0.519.

Second, as discussed in the data section, candidate nominations are largely controlled by regional party elites and influenced by state-list considerations, which limits local discretion in candidate selection. More generally, the political science literature highlights the relatively small pool of potential candidates and the constraints this places on candidate choice (see, e.g., [Höhne \(2023\)](#)). I provide more information on candidate nomination practices in [Section H.1](#).

I further restrict the sample to municipalities with fewer than 10,000 inhabitants. Because the marginal electoral return to targeting a locality is low when its electorate is small, candidates have little incentive to tailor their platforms to these voters. Observed preferences in small municipalities are therefore less likely to reflect strategic candidate positioning than those in larger, more contested urban centers.

I also demonstrate observationally two more pieces of evidence that strengthen this assumption. First, estimated candidates' positions are to a large extent uncorrelated with districts' socio-economic characteristics. Second, candidates' positions are uncorrelated with individuals' self-reported views (based on surveys). See [Figures F5, F6](#) and [Table F2](#) for these results respectively.

I assess the robustness of the identification assumption in three ways: (i) by restricting the analysis to small municipalities (smaller than 5k inh.), (ii) by excluding AfD candidates with stronger incentives to cater to local voters, (iii) by relaxing the exogeneity assumption and separating the districts into conservative and moderate (an approach similar to [Tebaldi, Torgovitsky and Yang \(2023\)](#)). The first approach follows logic similar to that in [Spenkuch and Toniatti \(2018\)](#), which suggests that politicians are more likely to target the preferences of voters in larger municipalities and cities. I therefore re-estimate the results using only municipalities with 5,000 or 10,000 inhabitants or fewer. The latter approach exploits the idea that politicians' incentives to cater to local voters vary with their electoral prospects, as competitive candidates are more likely to tailor their positions to constituency preferences.

6 Results

The main goal of this section is to provide the new insights into the rise of populism using the revealed-preference evidence from the approach developed in this paper. To do so, I first estimate voters' preferences: the weight on the anti-elite dimension relative to the Left–Right dimension, w_{AE} , and the distribution of ideal points in a Left–Right–Anti-Elite–Pro-Elite policy space separately for demographic groups d in markets m $f(x \mid m, d)$. I next use those estimates for the counterfactual analysis.

Throughout the chapter I often present the results separately for eastern and western Germany. There are many reasons for this distinction, such as different histories and present-day institutional environments, as well as different levels of AfD support.²³ Crucially, the results from both parts of the country, analyzed separately, could inform our understanding of the rise of populism across countries with different institutional environments, such as those in eastern and western Europe.

6.1 Relative importance of dimensions

I start by describing the evolution of relative importance of the two ideological dimensions, as weights define the form of the partition space used to evaluate the target parameters later on.²⁴

As discussed in the model section, only relative weights can be recovered; I normalize the weight on the Left–Right dimension to $w_{LR} = 1$ and search for the best-fitting value of w_{AE} . Precise estimation of weights requires especially rich data, as demonstrated in the simulations in the Appendix Section B. To increase the observational richness of the analysis, I estimate the weights separately for East and West Germany, and I allow ideal-point distributions to vary across federal states – that is, each demographic group d in each market m (federal state) has its own bliss point distribution $f(x \mid m, d)$; while I assume the same weight w_{AE} for all groups in a given year and region (East or West).

The constructed grid to search for the best weight includes relative weights across the two dimensions of $(0.2, 0.3, \dots, 0.9, 1)$ and the corresponding inverse ratios.²⁵ The best relative weight is

²³At the moment of writing this draft, the AfD tops the polls nationwide with almost 30% support and 6% winning margin. The within-country heterogeneity is striking. In eastern Saxony-Anhalt, which holds elections in September 2026, the AfD is expected to win an outright majority, whereas in western Rhineland-Palatinate and Baden-Württemberg the party obtained less than 20% in March 2026.

²⁴Note that in this section, by evolution, I refer to changes in the preferences of AfD voters, which may reflect both within-individual changes in preferences and compositional changes in the electorate. I am trying to decompose the two in the following subsections.

²⁵While it is possible to construct wider bounds, I do not do so for two reasons. First, wider bounds would amplify the measurement error in the estimated candidate positions. Second, the weights are identified only insofar as they meaningfully change the partition of the space, which for each pair of candidates depends on the weight and the ratio of their position differences, $\frac{|x_j^{LR} - x_k^{LR}|}{|x_j^{AE} - x_k^{AE}|}$. More than 80% of the values of this ratio across pairwise comparisons involving

defined then as a weight that yields the smallest error value for the model fit.

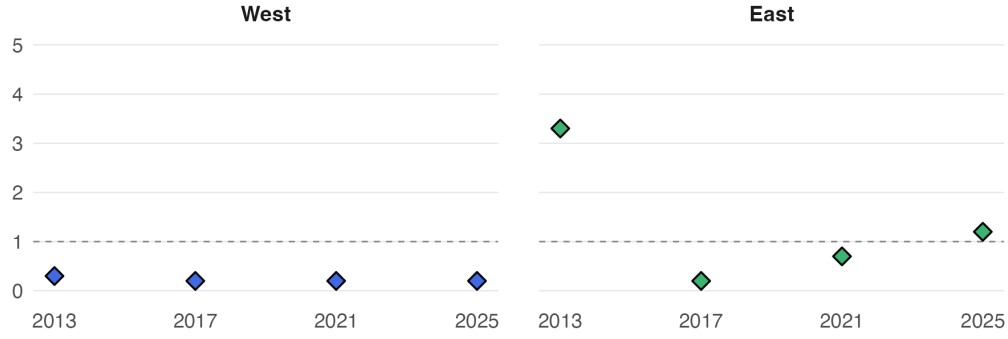
Figure 2 presents examples of the partition of the policy space for different weight configurations. It illustrates the intuition behind the weighting: a lower (higher) weight on the anti-elite dimension means that individuals tolerate a larger (smaller) distance along that dimension for a given distance along the left–right dimension between their bliss point and a candidate’s position.

The ability of the model to yield the best fit in the context of changing partitions (i.e. changing relative weights) depends on the number of cells available under each setup, which, in turn, depends on the patterns of co-location of candidate positions on anti-elite and left-right dimensions²⁶. To alleviate this concern, I fix the number of available cells at 500, restricting attention to the AfD attraction region. Specifically, I retain the 500 largest cells and merge all remaining smaller cells into the larger cells with which they share the longest boundary. Additionally, Figure F8 shows that the correlation between left–right and anti-elite positions differs little between Eastern and Western Germany, both within parties (Figure F8a) and across parties (Figure F8b); and Table F5 shows that the standard deviations also do not differ much across years and parts of the country. This suggests that the relative positioning of parties on these two dimensions is unlikely to account for the regional differences reported below and that the results are instead driven by demand-side differences in the relative importance attached to each dimension.

Figure 3 presents the results for the optimal weights. Points denote the point estimates corresponding to the best-fitting weight. Because the data are noisy, the optimization problem always yields a single best-fitting estimate. To reflect the fact that the weights are set identified, I also report intervals that allow for discrepancies of up to 1% in the objective-function error (Figure G1).

AfD candidates lie within [0.2, 5].

²⁶Note that this argument is analogous to a standard result on variance decomposition in linear regression: for a fixed regressor X_1 , the improvement in R^2 from adding a second regressor X_2 is smaller the larger the correlation between the two regressors.



Notes. $w_{LR} = 1$ is fixed throughout, w_{AE} should be interpreted as the relative weight placed on the Anti-Elite dimension. Separate exercises are run for East and West Germany. I allow ideal-point distributions to vary across federal states – that is, each demographic group d in each market m (federal state) has its own bliss point distribution $f(x | m, d)$; while imposing the same weight w_{AE} for all groups within a given year and region. To run this exercise, the grid of possible weights is constructed. The grid is symmetric around 1 and imply relative weights across the two dimensions of $(0.2, 0.3, \dots, 0.9, 1)$ and the corresponding inverse ratios. See G2 for full results over the grid. Points denote the point estimates corresponding to the best-fitting weight, while the shaded areas indicate intervals over which the objective-function error is less than 0.5% higher (light shading) or 0.25% higher (dark shading).

Figure 3. Best Weights for Anti-Elite dimension w_{AE} relative to Left-Right $w_{LR} = 1$ across times and parts of the country

The results suggest that, in Eastern Germany, the anti-elite dimension was more important than the left-right dimension in 2013 although this estimate is imprecise once allowing for a 1% difference in the objective-function error (see Figure G1). In 2017, 2021 the left-right dimension is more important in the East; in 2025 both dimensions carry relatively equal weight. Note that the evolution of weights is a combination of at least two factors – (i) genuine within-person evolution of relative importance of different dimensions, and (ii) compositional change. In Western Germany the left-right dimension remains consistently more important throughout the period, although the shaded intervals suggest some increase in the relative importance of the anti-elite dimension in 2021.

These results are in line with the differential role played by both dimensions in AfD support. A broad literature documents that Eastern Germans perceive themselves as neglected by federal elites, a perception grounded in measurable underrepresentation: as of 2024, Eastern Germans held only 12.1% of positional elite roles across twelve societal sectors, far below their population share of roughly 20% (Bundesministerium für Wirtschaft und Energie, 2025). This persistent elite gap plausibly heightens the salience of the anti-establishment dimension in the East relative to the classical left-right axis. The differential trend over time could be driven by time-specific factors (e.g. immigration crisis prior the 2017 elections). This anti-elite dimension is much more muted in the West, where the region-wide sentiment of elite underrepresentation is largely absent.

Figure G2 plots the objective-function value for different values of the weight on the second dimension. The error function is relatively smooth and displays a well-defined minimum, suggesting

that the estimates are stable. Figures G3, G4 demonstrate consistency of estimates across subsamples by plotting the objective-function value on different subsamples of electoral districts across years and parts of the country. I interpret these results as evidence supporting the validity of the exercise.

In the following analysis I use the values for the estimated weights to define the partition of the policy space across the two regions unless stated otherwise.

6.2 Distribution of ideal points

Next preference object is the distribution of voters' ideal points $f(x|m, d)$. To provide evidence on the evolution about this object, I focus in this subsection on the location of AfD voters in the weighted policy space.

6.2.1 Voters ideal points in fixed areas of policy space

The natural question to ask about voters' ideal points is *Where voters' ideal point are located in the policy space?* In particular, I ask this question in relation to the AfD voters in 2013-2025. The AfD median candidate position across the years is about (1.2, 0.8) and was around this area over the period.²⁷ To study the bliss points of AfD voters and their evolution I then first partition the attraction area of AfD in three regions: 1) Anti-Elite: Area with distinctively Anti-Elite Bliss points: those to the upper left from (0.5, 0.5); 2) Right: Area with distinctively Right Bliss points: those to the lower right from (0.5, 0.5); 3) Both: Area with both Anti-Elite and Right bliss points: those to the upper right from (0.5, 0.5). Figure 4a provide an illustration for these areas.

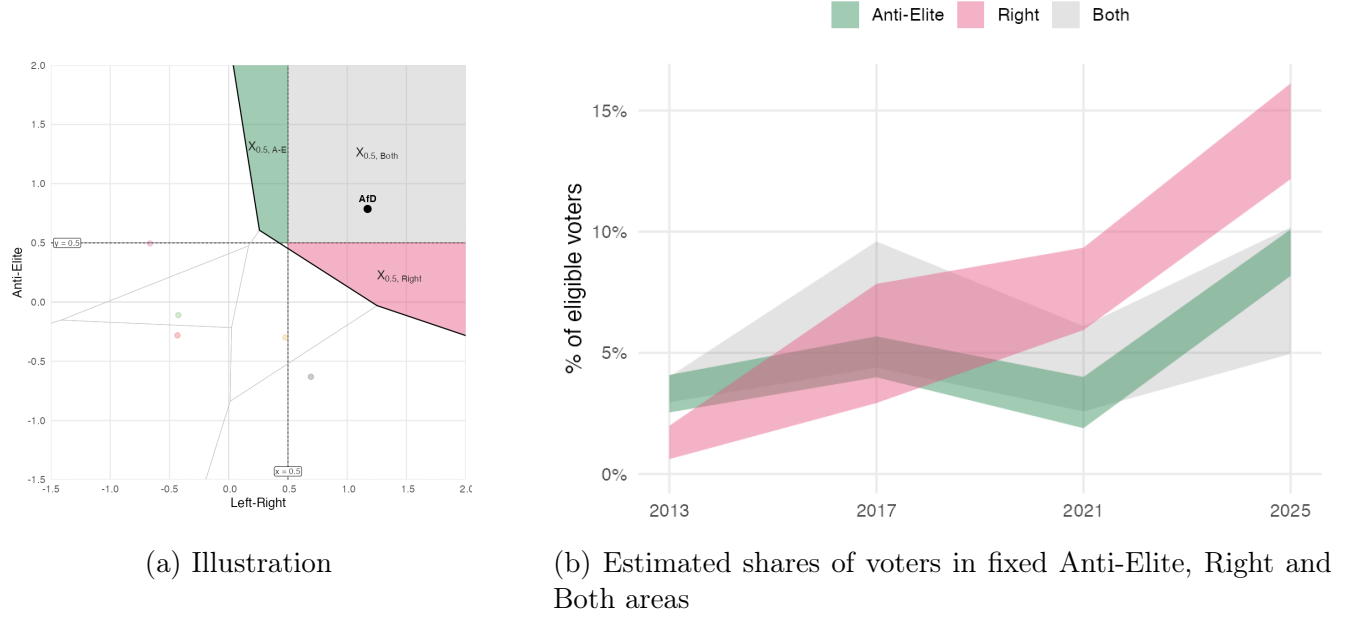
Let $f(x | m, d)$ be the density of ideal points among eligible voters in state m and demographic group d , and $F_{M,D}$ their joint (observed) distribution. The target parameter is the share of AfD voters in each region, e.g. for anti-elite:

$$\theta_{\theta_{\text{AE-Fixed}}} = \iint_{X_{\theta_{\text{AE-Fixed}}}} f(x | m, d) dx dF_{M,D}(m, d)$$

Because $f(x | m, d)$ is only partially identified from the municipal vote shares and demographic composition, I report bounds on $\theta_{\theta_{\text{AE-Fixed}}}$ and other target parameters; note that the bounds reported here and below are sharp (see Tebaldi, Torgovitsky and Yang (2023) for details).²⁸ Figure 4b plots the results.

²⁷The median positions of AfD candidates across the studied years are (0.96, 0.71), (1.25, 0.86), (1.3, 0.97), (1.09, 0.7).

²⁸Also note that the fixed masks are contingent, so the lower bounds are the most informative: they give the smallest possible share of voters located in the region (since the estimated objects are the *sharp* bounds on the target parameters). The upper bounds, by contrast, may reflect both the degree to which the problem is underidentified and the mass in the cells shared by contingent fixed masks.



Notes: Panel (a) illustrates party positions at their national median with an isotropic partition of policy space; the areas of AfD voters with distinctively by anti-elite bliss points (green, $X_{0.5,A-E}$), Right bliss points (pink, $X_{0.5,Right}$) or Both (gray, $X_{0.5,Both}$). Panel (b) plots bounds on the shares of voters whose bliss points lie in the green, pink and gray regions across years. The results are aggregated across the two parts of the countries.

Figure 4. Illustration of a target parameter (a) and results (b)

The overall increase in target-parameter values over time reflects the rise in AfD support from 2013 to 2025. This increase is heterogeneous across masks: a large and steady increase is observed for the share of AfD voters in the Right fixed mask, which contains up to 15% of all voters by 2025. This is evidence that the AfD's electoral base is shifting towards a more ideologically conservative, right-wing support base rather than a more anti-elite one.

This subsection has provided a first piece of evidence on the distribution of AfD voters' bliss points and its evolution. While instructive and intuitive, it does not pin down voters' motivations, which, beyond voters' locations, would also require accounting for the weights voters attach to the different dimensions. I now turn to the analysis that resolves this.

6.2.2 Voters ideal points and their motivations for AfD support

The AfD is distinctive in combining an ideologically radical-right stance with an anti-elite appeal, so a ballot cast for it is uninformative about why the voter chose it. Recovering that motivation matters, because it maps to the policy demands the vote expresses, and could allow to craft more appropriate responses. The analysis in this section resolves the ambiguity with a counterfactual: suppose the AfD split into two parties, one retaining its radical-right economic position but moving toward the establishment, the other retaining its anti-establishment appeal but moderating its eco-

nomic radicalism — what share of AfD voters would each attract? Equivalently, it measures the share of AfD voters whose ideal point in a weighted policy space leans anti-elite versus right, relative to where the AfD itself stands.

The estimated composition therefore has a behavioral interpretation: *a voter in the anti-elite region is one for whom protest against the elites, rather than economic ideology, is the pivotal reason to vote AfD, and conversely for the right region.*

Let $\bar{Y} = (\bar{Y}_{LR}, \bar{Y}_{AE})$ be the nation-wide median AfD position. Displacing the AfD by δ along each dimension defines two hypothetical successor parties and partitions the AfD win region $X_{AfD}(\bar{Y})$ into an anti-elite region X_{A-E} (voters nearer the party that retains the anti-elite appeal; green) and a right region X_{Right} (voters nearer the party that retains the economic-right position; pink), with $X_{AfD} = X_{AE} \cup X_{Right}$. Let $f(x | m, d)$ be the density of ideal points among eligible voters in state m and demographic group d , and $F_{M,D}$ their joint (observed) distribution. The target parameter is the share of AfD voters in each region:

$$\theta_{AE} = \frac{\iint_{X_{A-E}} f(x | m, d) dx dF_{M,D}(m, d)}{\iint_{X_{AfD}} f(x | m, d) dx dF_{M,D}(m, d)}$$

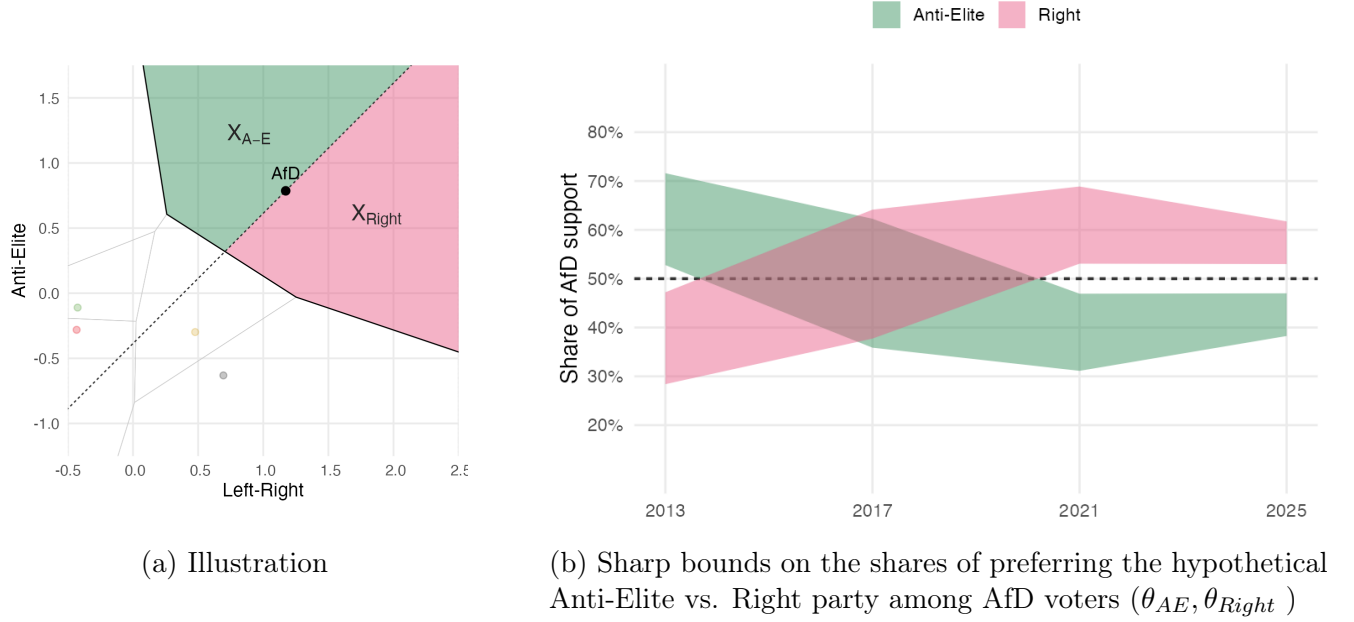
I now report bounds on $\theta_{AE}, \theta_{Right}$. Note that by construction, $\theta_{Right} = 1 - \theta_{AE}$.

Figure 5b provides the bounds for the target parameters over years. The results indicate that AfD support has changed over time. In earlier years the support of AfD was driven more by anti-elite motivations, while it flipped over time and in 2021 and 2025 more than 50% of AfD support comes from individuals with stronger ideological Right motivations. The evolution of AfD from party that is

These aggregate results mask substantial regional and subgroup heterogeneity. Figure G5 reports the estimates separately for Eastern and Western Germany. The pattern for Western Germany resembles the national-level pattern presented above; in turn, voters in Eastern Germany are more strongly driven by anti-elite attitudes than voters in Western Germany, especially in 2017.²⁹ The patterns are clearer and more precisely identified for Western Germany, reflecting the fact that it is larger in population and offers more electoral districts across which to exploit variation.

I now want to highlight two more subgroup heterogeneity results among Western voters over

²⁹Except in 2013, when anti-elite sentiment dominated in the West. The party's anti-elite appeal then was arguably of a different nature—rooted in opposition to the Eurozone bailouts rather than in the hostility towards the establishment and democratic institutions that came to define the party after 2015.

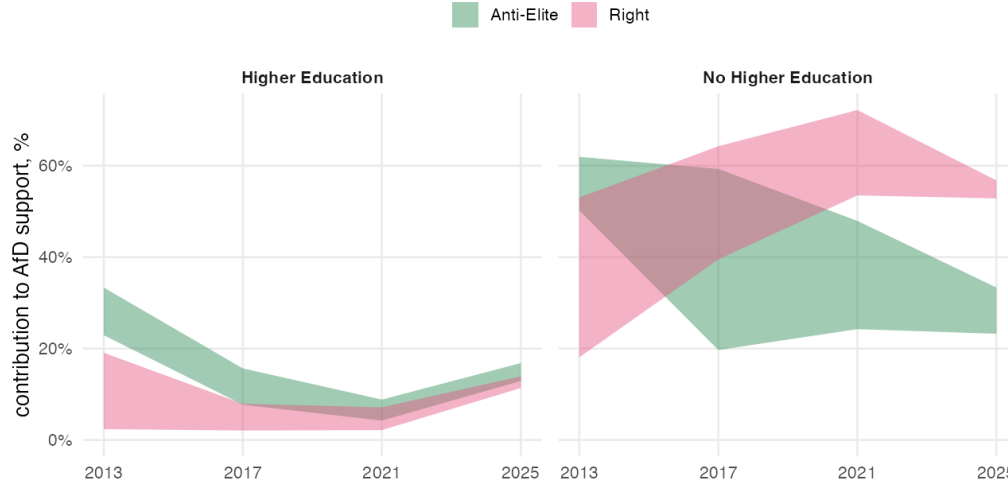


Notes: Panel (a) illustrates party positions at their national median with an isotropic partition of policy space; the areas of AfD voters motivated more by anti-elite (green, X_{A-E}) or right (pink, X_{Right}) ideological stance of AfD. The illustrated weight used is $w_2 = 1$, weights used in the estimation are the weights estimated in Subsection 6.1. Panel (b) plots bounds on the shares of voters whose bliss points lie in the green and pink regions across years. The results are aggregated across the two parts of the countries.

Figure 5. Illustration of a target parameter (a) and results (b)

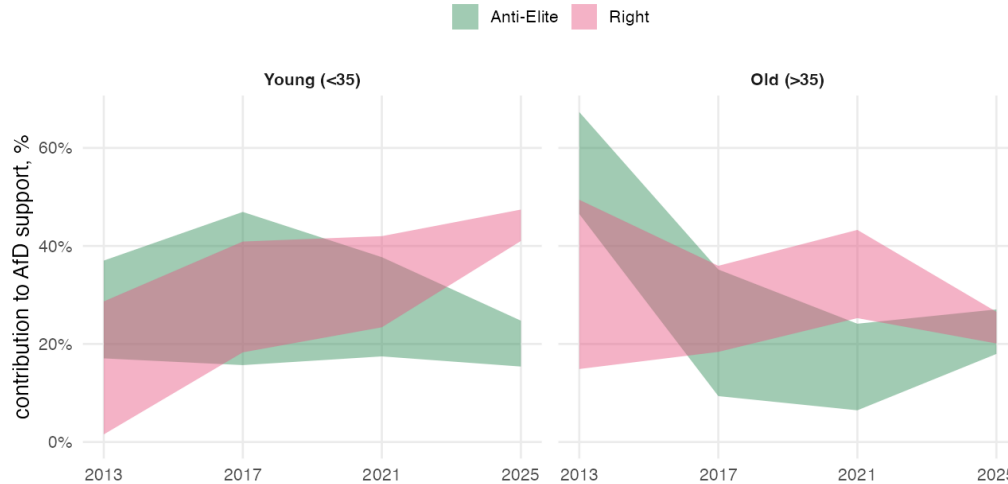
time, as Western Germany allows for detecting rich heterogeneity patterns in the data. Figure 7 demonstrates shares of AfD voters with and without higher education belonging to the Anti-Elite and Right areas of Figure 5a (these are the same target parameters as just above, but estimated at the subgroup levels); Figure 7 provides the same analysis for young (< 35 y.o.) and old (> 35 y.o.) individuals.

These figures are intended to highlight that individuals in different demographic groups have different rationales for supporting the same party, and that these rationales may evolve over time. In particular, for individuals without higher education the party's radical-right stance came to outweigh its anti-elite appeal only in later years. This could suggest that voters with lower educational attainment hold preferences that are more malleable, or more responsive to party messaging. Another important pattern is the rising share of young AfD voters whose preferences lie in the Right group. The growing support for radical (and in particular radical-right) political movements among young people has been documented in the political science literature, though the motivations behind these choices remain contested (van der Brug, Hobolt and Popa, 2026). The results of my analysis suggest that young individuals may increasingly support these parties because of their radical-right ideology rather than their anti-elite stance.



Notes. Bounds on the shares of voters whose bliss points lie in the green and pink regions across years are plotted. The results are only for the Western Germany.

Figure 6. Estimated shares of AfD voters in both subspaces of policy space by educational attainment



Notes. Bounds on the shares of voters whose bliss points lie in the green and pink regions on the Figure 5a across years are plotted. The results are only for the Western Germany.

Figure 7. Estimated shares of AfD voters in both subspaces of policy space by age

The results presented in this subsection speak to the changing nature of AfD support over time, across regions, and across demographic subgroups. As I regard these as one of the paper's main contributions, I verify that they are robust to relaxing the selection-on-observables assumption used to identify these patterns.

Robustness. I provide three robustness exercises that relax a selection-on-observables assumption on candidate positions. First, I run the main analysis on municipalities with fewer than 5,000 inhabitants (Figure G6). The idea is that parties are likely to target more populous, more electorally

relevant constituencies; small municipalities therefore face a set of candidates pre-selected for larger constituencies. The results remain qualitatively similar in the West and deliver a more stable Anti-Elite lean in the East across 2017–2025.

Second, I restrict the analysis to districts in which AfD candidates were not competitive for victory. Specifically, I exclude every district in which the AfD candidate trailed the winning candidate by no more than 5 (10) percentage points in the Eastern (Western) part of the country. This removes 12 districts in 2017 and 28 in 2021 in the East, and 22 in the West in 2025. The different thresholds reflect the party’s greater popularity in the East, where a uniform 10-point threshold would remove up to 80% of all districts in 2021. Moreover, for the East the analysis is restricted to 2017 and 2021: in 2013 there were no serious contenders, while in 2025 there were too many even under the 5-point threshold, as AfD candidates won almost every district.

The idea behind this analysis is that candidates who recognize they have little chance of winning will cater less to the voters of their particular district, while still shaping the party’s image through their more intrinsic ideological stances. Figure G7 presents the results separately for Eastern and Western Germany. The results for the presented years remain quantitatively similar.

Third, I exploit an instrument similar in construction to the one in [Tebaldi, Torgovitsky and Yang \(2023\)](#), which directly relaxes the assumption of preference similarities across districts with different candidates. In particular, I allow that the mass of voters of a given demographic group g within a state in each partition cell between conservative and moderate districts could differ by at most a factor of $1 + \kappa$ for some fixed κ . I set $\kappa = 1$, so that masses could be different no more than two three in both directions. Figure G8 presents the results. Relaxing the assumption on preference similarity does not quantitatively change the results, but provides somewhat wider bounds, especially in the East.

The analysis so far has focused on the evolution of AfD voters’ preferences. The next subsections turn to counterfactual exercises, examining how changes in the positions of the AfD and its center-right, pro-establishment rival (Christian Democrats, the CDU) would affect the AfD vote share.

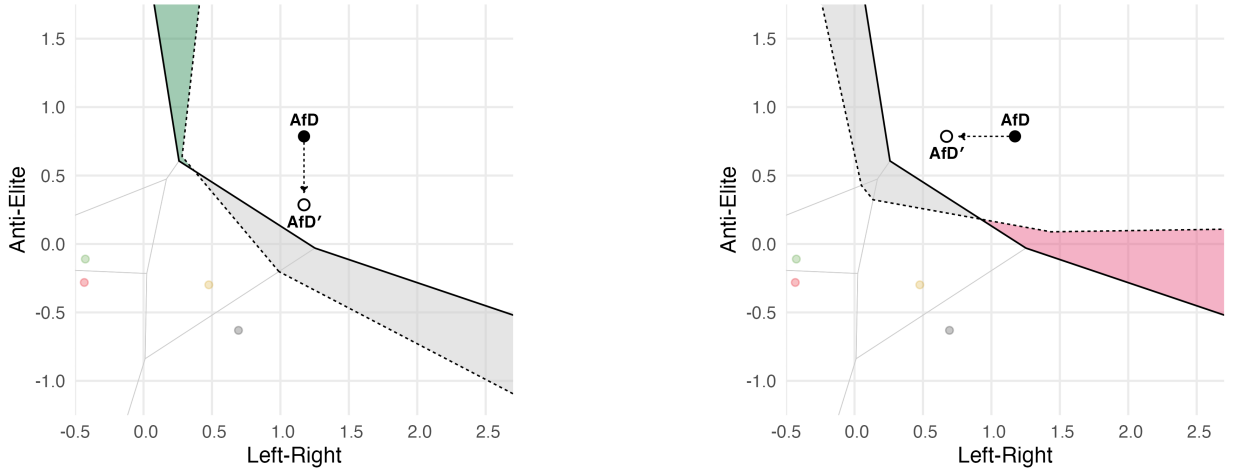
6.3 What explains the success of AfD in competition with other parties: conservative positions or anti-elite stance?

Answering the question of which dimension matters more for voters requires assessing the sensitivity of voters’ choices to counterfactual party positions. The party does not exist in isolation, but competes with other parties for voters.

To assess whether the AfD’s success is driven more by right-wing ideology or by anti-elite sentiment, I combine the estimated bliss-point distributions of AfD voters, the estimated relative

weights, and the median positions of all the other parties, and ask: *How sensitive are AfD voters to the party shifting its positions along Left-Right and Anti-Elite dimension?*. Put differently, the question is how many of the AfD’s baseline supporters the party would lose if it adopted a more pro-elite or a more left-wing position.

I keep the estimated preferences fixed and shift the AfD’s position relative to its national-level median, $\bar{Y}$, by 0.25/0.5/1 units along each dimension.³⁰ Figure ?? illustrates the shifts along both dimensions under median party positions across years and isotropic weights for clarity.



(a) AfD shift towards 0.5 more pro-elite position

(b) AfD shift towards 0.5 more left position

Notes: The panels present an illustration of the unweighted cropped policy space, with partitions determined by median party positions across years. Shaded dots mark the median positions of parties other than the AfD. **AfD** denotes the AfD’s median position across years, and **AfD’** denotes the shifted AfD position—either towards the pro-establishment position (Panel (a)) or towards a more left-wing position (Panel (b)). Solid lines determine the existing partition, the dashed line – counterfactual partition. The illustrated shift is 0.5 units along each dimension. Since positions are standardized by construction, a 0.5-unit shift is comparable across dimensions. The exercise estimates the share of voters the AfD would lose under the shift, i.e. those located in the green or pink areas in each panel.

Figure 8. Illustration of counterfactual moderating shifts by AfD by 0.5 units separately on two dimensions

Before presenting the results, I provide a substantive interpretation of what it means for a party to shift with a step of 0.25/0.5/1. To interpret these steps I explore in the used scoring pipeline what a shift of given steps would require from candidates in terms of the changes in their opinions. Note that the dimensions units (and the whole scoring process) are the same across the years, so those steps are consistent across time too.

A candidate’s position on each dimension is the average of her standardized answers to the relevant survey items, so moving a single question by one scale point shifts the dimension score by

³⁰Positions are standardized by construction, so a given shift is comparable across dimensions.

some σ , unique to each question. On the anti-elite dimension, the most influential single question—*Satisfaction with the way democracy works* (a 1–4 item)—moves the score by about 0.18 per scale point, and a typical five-point item such as *Legislation reflects citizens’ interests* by about 0.14. A 0.25 step therefore corresponds to revising one such question by roughly 1.5–2 scale points (say, from *rather agree* to *strongly disagree*), or two questions by 1 point each. Moving every anti-elite question by one scale point changes the score by about 0.95, so the 0.25/0.5/1 steps amount to softening (or hardening) a corresponding share of all anti-elite answers.

The economic Left–Right dimension behaves similarly: potent five-point items such as “*immigration is good for the economy*” or “*marriage only for heterosexual couples*” move the score by about 0.10 per point, and moving all economic items by one point shifts the score by about 0.69. In short, a 0.25 step is a modest, one-to-two-question adjustment; a 0.5 step is a substantial repositioning that must touch many questions at once; and a full 1-unit step amounts to broadly shifting one’s stance across an entire dimension. Table F6 provides score sensitivities across questions for 2017.

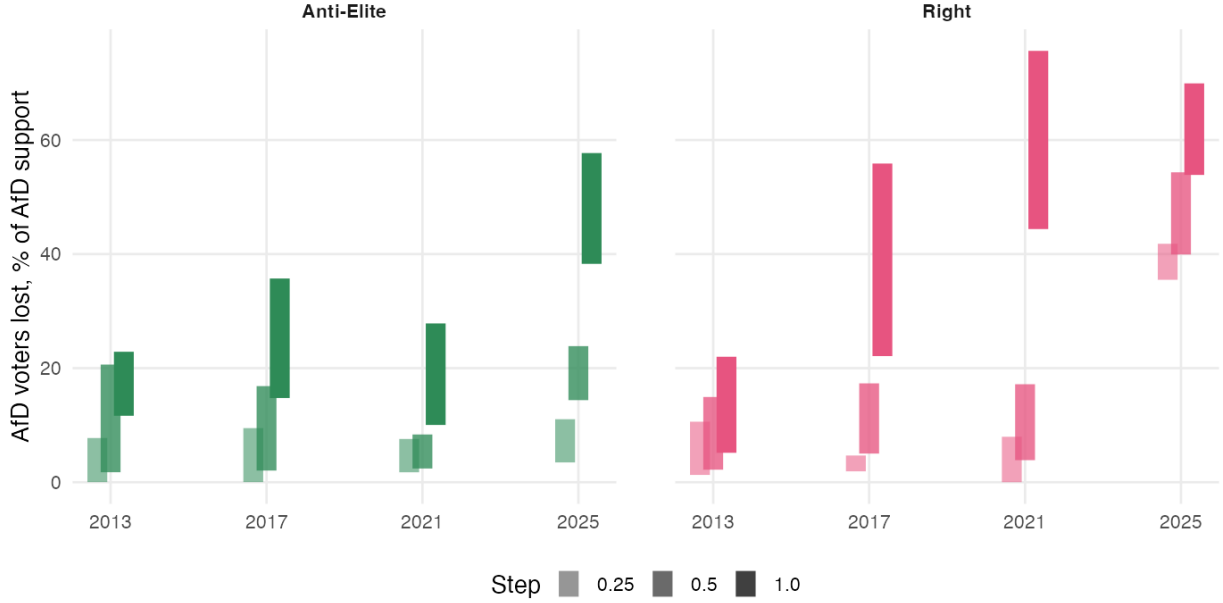
The target parameter is the share of baseline AfD voters who would abandon the party had it shifted to **AfD’**.³¹ Figure 9 plots the results separately for different values of steps and different directions of a shift.

The results suggest that under small steps of 0.25 (0.5), the share of voters abandoning the AfD is no more than 10% (20%) across years and dimensions. Shifts of a full step of 1 yield much larger losses, especially along the Right dimension, reaching up to 70%; importantly, for such large shifts the share of losses along the Anti-Elite dimension is much smaller, highlighting that there are just as many voters supporting the AfD in this area.

The results also suggest that the competition between AfD and other parties became much tighter by 2025, in the sense that even small AfD shifts (particularly along the Left-Right dimension) would produce large defections of voters to other parties. This was not the case before 2025, indicating that the party had previously existed in relative isolation.

If competition with the establishment parties has indeed intensified, those parties should be able to capture a large mass of AfD voters by moderating their positions only slightly. I now turn to such an analysis.

$$^{31}\text{Precisely, } \theta_{\text{Loss, A-E}} = \frac{\iint_{X_{\text{Loss, A-E}}} f(x \mid m, d) dx dF_{M,D}(m, d)}{\iint_{X_{\text{AfD}}} f(x \mid m, d) dx dF_{M,D}(m, d)}, \quad X_{\text{Loss, A-E}} = X_{\text{AfD}} \setminus X_{\text{AfD}'}$$



Notes. Each bar is the sharp identified interval for the share of *baseline* AfD voters who would abandon the party if it moderated by one step along the indicated dimension, holding all other parties at their observed positions. In the left panel (**Anti-Elite**) the AfD moves toward the establishment along the anti-elite dimension; in the right panel (**Right**) the AfD moderates along the left–right dimension. Within each year, shading denotes the size of the moderation step (0.25/0.5/1.0 in standardized policy-space units; see the discussion just above for its substantive interpretation). National figures aggregate the East and West regional bounds. Partitions are built under the weighted (anisotropic) spatial metric with the estimated relative weight w_{AE} .

Figure 9. Estimated shares of AfD voters loss under counterfactual AfD shifts

6.4 The response from the parties of the establishment

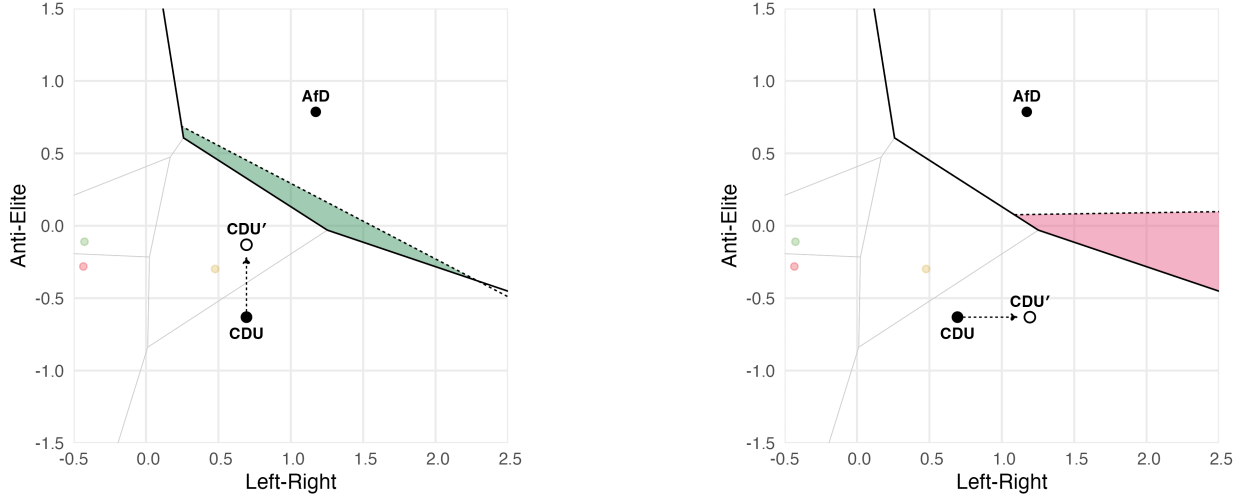
German establishment parties, which once commanded a large majority of the electorate, are now losing ground to the AfD, which leads the polls nationwide as of mid-2026. The development is by no means unique to Germany. Across Western democracies, populist and radical parties have risen sharply over the past two decades, in many cases entering national legislatures and governments. Examples include Italy, France, the Netherlands, Spain, Sweden among others. Beyond Europe, Donald Trump’s rise is widely regarded as part of the same populist wave.

This raises a natural question: how should establishment parties compete with populists? The answer to this question is far from obvious, as In this section I explore how effective it would be for them to adopt positions closer to the AfD in the two-dimensional ideological space defined by Left–Right and Anti–Elite dimensions. The main establishment party that could, in theory, compete with the AfD is the Christian Democratic Union (CDU), historically one of the two largest parties located on the right of the Left–Right spectrum.

In particular, I conduct an analysis similar to that in the previous subsection, but ask the reverse question: *if the CDU moves a given step towards the AfD along the Left–Right or Anti–Elite*

dimension, how many AfD voters would it capture?

Figure 10 provides the illustration of this counterfactual scenarios.



(a) CDU shift towards 0.5 more anti-elite position

(b) AfD shift towards 0.5 more right position

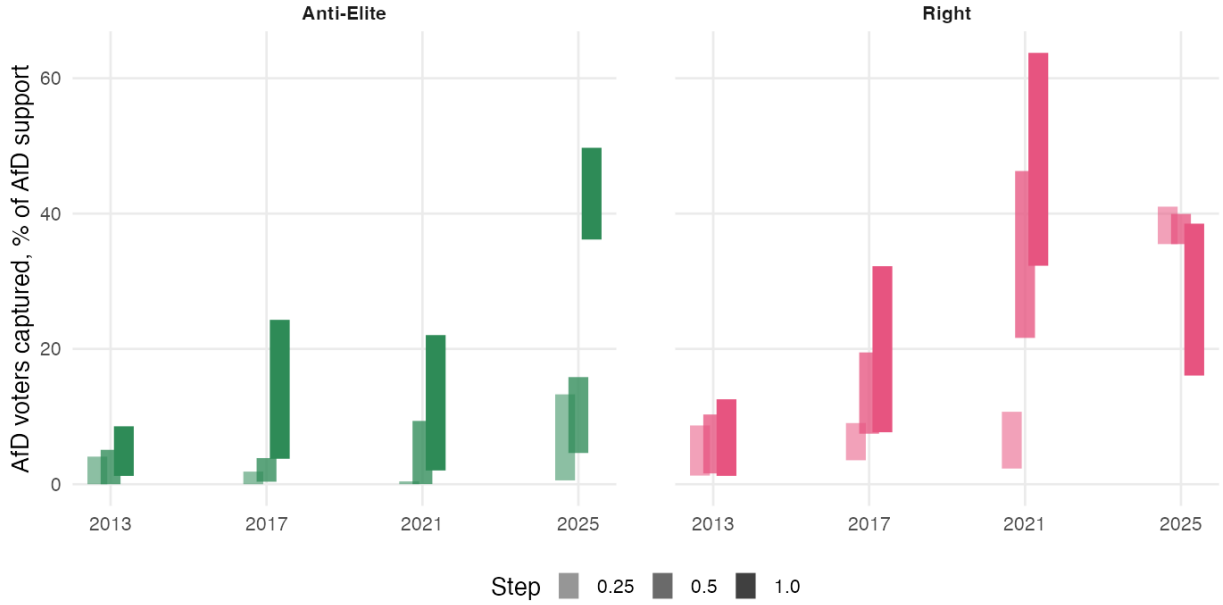
Notes: The panels present an illustration of the unweighted cropped policy space, with partitions determined by median party positions across years. Shaded dots mark the median positions of parties other than the AfD and CDU. **CDU** denotes the median CDU position across years, and **CDU'** denotes the shifted CDU position—either towards the anti-elite position (Panel (a)) or towards a more right-wing position (Panel (b)). Solid lines determine the existing partition, the dashed line – counterfactual partition. The illustrated shift is 0.5 units along each dimension. Since positions are standardized by construction, a 0.5-unit shift is comparable across dimensions. The exercise estimates the share of voters the AfD would lose under the shift, i.e. those located in the green or pink areas in each panel.

Figure 10. Illustration of counterfactual shifts by CDU by 0.5 units separately on two dimensions

Figure 11 plots the results separately for different step sizes and different directions of the shift. The results suggest that even a small shift (0.25) along the Left–Right dimension could allow the CDU to capture up to 10% of AfD support across all years. Shifts of size 0.5 and 1 would yield only marginally larger gains. However, as discussed above, such larger steps may be too costly for a party with a relatively stable policy platform over the years, leaving the 0.25 step as the only realistic option.

Note that in 2025 a 0.25 shift in the CDU’s position along the Left–Right dimension could allow it to capture between 35% and 40% of the entire AfD electorate. While this exercise is not designed to solve the CDU’s vote-share maximization problem, it is clear that by 2025 40% of AfD voters constituted a large mass of the electorate, and the CDU would be unlikely to lose a comparable mass of voters on the other (left) side of its attraction area.

Putting this result into perspective, it implies that if the CDU had slightly hardened its positions on immigration in the 2025 elections, it could have captured more than a third of the entire AfD electorate. Concretely, this could mean shifting from “Agree” to “Rather Agree” on the question



Notes. Each bar is the sharp identified interval for the share of *baseline* AfD voters who would abandon the party if CDU moved by one step along the indicated dimension, holding all other parties at their observed positions. In the left panel (**Anti-Elite**) the CDU moves toward the anti-elite positions; in the right panel (**Right**) the AfD moves towards the right positions. Within each year, shading denotes the size of the moderation step (0.25/0.5/1.0 in standardized policy-space units; see the discussion just above for its substantive interpretation). National figures aggregate the East and West regional bounds. Partitions are built under the weighted (anisotropic) spatial metric with the estimated relative weight w_{AE} .

Figure 11. Estimated shares of AfD voters loss under counterfactual CDU shifts

*“Immigration is good for the economy”, from “Agree” to “Strongly Agree” on “Immigrants should adapt to German culture”, and from 8 to 10 on “Self-Position on Immigration, 1–11”.*³²

These results are consistent with the experimental findings of [Guenther and Nunnari \(2025\)](#), who find that perceiving the CDU as one unit more liberal on immigration (on a 0–10 scale) raises the intention to vote for the AfD by an average of 5 percentage points – about 25% of the AfD vote. Putting the results on the same scale, my results suggest that the 2 units CDU hardening on immigration on the 0–10 scale would capture about 40% of the AfD electorate. They are also consistent with the documented defection of voters between parties in 2025: about 1.2 million CDU supporters in 2021 voted for the AfD in 2025, making this the largest such flow between any two parties in the country ([Vehrkamp and Posthofen, 2026](#)). The counterfactual is thus likely driven by the fact that the CDU could win these voters back with a slightly hardened Left–Right position.

This section was intended to provide some intuition for what the best strategy might be for establishment parties competing with populists. Notice that the party’s hardening of its anti-elite positions does not appear to be effective in any year—in particular, not even in earlier years, when

³²All illustrative answers here are given relative to the real median answers of a CDU candidate in the 2025 survey.

the AfD electorate was less pronouncedly right-wing. The party selected for this exercise was a large center-right establishment party, which could be affecting the results. Nonetheless, the finding that competition with populists is more effective along the Left–Right dimension at least for this party is a reassuring one, since competing on anti-elite rhetoric could, among other things, contribute to the erosion of democratic institutions.

7 Discussion

This paper develops a nonparametric approach to recovering voter preferences from aggregate election returns. Building on the spatial voting model, the method exploits candidate positions across choice situations to partition the policy space into regions of observationally equivalent choice behavior. Bounds on interpretable target parameters are then obtained by solving linear programs, without imposing assumptions on unobserved preference heterogeneity.

I apply the approach to study the evolution of preferences of the electoral base of the radical right populist party in Germany, Alternative for Germany (AfD), across the 2013, 2017, and 2021 Bundestag elections. Several main findings emerge. First, the bliss-point distribution of AfD voters became more closely aligned over time with the party’s right-wing ideological stance. Second, AfD voters consistently put more weight on the party’s conservative positions than on its anti-establishment stance. Taken together, Shapley decomposition exercise suggest that up to 80% of later AfD support is driven by its conservative left-right positions.

What lessons can be drawn from this exercise? A large political economy literature has sought to explain the rise of populism in Western democracies, either by identifying specific causal forces that led voters to abandon establishment parties or by modeling the defining features of populism in order to derive and test theoretical implications. This paper takes a different perspective and seeks instead to *understand the preferences of populist voters at large*. Doing so can help identify the most relevant underlying forces behind populist support and, in turn, inform the design of appropriate responses.

This question is especially important given widespread concern that populism may be harmful to economic performance and democratic institutions; indeed, [Funke, Schularick and Trebesch \(2023\)](#) provide the first robust evidence that populism can reduce economic growth. In this context, understanding how best to respond is crucial, and such responses must begin with an understanding of what populist voters demand. Broadly speaking, one can distinguish between two classes of responses: those addressing voters’ anti-establishment concerns, and those addressing their ideological preferences. The former may involve forms of commitment politics (see [Bellodi et al. \(2025\)](#) for a

formalization), whereas the latter call for more economically and culturally conservative policies.

The results suggest that populist voters’ demands are not fixed, but could evolve over time. In Germany, earlier AfD voters appear to have favored more stronger commitment politics from establishment parties, whereas later AfD voters increasingly prioritized more conservative policy positions. These shifts are broadly consistent with the responses of mainstream parties. In particular, after Friedrich Merz became CDU leader in early 2022, the party adopted a more distinctly conservative profile.

More generally, the paper provides a framework for unpacking the importance of the bundle of characteristics embodied in any political option. Each party or candidate can be understood as a product with multiple attributes, and identifying which of these attributes matter most to voters is far from straightforward. Voting differs from most settings studied in empirical industrial organization as, in a spatial voting model, individuals have unobserved ideal points along each dimension. As a result, one cannot impose the standard linear relationship between characteristics and utility often used in the literature: for example, the question *Do you prefer sweeter or less sweet cereal?* does not translate directly into *Do you prefer a more right-wing or a less right-wing candidate?*

The method is portable to other electoral settings in which variation in candidate choice sets provides identifying variation, offering a general tool for recovering multidimensional voter preferences from aggregate data. Such variation may arise, for example, from district-level differences in candidate characteristics, simultaneous elections, or ranked-choice voting. These settings account for a substantial share of existing electoral environments. The framework can also be extended to incorporate turnout, candidate valence, or additional ideological dimensions.

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A Identification of dimension weights

This section establishes that the relative weight in the two-dimensional weighted spatial voting model of Section 2 is identified up to an interval on $\mathbb{R}_+$.

The intuition behind the result is as follows. The weight w_2 determines which cells of the MRP are non-empty, and weights are therefore identified only up to the differences in the cells they make available. The argument has two steps. First, a single cell is non-empty only for weights in an interval. Second, since a partition is realized only where all of its cells are simultaneously non-empty, the weights yielding a given partition form an intersection of such intervals, which is again an interval.

A.1 Setting and notation

I use the weighted spatial voting model of Section 2 with the normalization $w_1 = 1$, so the single parameter is the relative weight $w_2 = w > 0$ on the second dimension. Recall that the cell chosen by candidate j under a position vector y is $\mathcal{X}_{j;w}(y) = \{x \in \mathcal{R} : d(x, Y_{ij})_W \leq d(x, Y_{ik})_W \ \forall k \neq j\}$, with $d(x, Y_{ij})_W^2 = (x_1 - Y_{ij,1})^2 + w(x_2 - Y_{ij,2})^2$.

Superimposing the partitions induced by every $y \in \mathcal{Y}$ at weight w yields the minimal relevant partition $\mathbb{X}_w$.³³ A cell of $\mathbb{X}_w$ is indexed by a *choice profile*

$$\mathbf{c} = (j(y))_{y \in \mathcal{Y}}, \quad j(y) \in \mathcal{J},$$

recording the candidate chosen under each environment, and equals the intersection

$$\mathcal{X}_w(\mathbf{c}) = \bigcap_{y \in \mathcal{Y}} \mathcal{X}_{j(y);w}(y). \quad (\text{A1})$$

The profile $\mathbf{c}$ is *available* at weight w when $\mathcal{X}_w(\mathbf{c}) \neq \emptyset$; only available profiles enter the prediction of observed vote choices (cf. Equation 5). A partition $\mathbb{X}_w$ is thus summarized by its set of available profiles,

$$\mathcal{C}(\mathbb{X}_w) = \{\mathbf{c} : \mathcal{X}_w(\mathbf{c}) \neq \emptyset\},$$

and two weights are observationally equivalent whenever they induce the same $\mathcal{C}(\mathbb{X}_w)$.

The cell as a system of linear constraints. By (A1), $\mathcal{X}_w(\mathbf{c})$ is an intersection of half-planes: one per environment y and rival $k \neq j(y)$, for a total of $M = \sum_{y \in \mathcal{Y}} (|y| - 1)$ constraints. The set of bliss points at least as close to an own position s as to a rival s' is the half-plane $\{x : d(x, s)_W^2 \leq d(x, s')_W^2\}$, which after expansion reads $2(s_1 - s'_1)x_1 + 2w(s_2 - s'_2)x_2 \leq (s_1^2 - s_1'^2) + w(s_2^2 - s_2'^2)$.³⁴ Indexing the constraints by $m = 1, \dots, M$, each takes the form

$$\alpha_m x_1 + w \beta_m x_2 \leq \gamma_m + w \delta_m, \quad m = 1, \dots, M, \quad (\text{A2})$$

with fixed constants $\alpha_m = 2(s_1 - s'_1)$, $\beta_m = 2(s_2 - s'_2)$, $\gamma_m = s_1^2 - s_1'^2$, $\delta_m = s_2^2 - s_2'^2$, where for constraint m arising from environment y and rival k the own position is $s = Y_{i,j(y)}$ and the rival is

³³This coincides with the coarsest partition of Section 2: superimposing the per-environment partitions merges exactly those bliss points that choose identically under every $y \in \mathcal{Y}$.

³⁴Ties (the bisector itself) occur with probability zero under the continuity assumption of Section 2, so the weak/strict inequality convention is immaterial for availability.

$s' = Y_{ik}$. The key structural feature is that w enters (A2) only linearly, both in the coefficient of x_2 and in the threshold, and the same scalar w is shared across all M constraints.

The *availability set* of a profile $\mathbf{c}$ is the set of weights at which its cell is non-empty:

$$\mathcal{W}(\mathbf{c}) = \{ w > 0 : \mathcal{X}_w(\mathbf{c}) \neq \emptyset \} = \{ w > 0 : \exists x \in \mathbb{R}^2 \text{ satisfying (A2) } \forall m \}. \quad (\text{A3})$$

A.2 Set-identification of weights

Lemma 1. *For every choice profile $\mathbf{c}$, the availability set $\mathcal{W}(\mathbf{c})$ is an interval.*

Proof. Fix $w > 0$ and apply the change of variables $(z_1, z_2) = (x_1, wx_2)$, with inverse $x_2 = z_2/w$. For each fixed $w > 0$ this is a bijection on $\mathbb{R}^2$, so $\mathbf{c}$ is available at w if and only if the transformed system

$$\alpha_m z_1 + \beta_m z_2 \leq \gamma_m + w \delta_m, \quad m = 1, \dots, M, \quad (\text{A4})$$

admits a solution $z \in \mathbb{R}^2$. Hence

$$\mathcal{W}(\mathbf{c}) = \{ w > 0 : \exists z \in \mathbb{R}^2 \text{ satisfying (A4)} \}. \quad (\text{A5})$$

The substitution linearizes the bilinear term $w\beta_m x_2$ into $\beta_m z_2$, so w now appears only in the thresholds. Define the augmented set in (z_1, z_2, w) -space

$$\Omega = \{ (z_1, z_2, w) \in \mathbb{R}^3 : \alpha_m z_1 + \beta_m z_2 - w \delta_m \leq \gamma_m, \quad m = 1, \dots, M \}. \quad (\text{A6})$$

Each defining inequality is linear in the joint variable (z_1, z_2, w) , so Ω is an intersection of M closed half-spaces of $\mathbb{R}^3$, hence a convex polyhedron.

Let $\pi : \mathbb{R}^3 \rightarrow \mathbb{R}$ be the projection $\pi(z_1, z_2, w) = w$. Comparing (A5) and (A6),

$$\mathcal{W}(\mathbf{c}) = \pi(\Omega) \cap (0, \infty).$$

The image of a convex set under a linear map is convex, so $\pi(\Omega)$ is convex; intersecting with the convex set $(0, \infty)$ preserves convexity. Therefore $\mathcal{W}(\mathbf{c})$ is an interval. $\square$

Theorem 1. *The set of weights inducing a given profile set $\mathcal{C}(\mathbb{X}_w)$ is an interval.*

Proof. Fix a partition and let $\mathcal{C}$ denote its set of available profiles. The partition is realized exactly at weights where every one of its cells is non-empty, so the set of such weights is

$$\bigcap_{\mathbf{c} \in \mathcal{C}} \mathcal{W}(\mathbf{c}).$$

By Lemma 1 each $\mathcal{W}(\mathbf{c})$ is an interval, and a finite intersection of intervals is again an interval. Since $\mathcal{Y}$ is finite, $\mathcal{C}$ is finite, which concludes the proof. $\square$

B Weights estimation: simulation

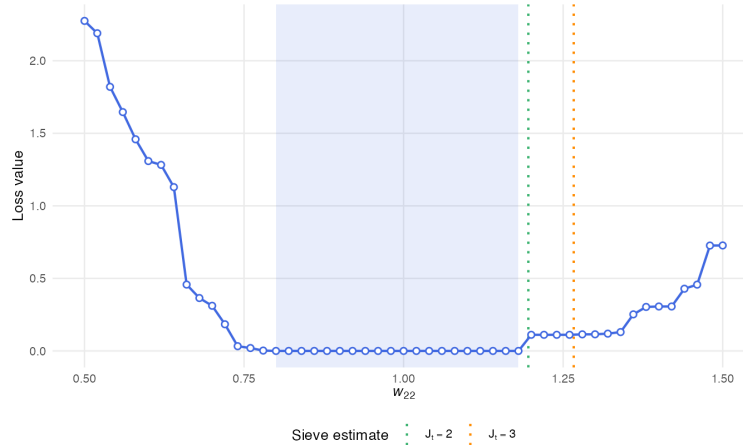
This section provides a simulation comparing the weight estimator based on 1) the method used in this paper, and 2) an approach that allows for a simultaneous recovery of bliss points and weights from Merlo and De Paula (2017).

I simulate a population of 10,000 voters voting in twenty-four elections with six candidates drawn at random. Bliss points are drawn from a mixture of two dominant clusters and a diffuse background. Voters choose the candidate closest to their bliss point under the weighted Euclidean metric with true weight $w_2 = 1$; the researcher observes only the resulting candidate vote shares in each elections.

For each candidate value of w_2 on a grid, the implied superimposed Voronoi tessellation determines the set of feasible cells (joint vote patterns). The question then is whether voter mass can be distributed across these cells so as to reproduce the observed shares exactly, subject to masses being non-negative and summing to one. The set of weights for which an exact fit exists is the identified set: since every cell is feasible only on an interval of weights, and concentrated vote shares force particular cells to carry mass, the weights consistent with the data form an interval around the truth.

Figure B1 plots the residual fit against w_2 . The loss is exactly zero on the interval $[0.80, 1.18]$, which contains the true value $w_2 = 1$, and rises outside it. The cell geometry alone thus set-identifies the weight without any assumption on the distribution of bliss points.

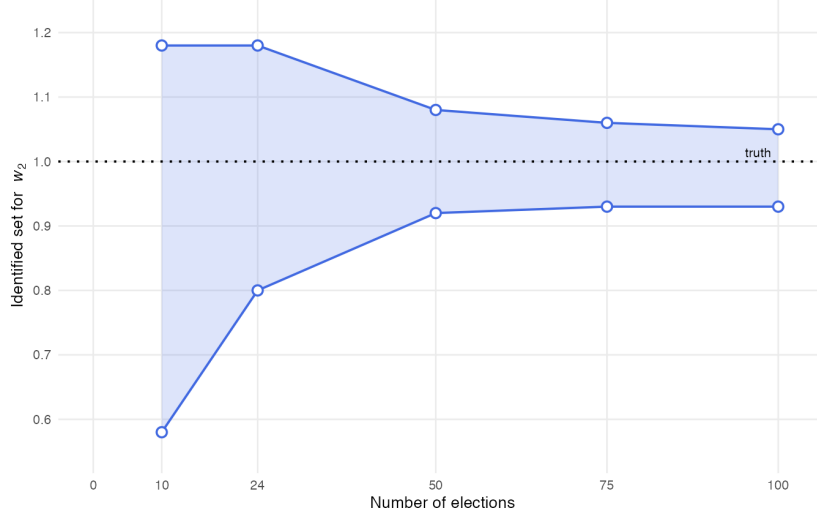
The dotted lines mark the estimates obtained from the sieve estimator at polynomial orders $J_t = 2, 3$, which impose a flexible parametric form on the bliss-point density. Because a low-order polynomial correction of a Gaussian base cannot reproduce the distribution of voters' bliss points, the resulting weight estimates are far from the truth as well, at 1.20 and 1.27 for $J_t = 2, 3$ respectively.



Notes: Monte Carlo with 10,000 voters, true $w_2 = 1$, twenty-four six-candidate elections; bliss points follow a two-cluster mixture with a diffuse background, and only vote shares are observed. The solid curve is the minimal L^1 distance between observed shares and shares generated by any distribution of voter mass over the cells of the superimposed tessellation at each candidate weight. The shaded region is the identified set $[0.80, 1.18]$, where the fit is exact; it contains the truth. The dotted lines mark the sieve estimates of Merlo and De Paula (2017) at $J_t = 2, 3$ ($\hat{w}_2 = 1.20, 1.27$), both of which lie outside the identified set.

Figure B1. Set identification of the weight via the method used in this paper, and sieve estimates from Merlo and De Paula (2017)

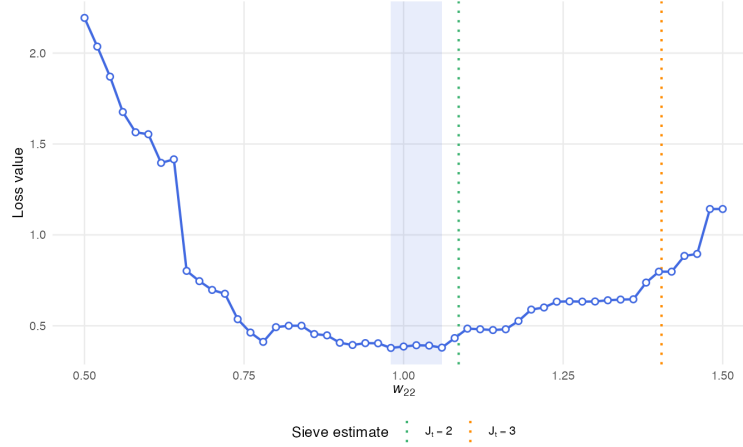
Number of Elections. The identified set shrinks as the number of elections grows. Because each additional election contributes a further set of cells that the observed shares may or may not be able to populate, more elections impose more restrictions on which weights admit an exact fit. Figure B2 traces the identified set for w_2 against the number of elections, holding the data-generating process fixed: the set narrows monotonically around the true value, from $[0.58, 1.18]$ with ten elections to $[0.93, 1.05]$ with one hundred. Note that this simulation illustrates that the size of the identified sets shrinks with the number of elections, the size of the identified sets itself at every elections level depends on the positions of the candidates, and the distribution of preferences.



Notes: Each point pair gives the lower and upper bounds of the identified set for w_2 (the weights admitting an exact fit), computed as in Figure B1 but varying the number of elections. The dotted line marks the true value $w_2 = 1$.

Figure B2. Identified set for the weight at different numbers of elections

Noisy perceptions of candidate positions. Realistically, however, there is always noise in the data. I next simulate the behaviour of both estimators under the assumption that each voter observes every candidate's position with an idiosyncratic mislocation error, distributed $\mathcal{N}(0, 0.1^2 I)$, while the estimation procedures remain unchanged. Under noise an exact fit is no longer attainable at any weight. The question, however, is where the minimum is located and how large the basin delivering a near-identical fit is. The region of near-minimal losses becomes *smaller*: the loss now develops a strict interior minimum, and the near-feasible set (weights whose fit is within one percentage point of share mass of the best attainable) shrinks from the noise-free interval $[0.80, 1.18]$ to $[0.98, 1.06]$, still containing the truth (Figure B3). Intuitively, the smoothed shares are best approximated by the cell structure of the true weight, so mild noise converts the flat exact-fit region into a well-defined basin centred near $w_2 = 1$. The sieve estimator yields $\hat{w}_2 = 1.09$ and 1.40 at $J_t = 2, 3$, both falling outside the near-feasible set. Note that in this simulation the results obtained with $J_t = 2$ are closer to the truth; this suggests that additional flexibility in the estimator does not necessarily translate into more precise estimates.



Notes: Same setup as Figure B1, except that each voter perceives every candidate’s position with an idiosyncratic $\mathcal{N}(0, 0.1^2 I)$ mislocation error. The solid curve is the minimal L^1 distance between observed shares and shares generated by any distribution of voter mass over the cells of the superimposed tessellation at each candidate weight; with noise the loss is strictly positive everywhere. The shaded region is the near-feasible set $[0.98, 1.06]$ (fit within 0.01 of the minimum), which contains the true value $w_2 = 1$. The dotted lines mark the sieve estimates of Merlo and De Paula (2017) at $J_t = 2, 3$ ($\hat{w}_{22} = 1.09, 1.40$), both outside the near-feasible set.

Figure B3. Set identification of the weight via the method used in this paper, and sieve estimates from Merlo and De Paula (2017) with voter perception noise

C Exploiting congressional district borders for identification

Longuet-Marx (2025) exploit differences in candidate positions across congressional district borders to identify voters’ preferences, but rely on a different model of voters’ choice. In this section I illustrate how this type of identifying variation could in principle be used to estimate the primitives of vote choice using the model considered in this paper.

For simplicity, I assume a one-dimensional model of choice, with the policy space normalized to the interval $[1, 5]$, and a population that is homogeneous in its observed characteristics across space (i.e. the entire population belongs to a single group d , for example educated females aged 30–35). This assumption simplifies the exposition without affecting the substantive argument.

Suppose that precincts with positions y and y' are paired. Then $f(x | y, m, d) = f(x | y', m, d)$ by the assumption that congressional district borders are exogenous. Likewise, for another pair y'' and y''' we have $f(x | y'', m, d) = f(x | y''', m, d)$. Crucially, this approach does not assume $f(x | y, m, d) = f(x | y'', m, d)$; that is, preferences may differ across distinct pairs of precincts.

Consider a two-party system, which is the most demanding case, as it provides the least variation. Each choice situation is then characterized by the positions of two candidates on a one-dimensional space. From the perspective of a spatial voting model, the object that uniquely defines this situation is the cut-point between the two candidates. Hence choice situation y yields a cut-point y_c , while y' yields a cut-point y'_c . The researcher observes vote shares for both candidates in both precincts, as illustrated in Figure C1.

Since the distribution of bliss points is identical within a pair of contiguous precincts, one can overlay the two sets of results to obtain a single, more finely partitioned distribution, as shown in Figure C2.



Figure C1. Two paired precincts. Candidate positions (A, B), the implied cut-point (thin vertical line), and observed vote shares differ across the pair, while the underlying distribution of bliss points is common by the exogenous-border assumption.

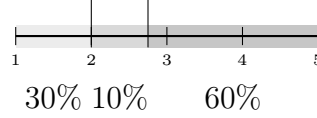


Figure C2. Overlaid distribution for the paired precincts of Figure C1. The two cut-points partition the line into three intervals, yielding refined shares of the common bliss-point distribution.

Repeating this exercise across many pairs of contiguous precincts yields a collection of refined distributions of bliss points. Note, however, that these distributions are assumed to differ across pairs and therefore cannot be overlaid with one another, even when, as in the case considered here, the demographics are identical. What the researcher can do instead is draw inferences from this collection of distributions, using the known population size of each pair of precincts.

For example, suppose the precinct pair above has a population of one thousand. The numbers of individuals lying in each of the three intervals are then 300, 100, and 600.

Now consider a second precinct pair of 1,500 individuals, with cut-points and shares as displayed in Figure C3. The corresponding numbers of individuals in each interval are 450, 300, and 750.

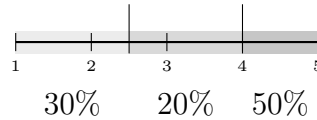


Figure C3. Overlaid distribution for a second precinct pair, with distinct cut-points and shares. This distribution is specific to the pair and cannot be combined with that of Figure C2.

What can be learned from the two distributions jointly? Although they differ and their electoral results cannot be pooled, one can still pose partial-identification questions in the spirit of this paper. For instance, what share of individuals of the given demographic hold left-wing views, defined as a bliss point below 2.5? This share necessarily includes the 30% from the first pair, that is 300 individuals; from the second pair it may include anywhere between 0 and 450 individuals, since the interval boundary at 2.5 coincides with a cut-point and the within-interval distribution is unrestricted. Hence, out of 2,500 individuals in total, the share holding left-wing views is set-identified within $[0.12, 0.30]$.

Note that identifying the dimension weights in this setup would not be possible without further assumptions. As discussed in the main text, the weights are identified through variation in the structure of the partition that candidate positions induce on the policy space, that is, in the composition of the resulting cells. Overlaying only two elections within a pair of contiguous electoral districts

generates little to no variation in this cell composition, and is therefore insufficient to pin down the weights.

D Model Estimation

This section describes how the empirical restrictions implied by the spatial voting model are translated into a finite-dimensional optimization problem, and how weights and bounds for target parameters are computed.

D.1 Weights estimation

This appendix details how the relative weight w_{AE} on the anti-elite dimension is computed. Throughout, the left–right weight is normalised to $w_{LR} = 1$, so the only free parameter is $w_{AE} \in \mathbb{R}_+$.

The exercise is run separately for each region $R \in \{\text{East}, \text{West}\}$ and year.³⁵ Within a region, the market m is the *state*, the submarket s is the *municipality* (with $s \in m$), and d indexes the 24 age×gender×education demographic groups, with observed demographic shares $q_{s,d}$, $\sum_d q_{s,d} = 1$. An *election* y is the candidate configuration of an electoral district; $\mathcal{Y}_R$ collects the elections of region R , and each municipality s belongs to one election $y(s)$ and one state $m(s)$. The choice set is the six parties $\mathcal{J}$.

For a candidate value w_{AE} , the weighting matrix is $W = \text{diag}(1, w_{AE})$. Since w_{AE} enters the distance $d(x, Y_{ij})_W = \sqrt{(x - Y_{ij})^\top W (x - Y_{ij})}$, it rescales the anti-elite axis relative to left–right and thus tilts the bisecting hyperplanes separating any two parties, reshaping each attraction region $\mathcal{X}_{j;w}(y)$ and hence which cells (joint choice sets) appear in the MRP.

Overlaying these regions across all elections in $\mathcal{Y}_R$ yields the minimal relevant partition (MRP) $\mathbb{X}_w$, and $\mathbb{X}_{j;w}(y)$ denotes the cells of $\mathbb{X}_w$ on which j wins under y . Because the number of feasible cells governs how well any given partition can fit the data, and this count varies with w_{AE} , I hold it fixed across weights: I restrict attention to the AfD attraction region, retain the $N = 500$ largest cells, and merge each smaller fragment into the neighbouring cell with which it shares the longest boundary. Let $\mathbb{X}_w$ denote the resulting 500-cell mesh.

The unknowns are the state-and-group mass functions

$$\phi(\mathcal{X} \mid m, d) \geq 0, \quad \sum_{\mathcal{X} \in \mathbb{X}_w} \phi(\mathcal{X} \mid m, d) = 1,$$

one per state m in region R and demographic group d . The weight w_{AE} is held common across all states and groups of the region–year. Given ϕ and w_{AE} , the predicted AfD vote share in municipality s is the finite-dimensional analogue of (9),

$$s_{\text{AfD}}(y(s), s; \phi, w_{AE}) = \sum_d q_{s,d} \sum_{\mathcal{X} \in \mathbb{X}_{\text{AfD};w}(y(s))} \phi(\mathcal{X} \mid m(s), d).$$

In finite samples the exact restriction (9) generically has no solution, so rather than testing solvability I measure how close it can be brought to holding. For each w_{AE} I choose the mass

³⁵I run the analysis separately for East and West Germany because precise estimation of the weights requires substantial variation in candidate positions; pooling electoral districts within each region helps achieve this.

functions that minimise the mean absolute municipal AfD-share error,

$$\mathcal{E}(w_{AE}) = \min_{\{\phi(\cdot|m,d)\} \in \Phi} \frac{1}{|\mathcal{S}_R|} \sum_{s \in \mathcal{S}_R} \left| s_{\text{AfD}}(y(s), s) - s_{\text{AfD}}(y(s), s; \phi, w_{AE}) \right|,$$

where $\mathcal{S}_R$ is the set of municipalities in region R and $s_{\text{AfD}}(y(s), s)$ the observed share. Introducing slacks $u_s \geq 0$, this is the linear program

$$\min_{\{\phi(\cdot|m,d)\}, u_s \geq 0} \sum_{s \in \mathcal{S}_R} u_s \quad \text{s.t.} \quad -u_s \leq s_{\text{AfD}}(y(s), s; \phi, w_{AE}) - s_{\text{AfD}}(y(s), s) \leq u_s,$$

solved over $\phi \in \Phi$.

The program is solved for every candidate weight on the grid

$$\mathcal{W} = \{0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 1.0\} \cup \{1/0.9, \dots, 1/0.4\} \quad (13 \text{ points}),$$

symmetric around 1 in log scale, and the estimate is the fit-minimising weight

$$\hat{w}_{AE} = \arg \min_{w_{AE} \in \mathcal{W}} \mathcal{E}(w_{AE}).$$

Values $\hat{w}_{AE} < 1$ mean voters tolerate larger distances on the anti-elite dimension (the left–right dimension weighs more); $\hat{w}_{AE} > 1$ the reverse. Because the data are noisy the minimizer is generically a single point.

Estimation is carried out independently by region, year. Each election contributes a random subsample of 50 municipalities, and the number of elections per state is capped at 20 (East) / 15 (West). The 500-cell mesh is rebuilt and re-truncated at every w_{AE} .

D.2 Target parameters estimation

This appendix describes the bounds used for the main Anti-Elite versus Right exercise. Estimation of bounds for the rest of the exercises follow the same logic. It follows the generic target-parameter LP and adds the two features specific to that exercise: the demographic-group masses are made *state specific* but are pooled onto a single region-wide partition, and they are tied together across states of a region by a *cross-state coupling* restriction. The exercise is run separately for each region $R \in \{\text{East}, \text{West}\}$ and year, with the relative weight fixed at the estimate $\hat{w}_{AE}$ from Appendix D.1.

Region-pooled partition and state-specific masses. Under the weight $\hat{w}_{AE}$ I overlay the elections of *all* states of region R to form one minimal relevant partition, and — exactly as in Appendix D — restrict it to the AfD attraction region and fix it at the $n_c = 500$ largest cells. Cells are indexed $c \in \{1, \dots, n_c\}$ and demographic groups (age $\times$ gender $\times$ education bins) by $d \in \{1, \dots, D\}$. Unlike the baseline of Appendix D, the masses are indexed by state m : for each state m in region R and each group d ,

$$\boldsymbol{\pi}_d^{(m)} = (\pi_{d1}^{(m)}, \dots, \pi_{dn_c}^{(m)})^\top \in \mathbb{R}_+^{n_c}, \quad \mathbf{1}^\top \boldsymbol{\pi}_d^{(m)} = 1,$$

so each state has its own bliss-point distribution over the shared region mesh. Municipalities are indexed $r \in \{1, \dots, R\}$, each belonging to one state $m(r)$; y_r is the observed AfD share and $w_{r,d}$ the turnout-adjusted share of group d among eligible voters in r . With $a_{r,c}^{\text{AfD}} \in \{0, 1\}$ the model-implied

AfD choice indicator in r and $\mathbf{a}_r^{\text{AfD}}$ their vector, the predicted AfD share is

$$\hat{y}_r(\boldsymbol{\pi}) = \sum_{d=1}^D w_{r,d} (\mathbf{a}_r^{\text{AfD}})^\top \boldsymbol{\pi}_d^{(m(r))}. \quad (\text{D1})$$

Counterfactual target regions. Let $(\bar{X}^{RR}, \bar{X}^{AE})$ be the AfD position (region median) on the left–right and anti-elite dimensions. I split the AfD attraction region into two target areas by which of two reference points is nearer under the weighted metric $W = \text{diag}(1, \hat{w}_{AE})$:

$$C^{AE} = (\bar{X}^{RR} - 1, \bar{X}^{AE}), \quad C^R = (\bar{X}^{RR}, \bar{X}^{AE} - 1).$$

The *Anti-Elite* region collects the cells nearer C^{AE} — a counterfactual that pulls on the economic axis and so retains the voters whose attachment to the AfD is anti-elite rather than economic; the *Right* region collects the cells nearer C^R , which pulls on the anti-elite axis and retains the economically right-wing voters.³⁶ Writing $b_c^{AE}, b_c^R \in \{0, 1\}$ for the resulting cell indicators, the target parameters are the shares of the region electorate who vote AfD and hold ideal points in each area,

$$\bar{\theta}^\bullet(\boldsymbol{\pi}) = \sum_m \sum_{d=1}^D (\mathbf{b}_d^\bullet)^\top \boldsymbol{\pi}_d^{(m)}, \quad \bullet \in \{AE, R\},$$

where $\mathbf{b}_d^\bullet$ weights the in-area AfD cells by the population mass of group d in state m ; the value is normalised by the subgroup electorate and reported in $[0, 0.5]$.

Constraints. The masses satisfy, besides non-negativity, normalisation, and the fit slacks $-u_r \leq \hat{y}_r(\boldsymbol{\pi}) - y_r \leq u_r$, the following moment restrictions (state-pooled survey targets $X_{m,b}$, group populations $T_{m,b}$):

$$\begin{aligned} (\text{AfD group share}) \quad & \left| \sum_{d \in b} (\mathbf{m}_{m,d}^{\text{AfD}})^\top \boldsymbol{\pi}_d^{(m)} - X_{m,b} T_{m,b} \right| \leq \tau_{\text{RS}} T_{m,b} & \forall m, \text{ base group } b, \\ (\text{Left 2nd choice}) \quad & \sum_d (\mathbf{v}_{m,d}^{\text{Left}})^\top \boldsymbol{\pi}_d^{(m)} \geq (\ell_m - \tau_{\text{SC}}) A_m & \forall m, \end{aligned}$$

with $A_m = \sum_b X_{m,b} T_{m,b}$ the state AfD mass and ℓ_m the observed AfD→Left second-choice rate. Here $\tau_{\text{RS}} = 0.1$ and the second-choice restriction is a one-sided floor with $\tau_{\text{SC}} = 0$; within-state education coupling is left off, so educated and non-educated masses are unrestricted relative to each other.

Cross-state coupling. The distinctive restriction of this exercise ties the states of a region together so that pooling is informative without forcing them to be identical. I coarsen the policy space into blocks B on a 0.5 grid and, for each coarse marginal $g \in \{\text{educ.}, \text{non-educ.}, \text{young}, \text{old}, \text{male}, \text{female}\}$ and block B , require every state’s within-marginal mass share on B to stay within a factor $(1 + \tau_{\text{state}})$ of the population-weighted regional mean. With $e_{m,g,B} = \sum_{d \in g} T_{m,d} \sum_{c \in B} \pi_{dc}^{(m)}$ the state’s block mass, $T_{m,g} = \sum_{d \in g} T_{m,d}$, and an auxiliary regional mean share $M_{g,B}$,

$$\sum_m e_{m,g,B} = \left(\sum_m T_{m,g} \right) M_{g,B}, \quad \frac{M_{g,B}}{1 + \tau_{\text{state}}} \leq \frac{e_{m,g,B}}{T_{m,g}} \leq (1 + \tau_{\text{state}}) M_{g,B} \quad \forall m.$$

³⁶In the code these masks carry the raw labels **Right** = C^{AE} and **AntiEst** = C^R ; the graphs relabel **Right**→“Anti-Elite” and **AntiEst**→“Right”. The display names used here are the final ones.

This is an anchor-to-mean band (not pairwise across states), imposed on the six coarse marginals and on policy blocks rather than individual cells, which keeps the coupling well conditioned. The example configuration uses $\tau_{\text{state}} = 2$ (each state’s block share within a factor of three of the regional mean).

Bounds. I first solve the best-fit program $\hat{Q} = \min_{\pi, \mathbf{u}} \mathbf{1}^\top \mathbf{u}$ subject to all of the above, with baseline mean error $t_{\text{bench}} = \hat{Q}/R$. The approximately feasible set caps the total fit slack additively, $\mathbf{1}^\top \mathbf{u} \leq \hat{Q} + R \max(\epsilon_{\text{abs}}, \epsilon_{\text{rel}} t_{\text{bench}})$ with $\epsilon_{\text{abs}} = \epsilon_{\text{rel}} = 10^{-5}$, and the bounds are the two LPs

$$\underline{\theta}^\bullet = \min_{\pi, \mathbf{u}} \bar{\theta}^\bullet(\pi), \quad \bar{\theta}^\bullet = \max_{\pi, \mathbf{u}} \bar{\theta}^\bullet(\pi),$$

subject to the same constraints and the fit cap. Bounds are computed for the combined electorate and for the education and age subgroups by restricting the objective and its normalising population to the relevant groups. The programs are sparse LPs solved with a dual/interior-point method (HiGHS); when a direct min/max solve fails to converge on the degenerate optimal vertex, the bound is bracketed by a sequence of feasibility LPs (bisection) instead.

Configuration. The reference run pools each region under $\hat{w}_{AE}$, samples 50 municipalities per election, caps elections per state at 20, fixes the mesh at 500 cells, and sets $\tau_{\text{RS}} = 0.1$, $\tau_{\text{SC}} = 0$ (one-sided), education coupling off, and cross-state coupling $\tau_{\text{state}} = 2$ on 0.5 policy blocks.

E Model Fit

Before turning to the target-parameter results, which help explain the rise of the AfD through changes in voters’ preferences, I discuss model fit. Table F3 presents summary statistics on model fit at the municipality level. In every election year, the mean residual is at most 0.003, while the median residual is exactly zero. In 2017 and 2021, the standard deviation equals 0.035, implying that about two thirds of municipalities have absolute residual value below 3.5% (relative to an average municipality-level vote share of about 15% in the later sample years)³⁷. The correlation between predicted and observed vote shares is above 0.8 in the later years, and about 0.4 in 2013. Overall, these results suggest a reasonable model fit, especially in the later years.

While the model does not permit an easy representation of the estimated underlying distributions of bliss points by demographic groups, it permits a reduced-form analysis of how the group composition at a municipal level contributes to the AfD vote share. To explore this, I run a regression separately for each year: Predicted AfD Share_{*m*} = $\sum_d \beta^d w_{m,d} + \gamma_{s(m)}$, where *m* denotes municipality, $\gamma_{s(m)}$ federal state fixed effects, and *d* – aggregated demographic group. Figure F7 plots main heterogeneity results by education, gender, age groups.

The results provide a picture of which groups were more likely to vote for the AfD over time. In particular, they show that in 2017 and 2021, individuals without higher education, men, and those under the age of 60 were, on average, substantially more likely to support the party. These findings are based on aggregate voting data and joint distributions of demographic characteristics, and are

³⁷Note that these values differ from the AfD’s national vote shares in 2017 and 2021 (12.6% and 10.4%, respectively) for two reasons. First, the statistic is calculated at the municipality level, and smaller municipalities (which are also often rural) tend to exhibit higher AfD vote shares. Second, the analysis does not include large cities that appear as a single municipality in the data.

consistent with the literature on populist support, which typically studies the composition of populist voters using survey data (see, e.g., [Gidron and Hall, 2020](#); [Danieli et al., 2023](#)). Moreover, between 2017 and 2021, the gap between individuals with and without higher education widened, and the same is true for the gender gap.

By contrast, the 2013 pattern points in the opposite direction, with more educated and older individuals being more likely to support the AfD. This is consistent with existing evidence on the evolution of the party, from a "party of professors" with conservative positions and relatively moderate criticism of the political establishment in 2013 to an openly radical-right populist party in 2017 and 2021 (see [Cantoni, Hagemeister and Westcott, 2019](#)).

To sum up, both the model fit and the observed heterogeneity in the data suggest that the model performs well in explaining observed AfD voting patterns.

F Tables & Figures

	<i>Dependent variable:</i>			
	<i>Perceived Party Position, L-R</i>			<i>Own L-R Position</i>
	All Parties	Chosen Parties	Chosen AfD	Chosen Parties
	(1)	(2)	(3)	(4)
Candidate, L-R	−0.042 (0.068)	0.392** (0.168)	1.843*** (0.582)	0.114 (0.164)
State F.E.	Yes	Yes	Yes	Yes
Party F.E.	Yes	Yes	Yes	Yes
Observations	49,176	7,023	269	6,878
Adjusted R ²	0.622	0.484	0.091	0.400

*p<0.1; **p<0.05; ***p<0.01

Standard errors clustered by respondent

Table F1. Correlation of LR Positions between Candidates (estimated) and Parties (reported)

	Candidate LR Position		
	All Parties	Chosen Parties	Chosen AfD
Self, LR	0.001 (0.001)	0.002 (0.002)	0.003 (0.010)
State F.E.	Yes	Yes	Yes
Party F.E.	Yes	Yes	Yes
Observations	22,715	3,236	124
R ²	0.883	0.859	0.213
Adjusted R ²	0.883	0.858	0.144

Note: *p<0.1; **p<0.05; ***p<0.01. Estimation is based on German Electoral Survey 2021 Data, which records each individual's electoral district. Self-reported views take values from 0 (very left) to 11 (very right).

Table F2. Candidates' LR positions are uncorrelated with voters' self-reported views

	2013	2017	2021
Mean	0.001	0.003	0.002
SD	0.015	0.035	0.035
P10	-0.016	-0.037	-0.037
Median	-0.000	-0.000	-0.000
P90	0.019	0.046	0.044
N	6,353	6,363	6,363

Note: Unit of observation is municipality. Residual = observed AfD vote share – model-predicted share.

Table F3. Distribution of Residuals Across Municipalities

	2013	2017	2021
Mean	0.051	0.154	0.148
SD	0.019	0.077	0.087
P10	0.029	0.077	0.068
Median	0.049	0.135	0.120
P90	0.076	0.266	0.290
N	6,353	6,363	6,363

Note: Unit of observation is municipality. Vote share expressed as fraction of valid votes cast.

Table F4. Distribution of Observed AfD Vote Share Across Municipalities

Year	Region	Total SD		Between-party		Within-party		AfD only	
		econ	anti-el.	econ	anti-el.	econ	anti-el.	econ	anti-el.
2013	East	0.63	0.60	0.66	0.58	0.21	0.29	0.22	0.11
2013	West	0.64	0.64	0.65	0.61	0.24	0.31	0.25	0.38
2017	East	0.66	0.58	0.67	0.56	0.24	0.28	0.29	0.26
2017	West	0.72	0.64	0.75	0.62	0.22	0.30	0.26	0.28
2021	East	0.72	0.63	0.71	0.56	0.32	0.36	0.48	0.43
2021	West	0.71	0.61	0.74	0.58	0.24	0.30	0.29	0.29
2025	East	0.69	0.42	0.72	0.33	0.22	0.29	0.23	0.32
2025	West	0.75	0.48	0.80	0.41	0.19	0.30	0.19	0.35

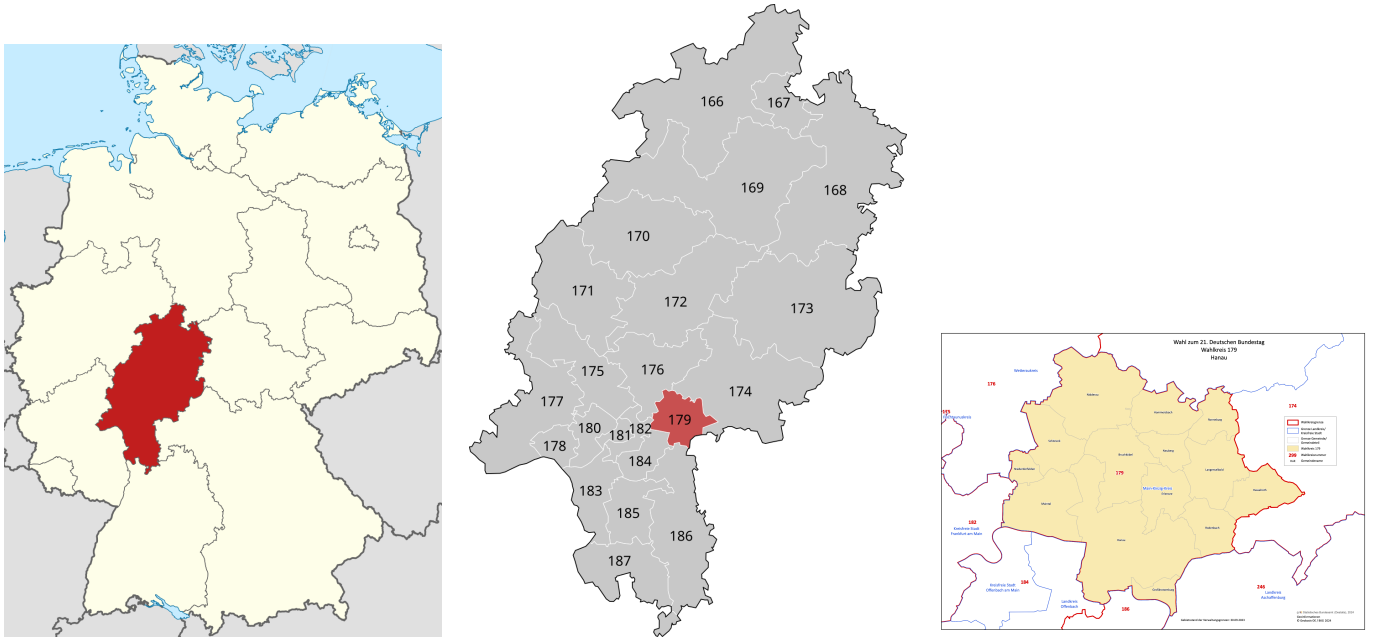
Notes: Standard deviations of district-level (Wahlkreis) candidate positions; between-party = SD of party means, within-party = SD of positions around the own-party mean. The between-party dispersion on the anti-establishment dimension compresses sharply in 2025, reflecting party convergence and the thinner 2025 item battery.

Table F5. Dispersion of estimated candidate positions by year and region

Survey item (question)	Theme	GESIS	Scale	σ (1 pt)
<i>Anti-elite dimension</i>				
Satisfaction with how democracy works in Germany	Current elites	c20	1–4	0.175
Parties are citizens’ central intermediaries	Current elites	d6c	1–5	0.144
Referendums lead to badly designed laws	Direct democracy	d6g	1–5	0.138
Legislation reflects citizens’ interests	Current elites	d6b	1–5	0.137
Citizens have lost trust in politics (rev.)	Current elites	d6h	1–5	0.137
There are enough opportunities to participate	Direct democracy	d6a	1–5	0.117
There should be federal-level referendums (rev.)	Direct democracy	d6f	1–5	0.098
<i>Left–right (economic) dimension</i>				
Same-sex marriage should be banned (rev.)	Culture	c2d	1–5	0.096
Immigrants are good for the German economy	Immigration	c2i	1–5	0.093
Women should decide themselves about abortion	Culture	c2j	1–5	0.085
Providing social security is a core government task	Econ. policy	c2g	1–5	0.081
Government spending on development aid	Econ. policy	c18h	1–5	0.080
Immigrants should adapt to German culture (rev.)	Culture	c2a	1–5	0.073
The government should stay out of the economy (rev.)	Econ. policy	c2b	1–5	0.071
The government should reduce income differences	Econ. policy	c2h	1–5	0.070
Government spending on defense (rev.)	Econ. policy	c18c	1–5	0.070
Own position on immigration (self-placement)	Immigration	c6	1–11	0.032

Notes. σ is the change in the standardized dimension score (Left–right or anti-elite) induced by moving a single item one scale point; it is identical across parties because the score is linear in the answers. To move a dimension score by a 0.25/0.5/1 step using a single item requires about $0.25/\sigma$, $0.5/\sigma$, $1/\sigma$ scale points: e.g. $\approx 1.4/2.9/5.7$ points on the strongest anti-elite item (satisfaction with democracy) and far more than the scale allows for the weak eleven-point immigration self-placement, so sizeable steps must come from moving many items at once. Units and item set are held fixed across years, 2017 survey items are provided as an example.

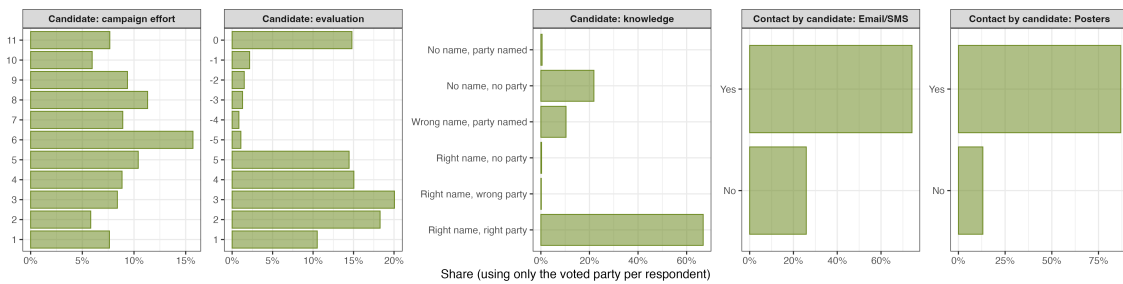
Table F6. Item-level potency (2017): shift in the dimension score from a one-point move on each survey item



(a) **Federal states.** Hessen is in red. (b) **Electoral districts** within Hessen. Hanau is in red. (c) **Municipalities** within Hanau.

Figure F1. Administrative Electoral hierarchy in Germany.

Municipalities are nested within electoral districts, which are nested within federal states. The figure illustrates these levels using one example state and one example electoral district.



Campaign effort (1–11): 1 = mostly about the candidate; 11 = mostly about the party; **Candidate knowledge:** respondents' ability to name the constituency candidate and their party; **Contact:** whether the respondent was contacted via the listed means

Figure F2. Voters knowledge and interaction with SMD candidates

Partei	Insgesamt		davon Wählerinnen und Wähler in der Geburtsjahresgruppe											
			1997 – 2003		1987 – 1996		1977 – 1986		1962 – 1976		1952 – 1961		1951 und früher	
			entspricht Altersgruppe etwa von ... bis ... Jahren											
			18 – 24		25 – 34		35 – 44		45 – 59		60 – 69		70 und mehr	
	1.000	%	1.000	%	1.000	%	1.000	%	1.000	%	1.000	%	1.000	%
Mecklenburg-Vorpommern														
Insgesamt														
Insgesamt	935,0	100	52,1	100	91,9	100	135,3	100	241,1	100	205,3	100	209,4	100
ungültig	17,0	1,8	0,9	1,7	0,9	1,0	2,2	1,6	3,2	1,3	4,1	2,0	5,8	2,8
gültig	918,1	98,2	51,2	98,3	91,0	99,0	133,1	98,4	237,9	98,7	201,2	98,0	203,7	97,2
CDU	180,7	19,7	5,3	10,4	11,2	12,3	22,2	16,7	48,9	20,6	42,9	21,3	50,0	24,6
SPD	261,4	28,5	9,7	18,9	18,8	20,7	29,0	21,8	59,6	25,1	64,4	32,0	80,0	39,3
AfD	170,0	18,5	7,4	14,4	19,2	21,1	33,5	25,2	52,7	22,1	35,7	17,7	21,5	10,6
FDP	66,3	7,2	8,4	16,4	9,1	10,0	11,5	8,6	18,8	7,9	11,5	5,7	7,0	3,4
DIE LINKE	116,0	12,6	6,7	13,1	9,7	10,7	12,5	9,4	25,3	10,7	28,3	14,1	33,5	16,4
GRÜNE	62,7	6,8	8,4	16,5	12,7	14,0	11,4	8,5	15,0	6,3	8,9	4,4	6,3	3,1
CSU	–	–	–	–	–	–	–	–	–	–	–	–	–	–
Sonstige	61,0	6,6	5,3	10,3	10,3	11,3	13,1	9,9	17,5	7,3	9,5	4,7	5,4	2,6
darunter:														
FREIE WÄHLER ¹	18,5	2,0	0,9	1,7	2,9	3,2	3,2	2,4	5,5	2,3	4,1	2,0	1,9	0,9
dieBasis ¹	15,4	1,7	0,8	1,5	1,9	2,1	3,8	2,8	5,3	2,2	2,2	1,1	1,4	0,7

Figure F3. Representative Electoral Statistics: Example

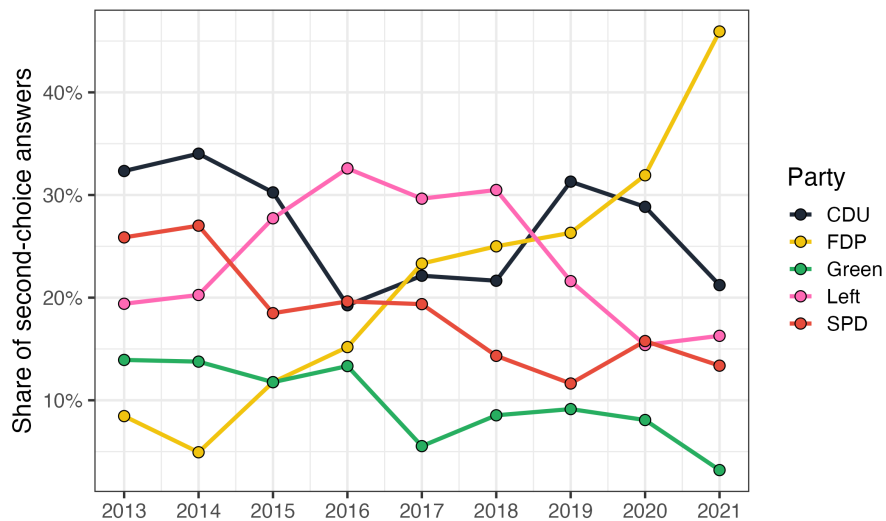


Figure F4. Second best party choices of AfD voters, federal level

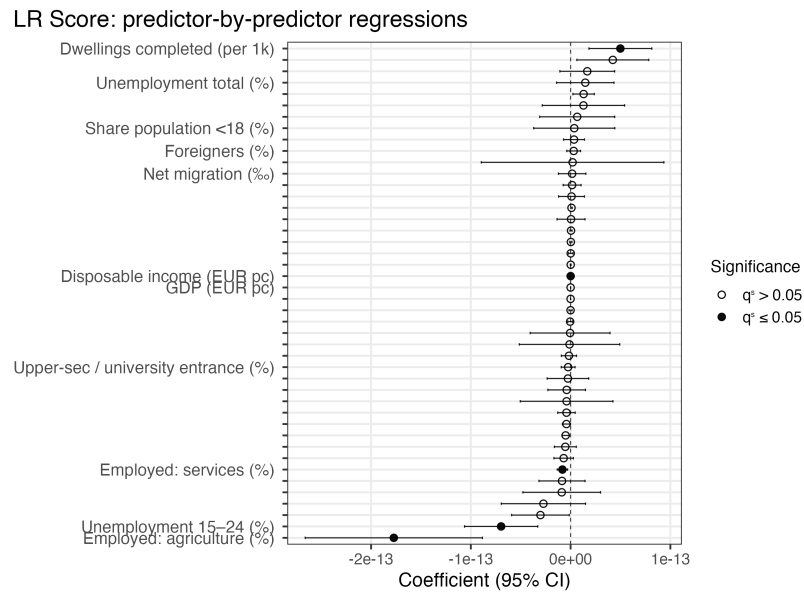


Figure F5. Correlation between candidates position on Left-Right dimension and districts' socio-economic characteristics.

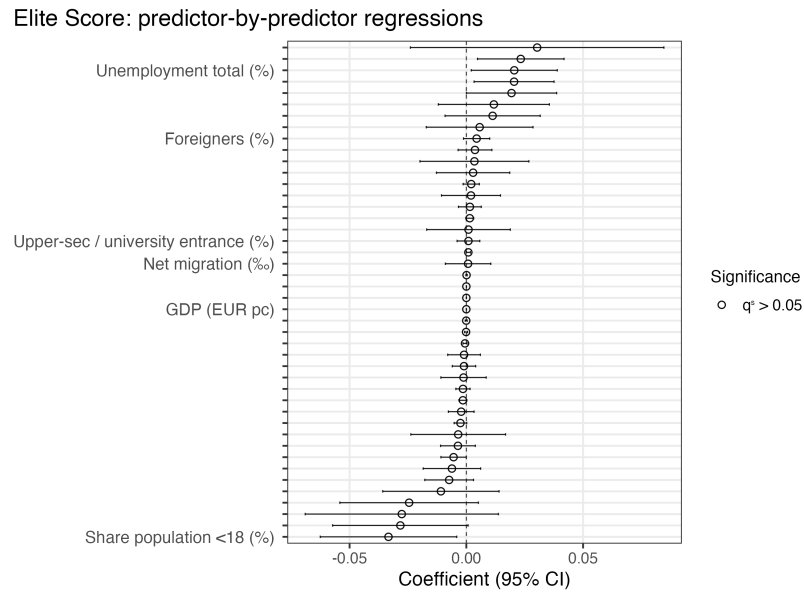
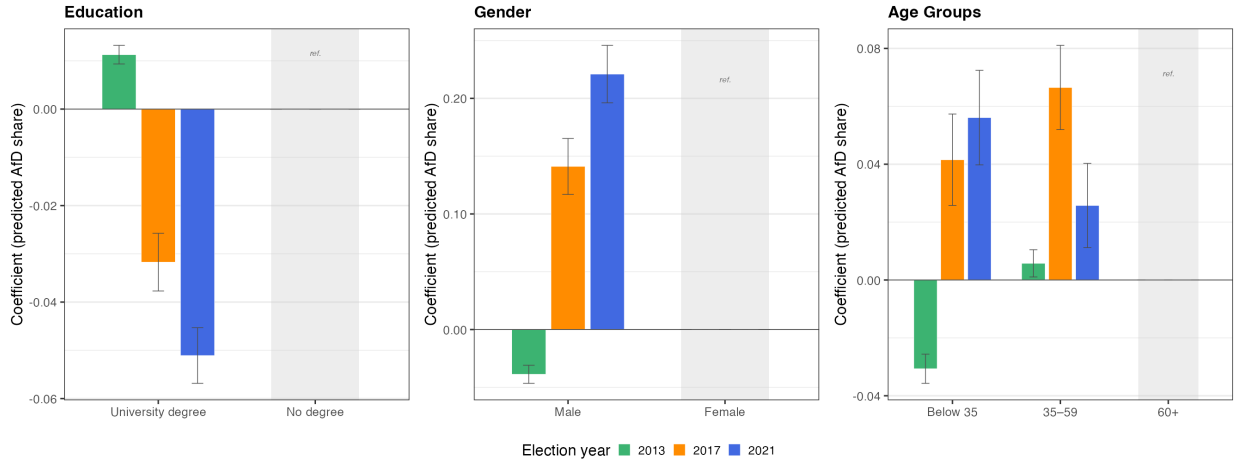
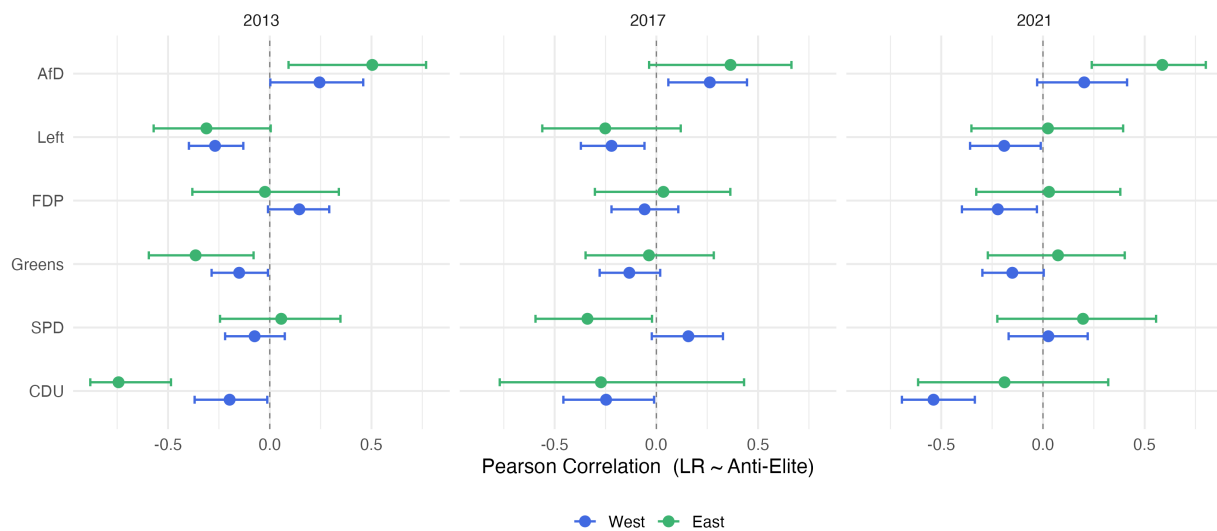


Figure F6. Correlation between candidates position on Anti-Elite dimension and districts' socio-economic characteristics.

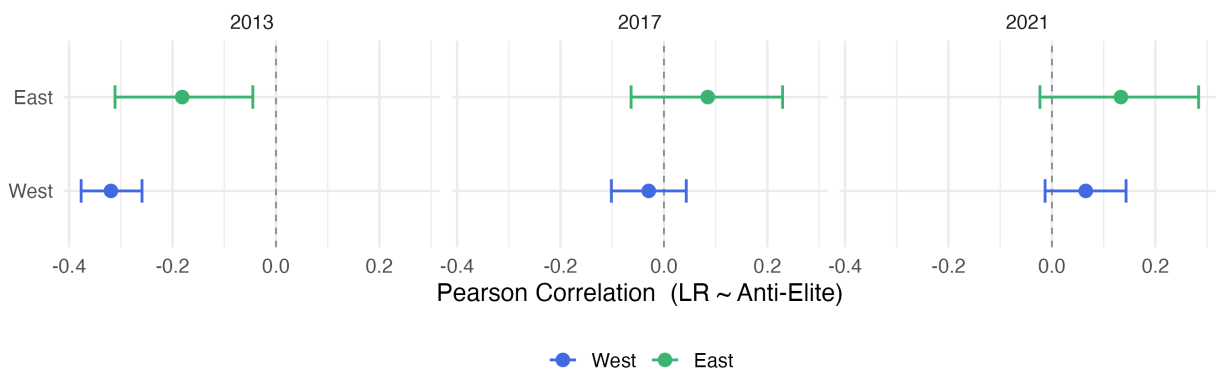


Notes. The panels plot the coefficients β^D from the specification $\text{Predicted AfD Vote Share}_m = \beta^D w_{m,D} + \gamma_{s(m)}$, where m indexes municipalities, $\gamma_{s(m)}$ – federal state fixed effects, and D denotes an aggregate demographic group. Here, $w_{m,D}$ is the share of group D in the population of municipality m . In each of the three plots for education, gender, and age, the omitted reference group is $D = \{\text{No Degree; Female; 60+}\}$, so that $\beta^D = 0$ for this category.

Figure F7. Estimated demographic coefficients by aggregated demographic groups by election year



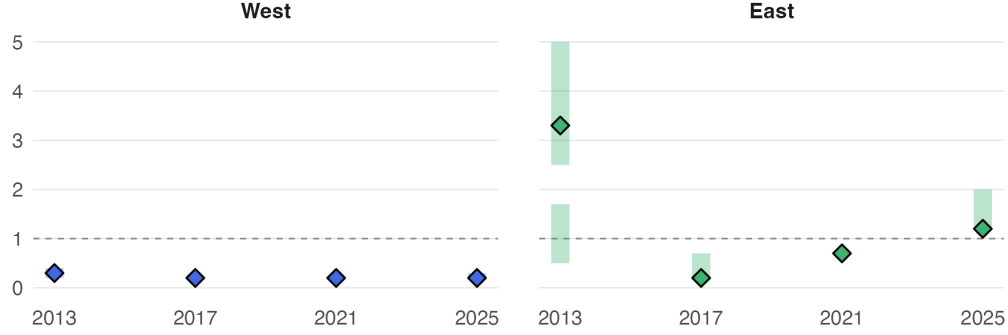
(a) Within parties



(b) Across parties

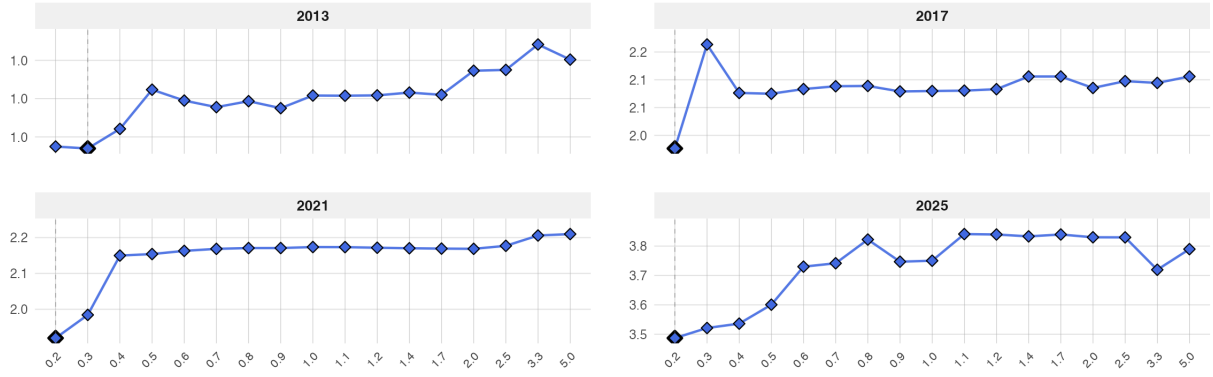
Figure F8. Correlation between candidate positions on Left-Right and Anti-Elite positions

G Additional Results

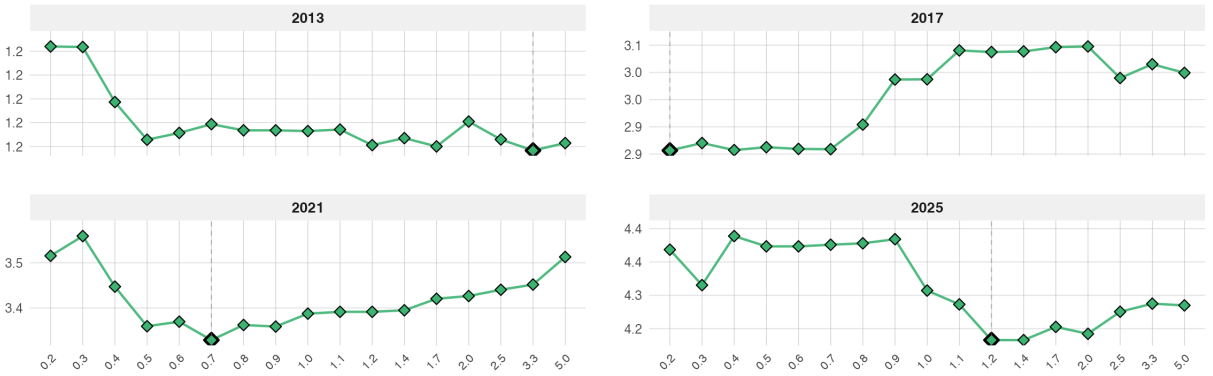


Notes. $w_{LR} = 1$ is fixed throughout, w_{AE} should be interpreted as the relative weight placed on the Anti-Elite dimension. Shaded intervals denote weights within 1% of min objective function value. Separate exercises are run for East and West Germany. I allow ideal-point distributions to vary across federal states – that is, each demographic group d in each market m (federal state) has its own bliss point distribution $f(x | m, d)$; while imposing the same weight w_{AE} for all groups within a given year and region. To run this exercise, the grid of possible weights is constructed. The grid is symmetric around 1 and imply relative weights across the two dimensions of $(0.2, 0.3, \dots, 0.9, 1)$ and the corresponding inverse ratios. See G2 for full results over the grid. Points denote the point estimates corresponding to the best-fitting weight, while the shaded areas indicate intervals over which the objective-function error is less than 0.5% higher (light shading) or 0.25% higher (dark shading).

Figure G1. Set Best Weights for Anti-Elite dimension w_{AE} relative to Left-Right $w_{LR} = 1$ across times and parts of the country



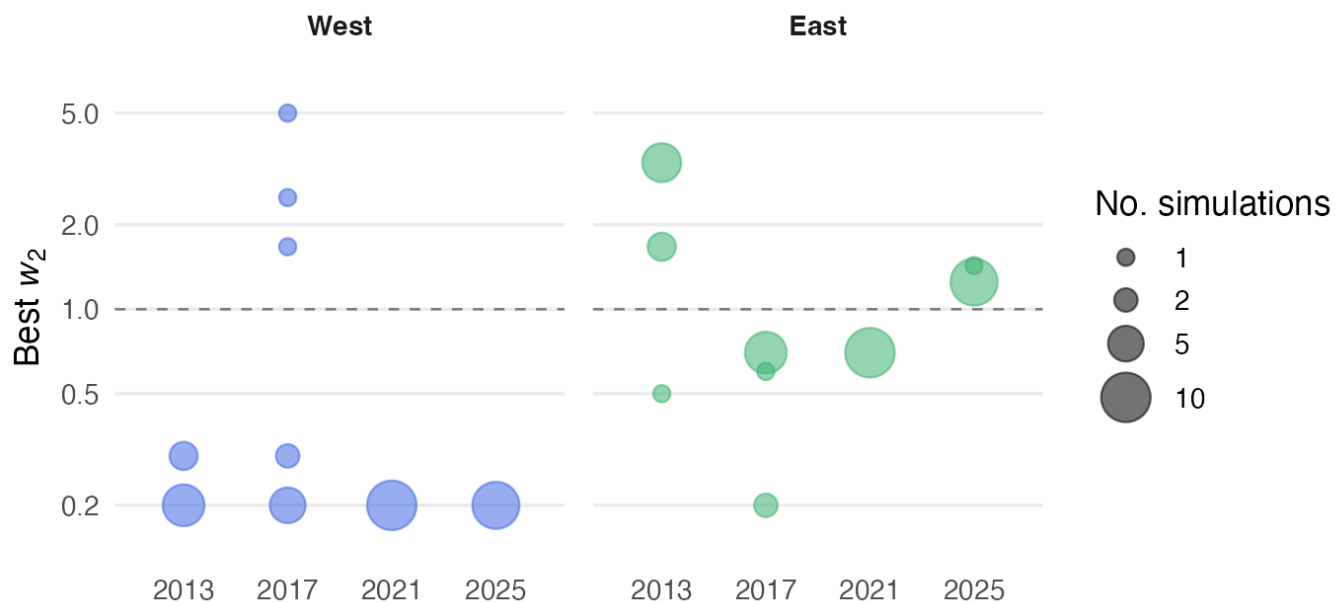
(a) West



(b) East

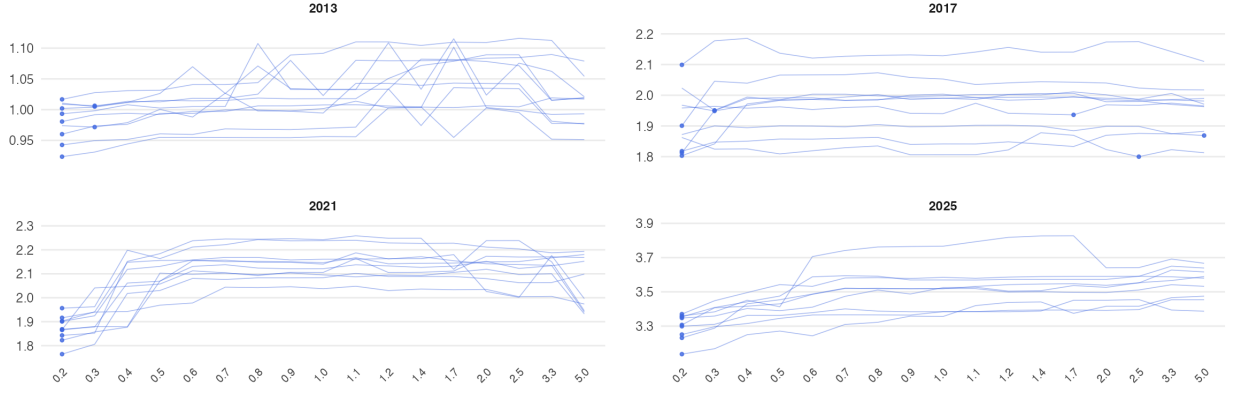
Notes. $w_{LR} = 1$ is fixed throughout, w_{AE} should be interpreted as the relative weight placed on the Anti-Elite dimension. Separate exercises are run for East and West Germany. I allow ideal-point distributions to vary across federal states – that is, each demographic group d in each market m (federal state) has its own bliss point distribution $f(x | m, d)$; while imposing the same weight w_{AE} for all groups within a given year and region. Note that the weights are symmetric around 1 and imply relative weights across the two dimensions of (0.2, 0.25, 0.4, 0.5, 0.75, 1) and the corresponding inverse ratios.

Figure G2. Model Error Value by different relative weight values, years and parts of the country

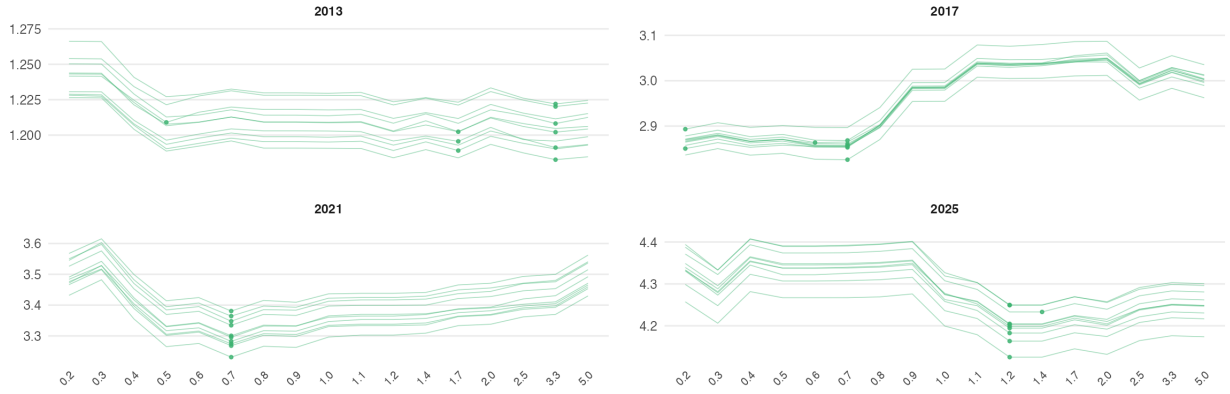


Notes. $w_{LR} = 1$ is fixed throughout, w_{AE} should be interpreted as the relative weight placed on the Anti-Elite dimension. Separate exercises are run for East and West Germany. I allow ideal-point distributions to vary across federal states – that is, each demographic group d in each market m (federal state) has its own bliss point distribution $f(x \mid m, d)$; while imposing the same weight w_{AE} for all groups within a given year and region. Note that the weights are symmetric around 1 and imply relative weights across the two dimensions of $(0.2, 0.25, 0.4, 0.5, 0.75, 1)$ and the corresponding inverse ratios. The exercise was repeated 10 times for each part of the country and each year by subsampling at most 10 electoral districts from each state, with at most 50 municipalities in each district.

Figure G3. Optimal relative weight values by years and parts of the country



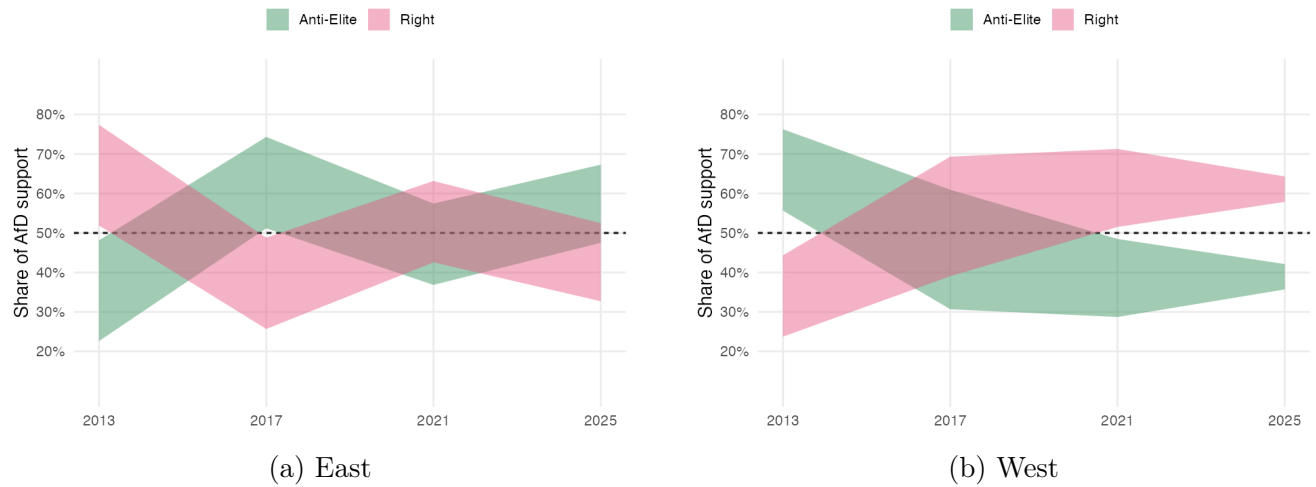
(a) West



(b) East

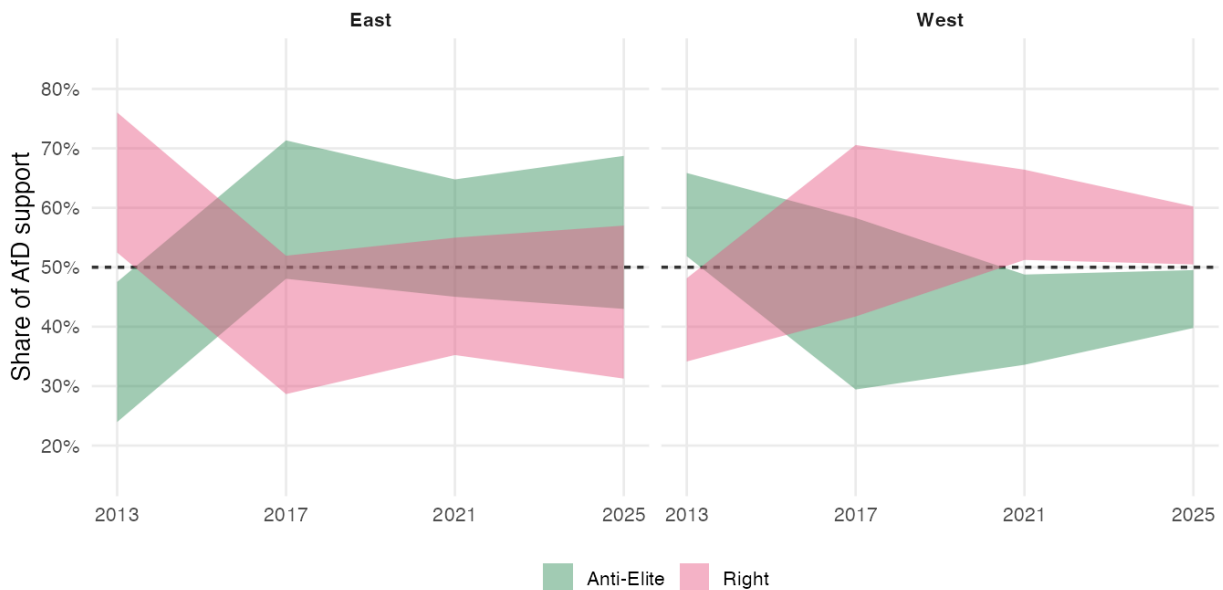
Notes. $w_{LR} = 1$ is fixed throughout, w_{AE} should be interpreted as the relative weight placed on the Anti-Elite dimension. Separate exercises are run for East and West Germany. I allow ideal-point distributions to vary across federal states – that is, each demographic group d in each market m (federal state) has its own bliss point distribution $f(x | m, d)$; while imposing the same weight w_{AE} for all groups within a given year and region. Note that the weights are symmetric around 1 and imply relative weights across the two dimensions of (0.2, 0.25, 0.4, 0.5, 0.75, 1) and the corresponding inverse ratios. The exercise was repeated 10 times for each part of the country and each year by subsampling at most 10 electoral districts from each state, with at most 50 municipalities in each district.

Figure G4. Model Error Value by different relative weight values, years and parts of the country by subsamples



Notes: Panel plots bounds on the shares of voters whose bliss points lie in the attraction areas of two hypothetical parties that retain the AfD's median position within the state on one dimension while adopting a more moderate position on the other: the *Anti-Elite* party is more left-wing (green), whereas the *Right* party is more pro-establishment (pink). Together, these areas recover the full AfD support. See Figure 5 for an illustration.

Figure G5. Regional Heterogeneity: Results separately for West and East Germany.



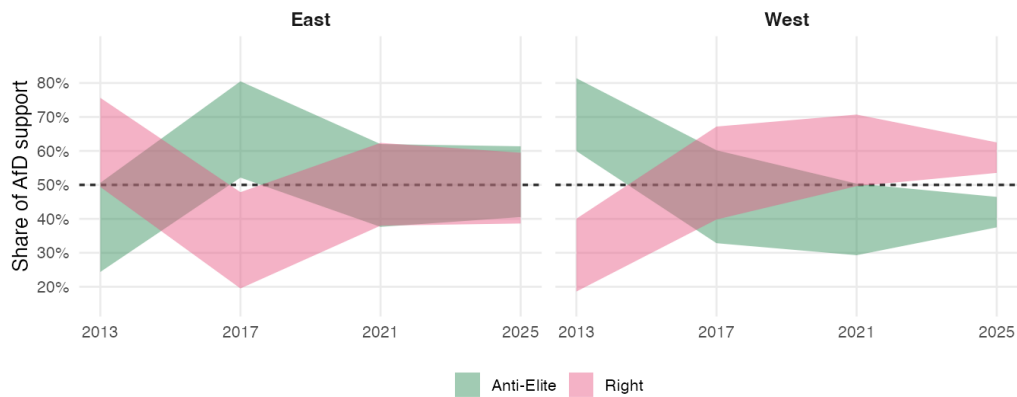
Notes. Bounds on the shares of voters whose bliss points lie in the green and pink regions on the Figure 5a across years are plotted.

Figure G6. Estimated shares of AfD voters in both subspaces on the sample of municipalities with population < 5 thous. inhabitants.



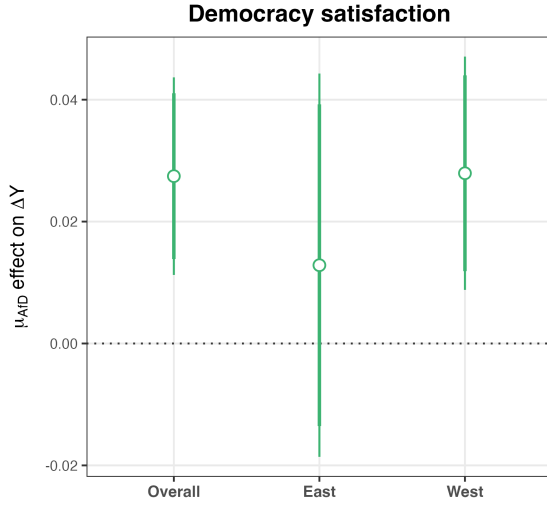
Notes: The figure plots bounds, across years, on the shares of voters whose bliss points lie in the green and pink regions of Figure 5a. Only districts in which the AfD candidate's vote share falls short of the winner's by more than 5% (10%) in the East (West) are retained, for the years 2017 and 2021 (2025). These restrictions drop 12 districts in the East in 2017 and 28 in 2021, and 22 districts in the West in 2025. The East in 2013 and the West in 2013–2021 are omitted because there were no serious contenders, so the results from the main text stand; the East in 2025 is omitted because the AfD won in almost every district, making the analysis infeasible.

Figure G7. Estimated shares of AfD voters in both subspaces on the sample of noncontender electoral districts.

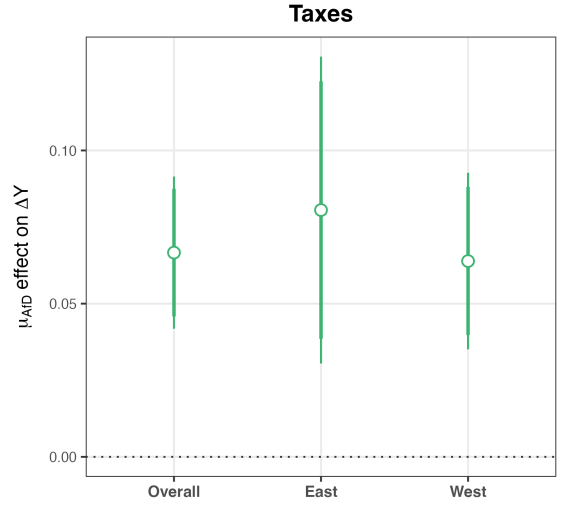


Notes: Bounds on the shares of voters whose bliss points lie in the green and pink regions on the Figure 5a across years are plotted. The analysis allows the estimated ideal-point mass of each demographic group, in each partition cell, to differ by up to a factor of two between conservative and moderate electoral districts of the same state — i.e. the conservative-district mass may range from one-half to twice the moderate-district mass in every cell. A district is classified as conservative if its AfD candidate is positioned above the within-state median AfD position on both the economic (Left–Right) and the anti-elite dimension, and moderate otherwise. This analysis is similar to an instrument in Tebaldi, Torgovitsky and Yang (2023).

Figure G8. Estimated shares of AfD voters in both subspaces allowing for the differences in ideal points distributions between *conservative* and *moderate* electoral districts.



(a) Anti-Elite



(b) Left-Right

Notes: The panels plot an average yearly within-person evolution of preferences of repeat AfD voters relative to voters of other parties. Based on self-reported positions $Y_{i,t}$ I construct $\Delta Y_{i,t} = Y_{i,t} - Y_{i,t-1}$ for consecutive years $t-1, t$, and drop any respondent-year pair for which AfD status changes across the two waves. That is, keep only pairs where either $AfD_{i,t-1} = AfD_{i,t} = 1$ (consistent AfD voters) or $AfD_{i,t-1} = AfD_{i,t} = 0$ (consistent non-AfD voters). Plotted values are μ_{AfD} from: $\Delta Y_{i,t} = \mu + \mu_{AfD} \cdot \mathbf{1}\{AfD_{i,t-1} = AfD_{i,t} = 1\} + \delta_t + \varepsilon_{i,t}$. Higher values correspond to positions closer to the AfD. The point estimate of μ_{AfD} together with 90% (thick) and 95% (thin) confidence intervals in the Overall, East, and West samples are reported. Standard errors clustered at the respondent level.

Figure G9. Results for within-person preference evolution for repeat AfD voters relative to the voters from other parties

	Overall		East		West	
	No FE	Year FE	No FE	Year FE	No FE	Year FE
μ	0.0229*** (0.0047)	0.0260*** (0.0048)	0.0165* (0.0099)	0.0183* (0.0102)	0.0249*** (0.0053)	0.0282*** (0.0054)
Year FE	No	Yes	No	Yes	No	Yes
Obs. (ΔY)	4806	4806	1155	1155	3651	3651
Individuals	2182	2182	506	506	1688	1688

Notes: Sample: AfD voters only (from their first AfD vote onward). For each year t we compute the *median* AfD voter's position in the normalized (taxes, democracy-satisfaction) space, where both outcomes are oriented so that higher values are closer to the AfD, and draw a 45 degree line through the point, and define a dummy $\text{Below45}_{i,t}$ for being 'below' the 45 degree line. Estimate, on consecutive-year pairs in which i is an AfD voter in t , $\text{Below45}_{i,t+1} - \text{Below45}_{i,t} = \mu + \delta_t + \varepsilon_{i,t}$. Standard errors are clustered at the respondent level. Columns are estimated separately on the Overall, East, and West subsamples.

Table G1. Annual drift in AfD voters "below 45°" dummy

H Candidate data

H.1 Candidate nominations across state lists and single member districts

The data on all the nominated politicians with their names and party affiliations was obtained by scraping the non-partisan platform *Wen Wählen? (For Whom To Vote For?)*. This platform sets as its goal providing comprehensive and easy-access information about federal elections in Germany. In particular, it compiles a comprehensive set of candidates participating in elections (which would not be available otherwise), as well as reaches to the candidates to ask them to participate in an electoral survey about their positions. While the portal provides information on the universe of nominated candidates, I focus just on the candidates from the main parties considered in the text of the paper (AfD, CDU/CSU, FDP, Left, Green, SPD).

The nominations done by parties are, given mixed electoral voting system, separated into two parts. In Bundestag elections, the nomination of direct candidates is formally decentralized to constituency-level members' or delegates' assemblies. State/Land party actors do not normally nominate direct candidates themselves, though they possess limited gatekeeping powers. Their strategic influence is stronger at Land level because Land-list rankings determine many actual Bundestag careers, and local selectorates often choose district candidates with Land-list prospects in mind. In practice, around 64% of candidates who appear on the state lists, also run in single member districts. This implies that in many ways the state lists, chosen by state leadership, is at least constraining the choice of the candidates to the electoral districts by local electoral units. These values differ across parties and years – Figure H2 illustrates that.

Importantly for the purpose of this study, the share of AfD candidates who are nominated both ways is very high, especially in 2021 (78%). Suggesting that most of the candidates were nominated, or at least approved, by the state level leadership. Such patterns could also indicate lack of candidates who would like to be balloted. It is also clear from the data that successful candidates in the SMD are more likely to be added to the state lists – Figure H3. Overall, politicians who are on the state lists are more successful, and heterogeneously so – those on higher positions are more successful, (Figure H4).

The presence of a candidate on both state and single member districts lists lower motives

for these politicians to target their constituency, as they could reasonably expect to be elected to Bundestag no matter the electoral outcome in the SMD. One example could be the leader of AfD, Alice Weidel. She is being nominated from the Electoral district 293, Bodensee in Baden-Wurtemberg. She does not have realistic chances of winning the district, but she is being placed first on the state list, and, hence, gets reelected.

I use this data to explore the patterns of candidate nominations to single member districts. In particular, I ask how likely a candidate is to be 1) renominated, 2) renominated to the same district, and how it depends on his electoral performance.

Across 2013, 2017, 2021 there were 5215 unique nominations³⁸ (at the level of elections-candidates). Among them, 985 were repeated nominations from one electoral cycle to another – e.g. the same candidate could be nominated in 2013 and 2017, or in 2017 and 2021. Out of all the nominations in 2017 and 2021 repeated nominations make up 28.7% of all the nominations. Among the renominations only 58 (5.8%) happen to be in another electoral district, and only 4 (< 0.5%) in another federal state. This suggests that when a candidate is renominated, it is usually within the same federal state, which is consistent with strong influence of state-level party organizations over nominations.

Renomination patterns are different by electoral success of the candidate. Among those candidates who win their electoral district, 58% get renominated; among those who lose – 21.7%. Figure H1 demonstrates this pattern in more details – for each of the observed largest parties the probability of renomination is positively correlated with first-vote share.

Upon a victory and renomination, a candidate gets renominated to the same district in 100% of cases³⁹. Among those who lose and get renominated 87.4% get nominations in the same electoral district.

The strong attachment of candidate to their electoral districts (and federal states) is in part due to where these candidates reside at the time of the elections. To take a closer look at this question, I exploit the question of the GLES Candidate Survey which asks Bundestag Candidates how long they have been residing in this electoral district. Importantly, participation in the survey is voluntary, so not all candidates respond, and the information is self-reported. That said, the survey is conducted for research purposes and is not released before the election, which should limit any “demand-side” effects.

The analysis of this data reveals that share of candidates born in the electoral district in which they run was 28%, 16%, 31% in years 2013, 2017, 2021 respectively. Mean number of years of residence in a given electoral district are 25, 25.3, 25.9 across the same years. The share of candidates who lived in these districts for less than 10 years is about 20% across all the years.

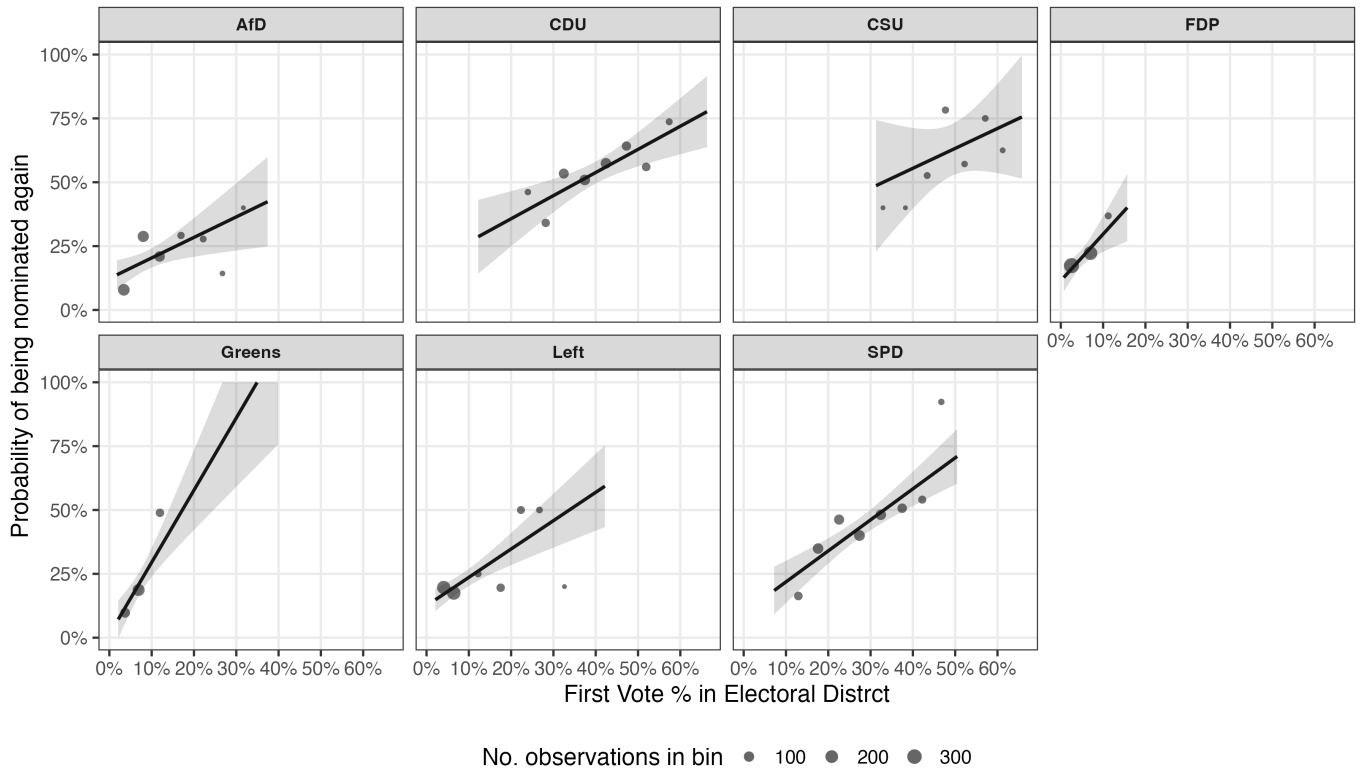
This highlights that while almost one third of candidates are local, there also is a non-negligible share of those who moved not that long before the elections. This could happen for strategic reasons related to the career of a politician, as well as for completely unrelated issues.

³⁸Note that the max number is 5382. It is slightly because of two parties – AfD and Greens – not nominating candidates to all the districts in some years.

³⁹There is 1 case in which a candidate was nominated to *another* district due to the change in districts’ boundaries. This was Albert Helmut Weiler who won the district of Gera – Jena – Saale-Holzland-Kreis in 2013 which was dissolved by 2017 elections. He was nominated and won 2017 elections in a district Saalfeld-Rudolstadt – Saale-Holzland-Kreis – Saale-Orla-Kreis, which was partly composed from the district in which he ran in 2013.

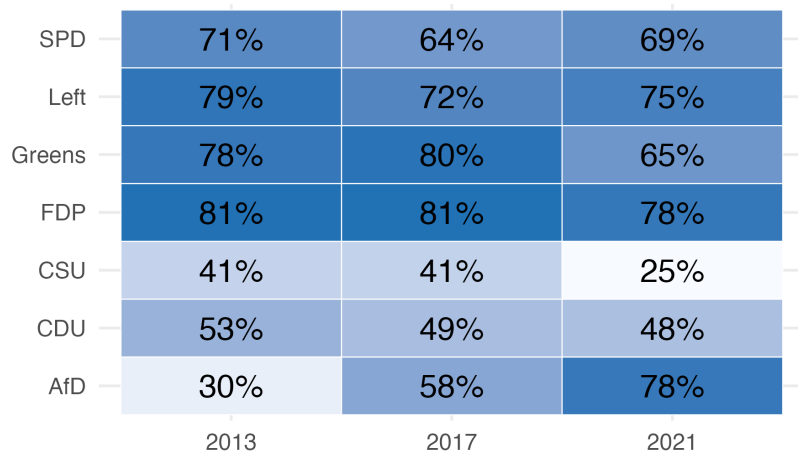
In light of the main analysis in the paper, I interpret this as following. When nominating candidates at the federal level, parties take into account candidates' personal histories – this could be for the logistics reason, as well as for the reasons related to the fact that such candidates could be more recognizable and likable by voters. In practice, candidates who are elected via a direct vote are expected to spend some time in their electoral district, and communicate with their constituency. This would often require easier presence and connection with the district. My identification claims rests on the assumption that this mechanism of candidate selection could be happening even at the expense of ideological congruence.

The picture of renomination I highlighted before draws a similar picture – many candidates get renominated to the same districts, even if they lose their elections. I claim that this happens mostly because it is logistically easier for the parties to keep going with the same candidate, instead of trying to find a new, and better one.



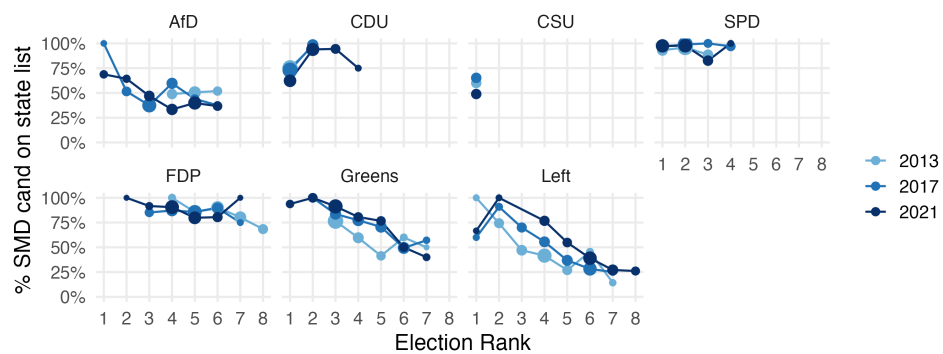
Notes. The data for this figure are drawn from the non-partisan candidate platform Wen Wählen?, which provides information on all candidate lists for federal elections. Vote share refers to the candidate's first-vote (Erststimme) share in their electoral district. The spline plots fitted values from an OLS regression of the form $Renominated_{i,t+1} = \beta_0 + \beta_1 Erststimme_{i,t} + \epsilon_{i,t}$.

Figure H1. Probability of renomination by candidate vote shares across main parties.



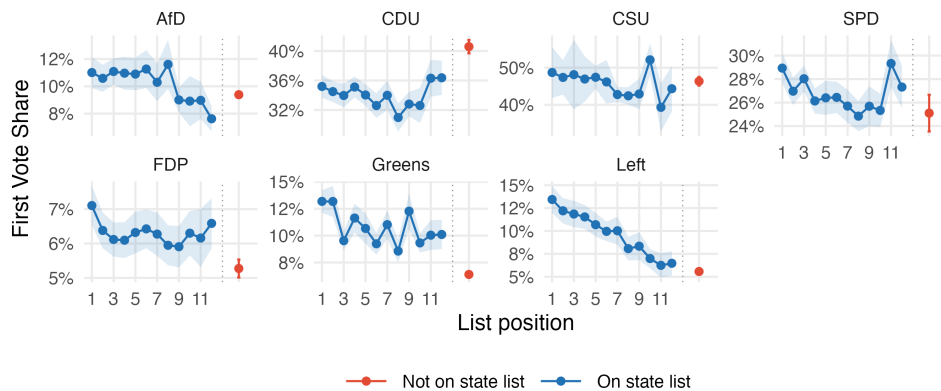
Notes. The data for this figure are drawn from the non-partisan candidate platform Wen Wählen?, which provides information on the candidate lists and state lists for federal elections. This data provides the universe of all party nominations.

Figure H2. Share of candidates from the state lists nominated in single member districts



Notes. The data for this figure are drawn from the non-partisan candidate platform Wen Wählen?, which provides information on the candidate lists and state lists for federal elections. This data provides the universe of all party nominations. Only dots representing more than 1 candidate are plotted.

Figure H3. Share of SMD candidates nominated in state lists by their electoral success



Notes. The data for this figure are drawn from the non-partisan candidate platform *Wen Wählen?*, which provides information on the candidate lists and state lists for federal elections. This data provides the universe of all party nominations.

Figure H4. First Vote Share of SMD candidates by their list position

H.2 Estimation of candidates' positions

1. Estimation of candidates' positions using GLES Candidate Survey Data Candidate positions on two dimensions — left-right and anti-elite — are estimated from the GLES Candidate Studies for 2013, 2017, and 2021. For each dimension, a set of survey items is first harmonized (see Table H1 for the full list of questions). Each item is then z-scored using a pooled mean and standard deviation computed across all waves in which it appears. This strategy places all items on a common metric and preserves comparability across years.

Each candidate's score on a dimension is the unweighted mean of all available z-scored items within that dimension. The anti-elite dimension comprises seven questions on satisfaction with democracy and current institutions; the economic left-right dimension uses questions covering redistribution, state intervention, immigration, and social issues, such as abortion and same-sex marriages. It is also supplemented by a left-right self-placement scale. Missing scores are imputed with the within-state-wave median; if that median is also unavailable, the global within-wave median serves as a fallback.

Because of the low response rate to this survey (around 35% across all years and parties), the survey was complemented with a non-partisan candidate survey *Wen Wählen*, which also asks candidates about their positions on different issues in each electoral year.

2. Complementing GLES Candidate Survey Data with non-partisan candidate survey *Wen Wählen*. For each year, candidates from the GLES Candidate Survey are merged with candidate data from the non-partisan platform *Wen Wählen*. These three yearly data frames are then stacked into one pooled data set with year retained as a predictor. The elastic net procedure is then used to impute ideological scores for those candidates missing in the GLES Survey, but present in *Wen Wählen*.

Two separate elastic net models are estimated, one for each ideological dimension. The outcome y_i is the GLES-derived score for the candidate. Only candidates who are matched to the GLES candidate survey enter the training set.

The feature matrix $\mathbf{X}$ contains, for candidate i :

1. The encoded responses to the relevant policy questions (17 questions for the economic model, 12 for the anti-elite model), scored on the five-point scale $\{Ja! = +2, Eher\ ja = +1, Unentschieden = 0, Eher\ nicht = -1, Nein! = -2\}$.
2. One-hot indicators for party affiliation (CDU/CSU, SPD, Greens, FDP, Left, AfD).
3. One-hot indicators for election year (2013, 2017, 2021).

The elastic net minimises a penalised least-squares criterion that interpolates between Ridge (ℓ_2) and Lasso (ℓ_1) regularisation:

$$\hat{\beta}(\lambda, \alpha) = \arg \min_{\beta} \frac{1}{n} \sum_{i=1}^n (y_i - \mathbf{x}_i^\top \beta)^2 + \lambda \left[\alpha \sum_j |\beta_j| + \frac{1-\alpha}{2} \sum_j \beta_j^2 \right]$$

where n is the number of candidates in the training set, $\lambda \geq 0$ controls the overall amount of regularisation, and $\alpha \in [0, 1]$ determines the mix between the two penalty types. Setting $\alpha = 1$ recovers the Lasso, which drives many coefficients to exactly zero (implicit variable selection); $\alpha = 0$ recovers Ridge, which shrinks all coefficients towards zero while retaining all predictors. Intermediate values of α combine both effects.

Both λ and α are selected jointly by 10-fold cross-validation on the labelled candidates (those with a GLES score). The procedure is:

1. The labelled training set is randomly partitioned into 10 roughly equal folds
2. A full Cartesian grid of 330 hyperparameter combinations is constructed:
 - $\lambda \in \{10^{-4}, 10^{-3.69}, \dots, 10^1\}$ (30 values, log-uniform spacing);
 - $\alpha \in \{0.0, 0.1, 0.2, \dots, 1.0\}$ (11 values).
3. For each of the $330 \times 10 = 3\,300$ fold-hyperparameter combinations fit is run exclusively on the nine training folds and then applied to the held-out validation fold
4. The root mean squared error (RMSE) between predicted and observed scores is recorded for each held-out fold.
5. The pair $(\hat{\lambda}, \hat{\alpha})$ that minimizes the mean RMSE across the 10 folds is selected.

The winning model is then refit on the full labelled training set (all three election years pooled) using $(\hat{\lambda}, \hat{\alpha})$, and this final fit is used to predict scores for unlabelled candidates.

The model performs well, explaining large share of variation in observed GLES-based economic position scores ($R^2 = 0.789$), with a root mean squared error of 0.316 against a target standard deviation of 0.689 (RMSE/SD = 0.459). This indicates that the elastic net predictions are able to capture meaningful within-party variation in candidate positions across districts.

Performance is somewhat lower for the anti-elite score, where the elastic net achieves an RMSE of 0.430 against a target standard deviation of 0.668 (RMSE/SD = 0.644, $R^2 = 0.585$). This is expected, as anti-elite score is harder to be mapped by the policy-stance questions available in Wen Wählen. Nonetheless, the model still explains more than half the variance in observed anti-elite

scores, representing a clear improvement over mean imputation.

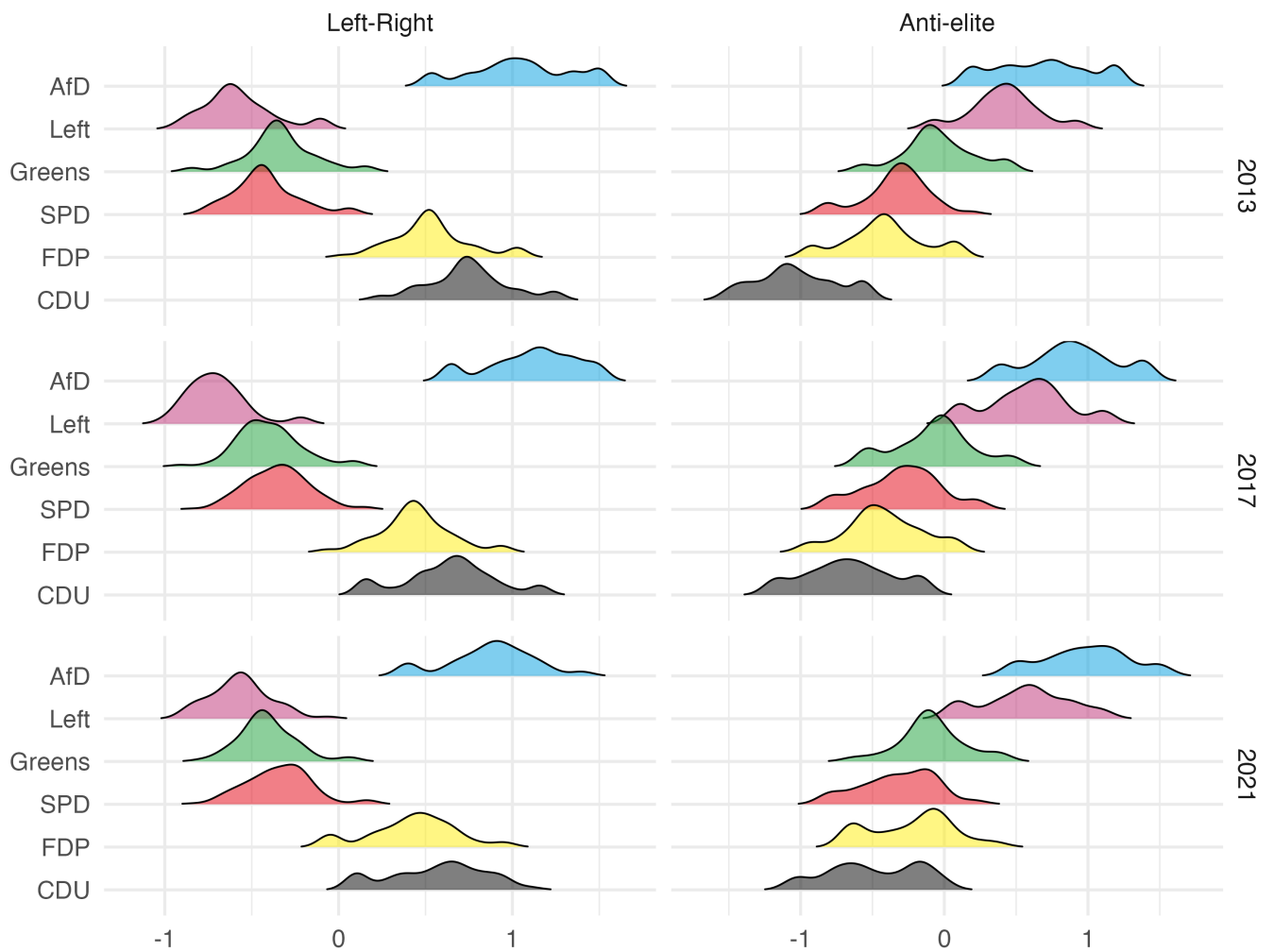
Both models are trained on a pooled sample spanning 2013, 2017, and 2021, using harmonized question numbering. To mitigate overfitting, performance metrics are computed from out-of-fold cross-validation predictions. Combining the two candidate surveys allows me to recover policy positions for about 65% of all candidates.

Dimension	Question	Values
Anti-Elite	Satisfaction with democracy in Germany	1–4
Anti-Elite	Democracy offers enough opportunities for citizens to participate	1–5
Anti-Elite	Laws reflect citizens’ interests	1–5
Anti-Elite	Political parties are the central intermediary between citizens and government	1–5
Anti-Elite	There should be nationwide referendums (direct votes) at the federal level	1–5
Anti-Elite	Referendums tend to lead to bad laws	1–5
Anti-Elite	Citizens have lost trust in politicians	1–5
Left–Right	In a crisis, the government should spend to stimulate economic growth	1–5
Left–Right	In a crisis, Germany should provide financial support to other countries/EU partners	1–5
Left–Right	Immigrants should adapt to German culture	1–5
Left–Right	Government should stay out of the economy	1–5
Left–Right	Same-sex marriage should be banned	1–5
Left–Right	Providing social security should be a key goal of government	1–5
Left–Right	Government should reduce income differences	1–5
Left–Right	Immigration is good for the German economy	1–5
Left–Right	Women should be free to decide about abortion	1–5
Left–Right	Self-placement on the left–right scale	1–11
Left–Right	Immigration policy: make immigration easier vs. restrict it	1–11

Table H1. Survey questions used to estimate candidate positions (GESIS Candidate Survey Data)

Dimension	Question	Values
Anti-Elite	Deploy the Bundeswehr domestically! (Q7)	-2 to +2
Anti-Elite	Less anonymity on the internet! (Q11)	-2 to +2
Anti-Elite	Germany into the UN Security Council! (Q27)	-2 to +2
Anti-Elite	More video surveillance! (Q28)	-2 to +2
Anti-Elite	A more transparent state! (Q31)	-2 to +2
Anti-Elite	Re-introduce the D-Mark! (Q40)	-2 to +2
Anti-Elite	Leave the EU! (Q41)	-2 to +2
Anti-Elite	Stay out of international conflicts! (Q44)	-2 to +2
Anti-Elite	Disclose MPs' secondary activities more clearly! (Q45)	-2 to +2
Anti-Elite	Stop Russia sanctions! (Q48)	-2 to +2
Anti-Elite	We have too much data protection! (Q54)	-2 to +2
Anti-Elite	Leave NATO! (Q55)	-2 to +2
Left-Right	Not too many changes! (Q1)	-2 to +2
Left-Right	Significantly increase the minimum wage! (Q5)	-2 to +2
Left-Right	Introduce a digital tax! (Q6)	-2 to +2
Left-Right	Statutory health insurance for all! (Q8)	-2 to +2
Left-Right	Mandatory pension insurance for the self-employed! (Q9)	-2 to +2
Left-Right	Tax wealth again! (Q10)	-2 to +2
Left-Right	Growth is more important than debt reduction! (Q17)	-2 to +2
Left-Right	The welfare state is more important than debt reduction! (Q18)	-2 to +2
Left-Right	Introduce an unconditional basic income! (Q19)	-2 to +2
Left-Right	Increase Hartz-IV benefits! (Q23)	-2 to +2
Left-Right	Privatise more! (Q26)	-2 to +2
Left-Right	Abolish same-sex marriage! (Q30)	-2 to +2
Left-Right	Regulate rents more strongly! (Q32)	-2 to +2
Left-Right	Better integrate foreigners! (Q33)	-2 to +2
Left-Right	Strengthen rights of trans and non-binary people! (Q35)	-2 to +2
Left-Right	Fewer Muslims! (Q47)	-2 to +2
Left-Right	Strengthen the traditional family! (Q52)	-2 to +2

Table H2. Survey questions used to estimate candidate positions (Wen Wählen)



Notes. The data is based on the estimated single-member candidate positions using GLES Candidate Survey and non-partisan platform Wen Wählen.

Figure H5. Distributions of left-right and anti-elite positions across candidates over years

I Representative Voting Statistics

State	18-25				25-35				35-45				45-60				60-70				70+			
	'17	'21	Δ_{pp}	$\Delta_{\%}$	'17	'21	Δ_{pp}	$\Delta_{\%}$	'17	'21	Δ_{pp}	$\Delta_{\%}$	'17	'21	Δ_{pp}	$\Delta_{\%}$	'17	'21	Δ_{pp}	$\Delta_{\%}$	'17	'21	Δ_{pp}	$\Delta_{\%}$
Schleswig-Holstein	3.7	3.8	+0.1	+3	6.5	7.7	+1.2	+18	6.3	7.1	+0.8	+13	6.7	5.5	-1.2	-18	5.9	4.5	-1.4	-24	3.6	2.7	-0.9	-25
Niedersachsen	4.5	3.9	-0.6	-13	7.7	6.1	-1.6	-21	8.8	8.4	-0.4	-5	7.4	6.6	-0.8	-11	6.5	5.0	-1.5	-23	3.6	2.7	-0.9	-25
Nordrhein-Westfalen	4.5	3.4	-1.1	-24	7.3	5.7	-1.6	-22	8.9	7.8	-1.1	-12	8.2	6.5	-1.7	-21	6.9	5.6	-1.3	-19	3.8	2.6	-1.2	-32
Hessen	5.7	4.0	-1.7	-30	8.6	7.2	-1.4	-16	10.7	8.6	-2.1	-20	10.0	8.3	-1.7	-17	9.1	7.5	-1.6	-18	6.0	4.0	-2.0	-43
Rheinland-Pfalz	5.5	4.3	-1.2	-22	9.5	8.2	-1.3	-14	10.4	10.5	+0.1	+1	9.9	8.2	-1.7	-17	8.5	6.1	-2.4	-28	4.2	3.4	-0.8	-19
Baden-Württemberg	5.6	4.0	-1.6	-29	8.6	7.1	-1.5	-17	10.2	9.9	-0.3	-3	10.3	9.0	-1.3	-13	9.9	8.0	-1.9	-19	5.9	4.0	-1.9	-32
Bayern	6.5	5.1	-1.4	-22	8.8	7.6	-1.2	-14	10.2	8.8	-1.4	-14	10.9	8.3	-2.6	-24	9.6	6.5	-3.1	-32	5.8	3.6	-2.2	-38
Saarland	6.7	4.2	-2.5	-37	6.9	9.6	+2.7	+39	11.1	11.6	+0.5	+5	9.2	8.9	-0.3	-3	6.7	7.4	+0.7	+10	3.0	3.6	+0.6	+20
Brandenburg	11.5	10.6	-0.9	-8	19.1	17.7	-1.4	-7	18.4	19.9	+1.5	+8	17.1	16.9	-0.2	-1	14.2	12.3	-1.9	-13	8.2	7.3	-0.9	-11
Mecklenburg-Vorpommern	9.7	10.8	+1.1	+11	15.8	15.8	0.0	0	17.0	18.7	+1.7	+10	15.9	16.3	+0.4	+3	13.0	13.5	+0.5	+4	8.6	7.6	-1.0	-12
Sachsen	17.3	13.0	-4.3	-25	21.7	19.7	-2.0	-9	23.3	22.9	-0.4	-2	26.2	24.7	-1.5	-6	21.9	22.5	+0.6	+3	14.1	13.2	-0.9	-6
Sachsen-Anhalt	8.1	11.1	+3.0	+37	18.7	19.2	+0.5	+3	18.8	22.0	+3.2	+17	17.0	19.1	+2.1	+12	14.1	13.8	-0.3	-2	8.3	8.0	-0.3	-4
Thüringen	13.1	14.5	+1.4	+11	20.0	23.0	+3.0	+15	22.1	26.6	+4.5	+20	20.5	23.6	+3.1	+15	16.0	17.8	+1.8	+11	10.9	10.4	-0.5	-5

Notes: AfD vote shares from Representative Statistics for the 2017 and 2021 federal elections. Δ_{pp} = absolute change in percentage points (2021 – 2017); $\Delta_{\%}$ = relative change in percent, i.e. $(s_{21} - s_{17})/s_{17} \times 100$. States are ordered West (above rule) then East (below rule).

Table II. AfD vote shares by federal state and age group: **Female voters**

State	18-25			25-35			35-45			45-60			60-70			70+								
	'17	'21	Δ_{pp}	$\Delta_{\%}$	'17	'21	Δ_{pp}	$\Delta_{\%}$	'17	'21	Δ_{pp}	$\Delta_{\%}$	'17	'21	Δ_{pp}	$\Delta_{\%}$	'17	'21	Δ_{pp}	$\Delta_{\%}$				
Schleswig-Holstein	6.4	5.5	-0.9	-14	12.3	11.1	-1.2	-10	14.9	12.9	-2.0	-13	12.9	10.6	-2.3	-18	9.9	7.7	-2.2	-22	7.5	4.5	-3.0	-40
Niedersachsen	8.9	6.0	-2.9	-33	13.7	9.9	-3.8	-28	16.8	14.4	-2.4	-14	13.7	11.6	-2.1	-14	10.8	9.0	-1.8	-17	7.5	4.9	-2.6	-35
Northrhein-Westfalen	8.2	5.6	-2.6	-32	12.6	8.9	-3.7	-29	15.4	13.5	-1.9	-12	14.9	11.6	-3.3	-22	12.2	9.5	-2.7	-22	7.8	5.1	-2.7	-35
Hessen	8.7	5.2	-3.5	-40	15.0	10.2	-4.8	-32	18.5	13.6	-4.9	-26	17.9	14.1	-3.8	-21	16.3	11.4	-4.9	-30	11.6	7.4	-4.2	-36
Rheinland-Pfalz	9.1	7.5	-1.6	-18	15.4	12.4	-3.0	-19	18.6	17.6	-1.0	-5	16.6	14.7	-1.9	-11	14.1	10.5	-3.6	-26	9.8	6.6	-3.2	-33
Baden-Württemberg	10.0	7.2	-2.8	-28	14.8	10.9	-3.9	-26	19.0	16.0	-3.0	-16	18.6	15.3	-3.3	-18	16.5	11.9	-4.6	-28	11.5	7.4	-4.1	-36
Bayern	11.0	8.0	-3.0	-27	16.1	11.5	-4.6	-29	18.8	14.9	-3.9	-21	19.0	13.7	-5.3	-28	16.5	11.0	-5.5	-33	10.8	6.1	-4.7	-44
Saarland	7.3	6.8	-0.5	-7	14.4	15.2	+0.8	+6	19.3	19.1	-0.2	-1	15.1	15.6	+0.5	+3	12.3	12.3	0.0	0	8.9	7.1	-1.8	-20
Brandenburg	16.4	14.5	-1.9	-12	28.2	25.5	-2.7	-10	31.5	29.0	-2.5	-8	30.3	29.0	-1.3	-4	25.2	20.0	-5.2	-21	16.6	12.5	-4.1	-25
Mecklenburg-Vorpommern	13.3	15.3	+2.0	+15	24.8	23.2	-1.6	-6	29.2	29.2	0.0	0	28.8	27.4	-1.4	-5	24.1	21.8	-2.3	-10	16.4	14.4	-2.0	-12
Sachsen	22.2	17.4	-4.8	-22	31.9	24.4	-7.5	-24	36.0	34.4	-1.6	-4	39.3	37.1	-2.2	-6	35.1	32.7	-2.4	-7	23.8	21.2	-2.6	-11
Sachsen-Anhalt	15.7	17.1	+1.4	+9	26.4	25.9	-0.5	-2	30.3	33.4	+3.1	+10	30.6	31.0	+0.4	+1	24.9	23.1	-1.8	-7	16.1	14.4	-1.7	-11
Thüringen	20.3	21.1	+0.8	+4	29.9	30.8	+0.9	+3	35.1	36.4	+1.3	+4	34.0	37.0	+3.0	+9	27.3	27.8	+0.5	+2	17.4	16.6	-0.8	-5

Notes: AfD vote shares from Representative Statistics for the 2017 and 2021 federal elections. Δ_{pp} = absolute change in percentage points (2021 – 2017); $\Delta_{\%}$ = relative change in percent, i.e. $(s_{21} - s_{17})/s_{17} \times 100$. States are ordered West (above rule) then East (below rule).

Table I2. AfD vote shares by federal state and age group: Male voters